

COURSE SLIDES

Certified Kubernetes Application Developer (CKAD)

WSQ Course Code: TGS-2025053212

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version 1.0

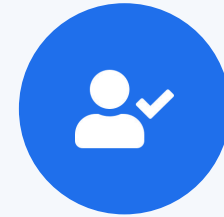


COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your phone camera and submit your attendance.
- A minimum of 75% attendance is required for assessment and funding.



**Minimum 75%
attendance required**

YOUR TRAINER

About the Trainer



Mohan Pothula

Kubernetes & Cloud-Native Trainer



Role

Principal Kubernetes & DevOps trainer at Tertiary Infotech Academy.



Qualifications

Certified Kubernetes Application Developer (CKAD) · CKA · Docker Certified Associate · AWS Solutions Architect.



Expertise

Kubernetes · Docker · CI/CD · Cloud-native platform engineering.



Experience

20+ years across DBS Bank, SingTel, Mediacorp and SPH Media.

YOUR TRAINER

About the Trainer



[Trainer Name]

[Title / Role]



Qualifications:



Expertise:



Experience:



Contact / Profile:

Let's Know Each Other

Take a minute to introduce yourself to the class:



Your role

Your name and organisation / role.



Your experience

Any experience with Docker, Kubernetes or the cloud.



Your goal

Whether you are targeting the CKAD exam or production skills.

Ground Rules



Phones on silent

Set your phone to silent mode.



Participate actively

No question is too small — ask freely.



Mutual respect

One conversation at a time.



Be punctual

Return from breaks on time.



Step out quietly

For calls or short breaks.



75% attendance

Required for funding eligibility.

SCHEDULE

Lesson Plan — 3½ Days, 8 hours/day

1

Day 1 · Domain 1

Application Design & Build

- Container images & multi-stage builds (Labs 1–2)
- Pods, Jobs & CronJobs (Labs 3–5)
- Multi-container Pods & Volumes (Labs 6–8)

2

Day 2 · Domain 2

Application Deployment

- Deployments, ReplicaSets & rollouts (Labs 9–10)
- Blue/green & canary (Labs 11–12)
- Helm, Kustomize, Daemon/StatefulSet (Labs 13–15)

3

Day 3 · Domains 3 & 4

Observability + Config & Security

- Probes, logging, metrics, debug (Labs 16–20)
- ConfigMaps, Secrets, SecurityContext (Labs 21–23)
- ServiceAccounts, RBAC, Quotas (Labs 24–26)

4

Day 4 (½) · Domain 5

Services & Networking + Assessment

- Services & Service DNS (Labs 27–28)
- Ingress with TLS & NetworkPolicy (Labs 29–30)
- 1:00 PM mock-exam practice, then final assessment from 2:00 PM

Daily timing: 9:00am – 6:00pm · 1-hour lunch · short tea breaks within each day

Learning Outcomes

1

LO1 Build and optimise container images with single- and multi-stage Dockerfiles.

2

LO2 Create and manage Pods imperatively and declaratively with kubectl and --dry-run.

3

LO3 Run batch workloads with Jobs and scheduled workloads with CronJobs.

4

LO4 Design multi-container Pods using the sidecar and init-container patterns.

5

LO5 Persist and share Pod data with emptyDir, tmpfs and hostPath volumes.

LO6 Deploy, scale, observe, secure and network applications across the five CKAD domains.

Assessment



Final Assessment

- Written / MCQ assessment on the LMS
- Mock-exam practice from 1:00 PM; final assessment from 2:00 PM
- Covers all five CKAD domains and the hands-on labs
- Open book — slides, Learner Guide & kubernetes.io
- Must be attempted for SSG funding



Funding & Competency

Minimum 75% attendance

Based on SSG digital attendance records.

Assessed as Competent

Pass the written / MCQ assessment.

Briefing for Assessment



Clear your space

Keep only approved materials on the table.



No recording

No photos of the assessment.



Work individually

No discussion during the assessment.



Use the LMS

The assessment is delivered online.



Open book

Slides, Learner Guide & kubernetes.io are allowed.



Time's up

Submit when time is up.

DAY 1

Application Design and Build

Images · Pods · Jobs · CronJobs · Multi-container · Volumes (CKAD Domain 1 — 20%)



TOPIC 1

Container Images

Dockerfiles, image layers & multi-stage builds

Why Container Images?

- An image is a read-only blueprint that packages your app with all its dependencies.
- A container is a running instance of an image — it starts in milliseconds.
- Kubernetes never builds images; it pulls them from a registry by name:tag and runs them.
- CKAD expects you to read, write and fix Dockerfiles under time pressure.

What is Kubernetes?

Kubernetes (k8s)

- Open-source container orchestration platform
- Originally designed by Google
- CNCF graduated project since 2016
- Automates deployment, scaling, and management of containerised workloads
- Declarative: describe desired state; K8s makes it so

Why Container Orchestration?

- Containers are ephemeral — orchestration ensures availability
- Scale to thousands of containers across many nodes
- Self-healing: restart failed containers automatically

Key Capabilities

Auto-scaling

Self-healing

Load balancing

Rolling updates

Virtual Machines vs Containers

Virtual Machines

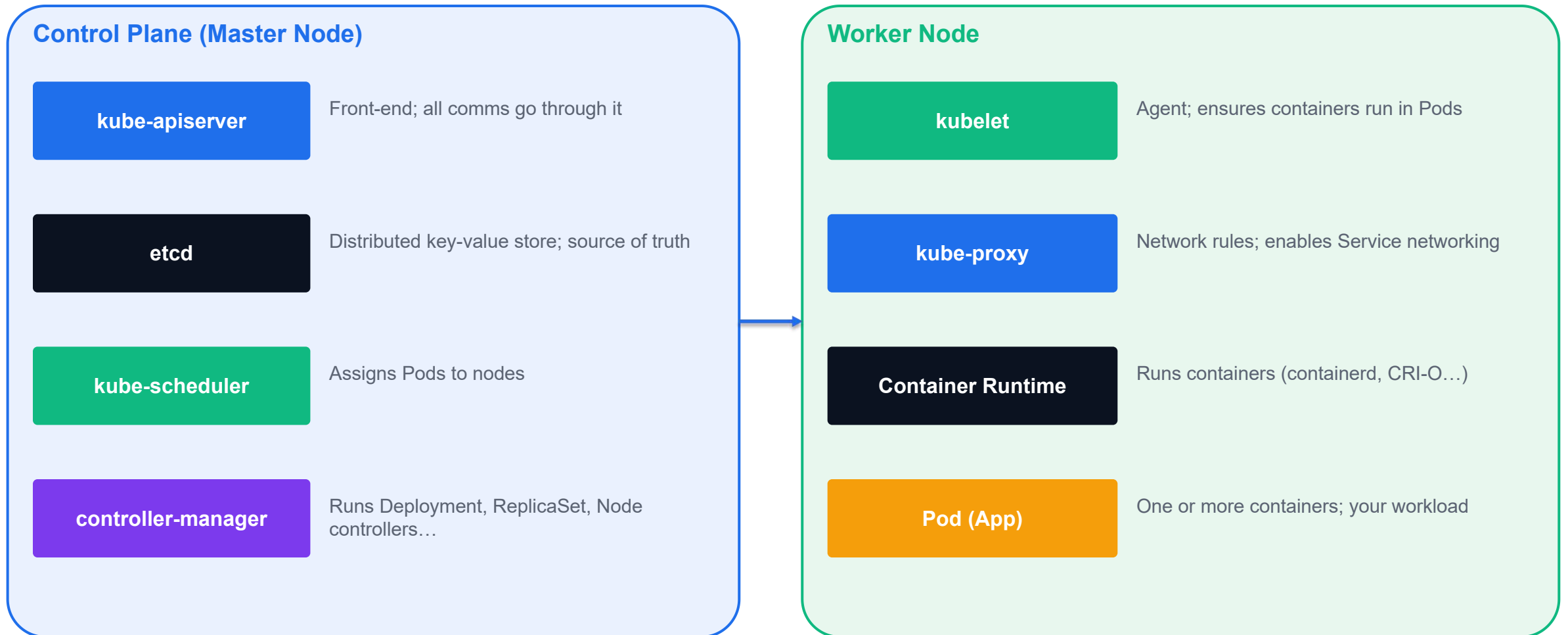
- Include full OS (kernel + libraries + app)
- GBs in size; minutes to boot
- Strong isolation via hypervisor
- Each VM has its own kernel
- Good for: legacy apps, OS-level testing

VS

Containers

- Share host OS kernel; package only app + libs
- MBs in size; milliseconds to start
- Process-level isolation via namespaces + cgroups
- Very low overhead; high density per host
- Good for: microservices, CI/CD, cloud-native apps

Kubernetes High-Level Architecture



kubectl — The Kubernetes CLI

Structure: verb + resource + flags

```
kubectl <verb> <resource> [name] [flags]
```

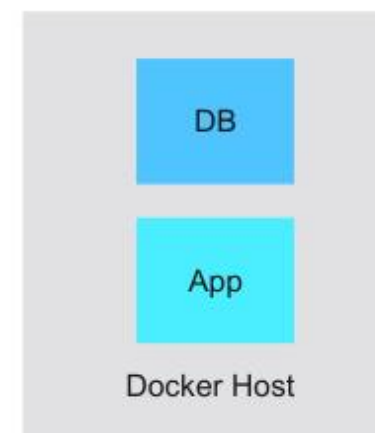
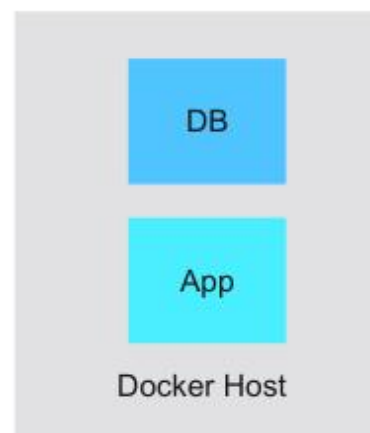
```
kubectl get          # list resources
kubectl describe     # detailed info
kubectl apply        # create/update from YAML
kubectl delete       # remove a resource
kubectl exec         # run command in container
kubectl logs         # view container logs
```

Useful flags for the exam

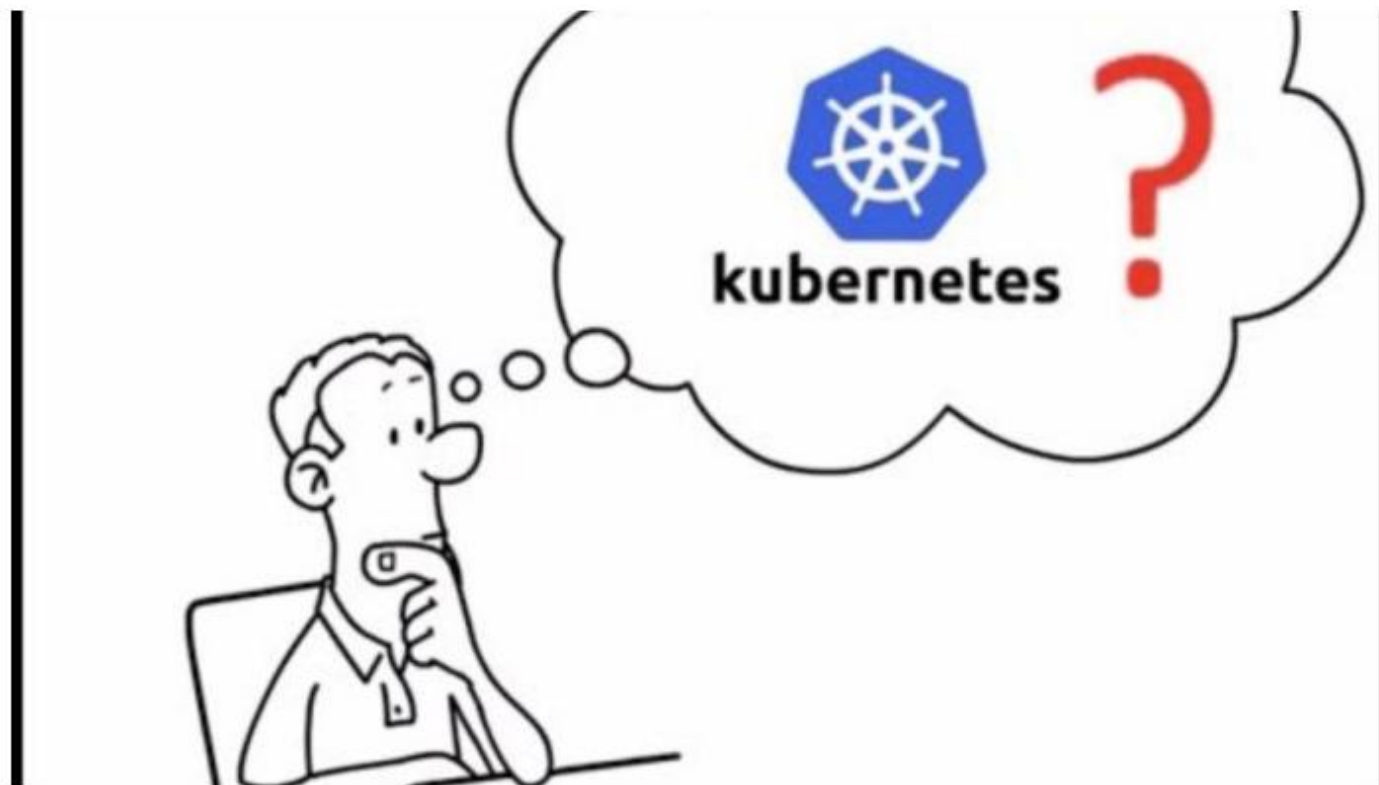
```
kubectl get pods -n <namespace>          # specify namespace
kubectl get all --all-namespaces          # across all ns
kubectl run nginx --image=nginx --dry-run=client -o yaml # generate YAML
kubectl explain pod.spec.containers        # inline API docs
```

Container Orchestration Tool

Orchestration



What is Kubernetes?



The founding of “κυβερνήτης”

History

- Greek for “Helmsman”
- Founded by Joe Beda, Brendan Burns and Craig McLuckie in Google
- First announced to public in mid-2014
- Version 1 released on July 2015
- We call it k8s for short



Orchestration tools

Kubernetes became the market leader after mass support from the community and support from key market players in the cloud space



Docker Swarm



Kubernetes

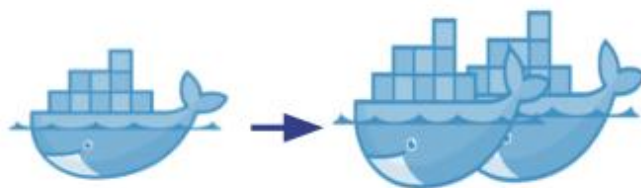


Mesos

Kubernetes (k8s) - Key Benefits



Portable

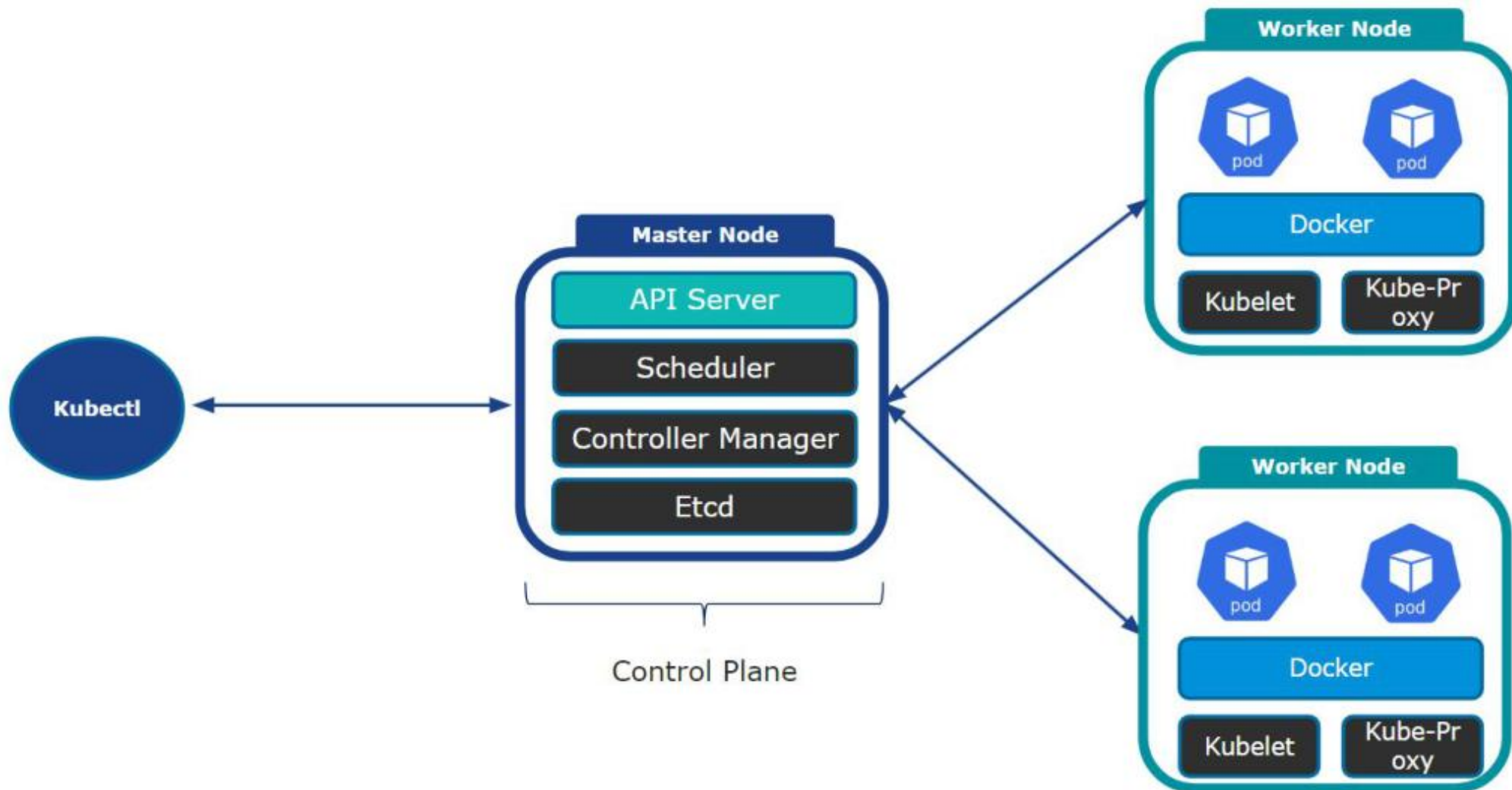


Scalable

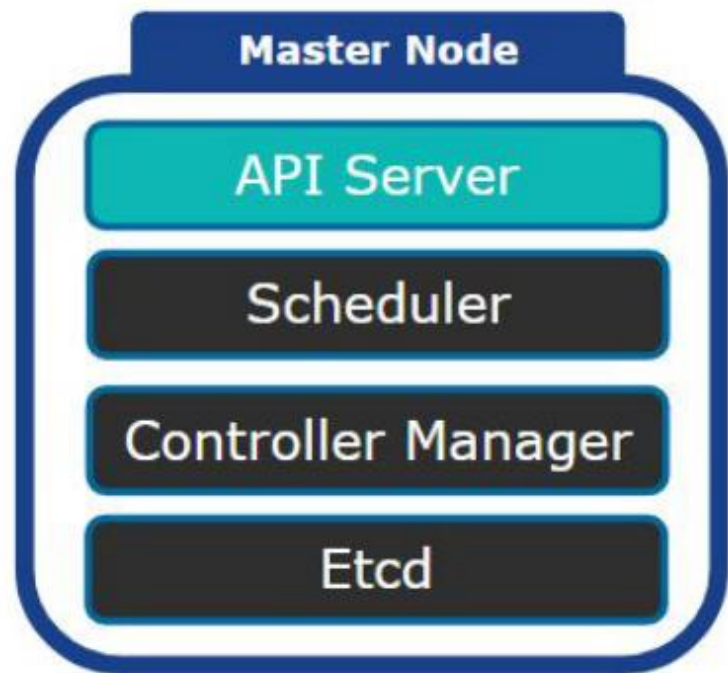


Highly Available

Kubernetes High-Level Architecture

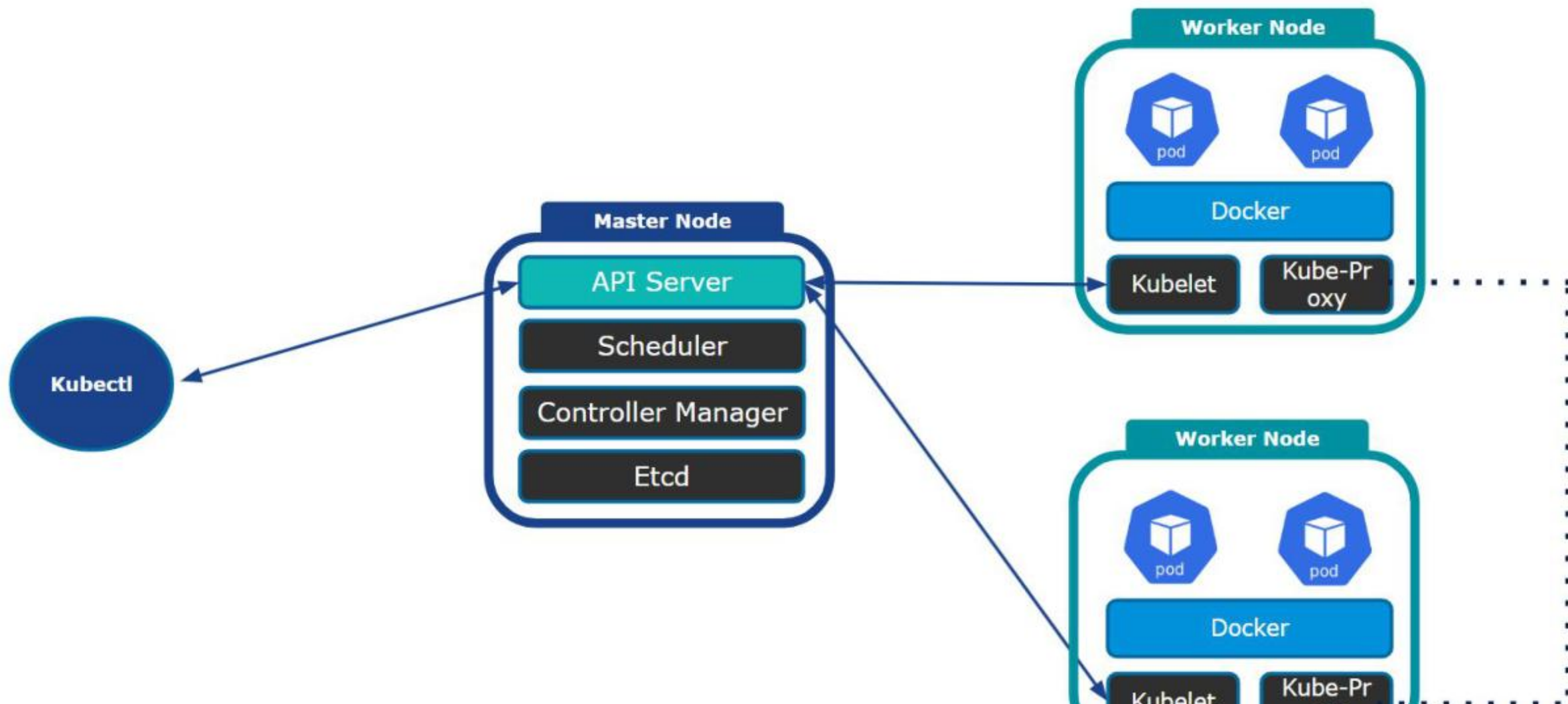


Master Node



- API Server (kube-apiserver): Orchestrating all operations within the cluster
- Scheduler (kube-scheduler): Selects optimal node to run pods
- Controller Manager (kube-controller-manager): Different controllers with different functions
- Etcd: Stores data in key-value format

Kubernetes High-Level Architecture



Declarative Configurations

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: redis-pod1
5  spec:
6    containers:
7    - name: myapp-container
8      image: redis
```

- YAML
 - Lines and white spaces delimiters

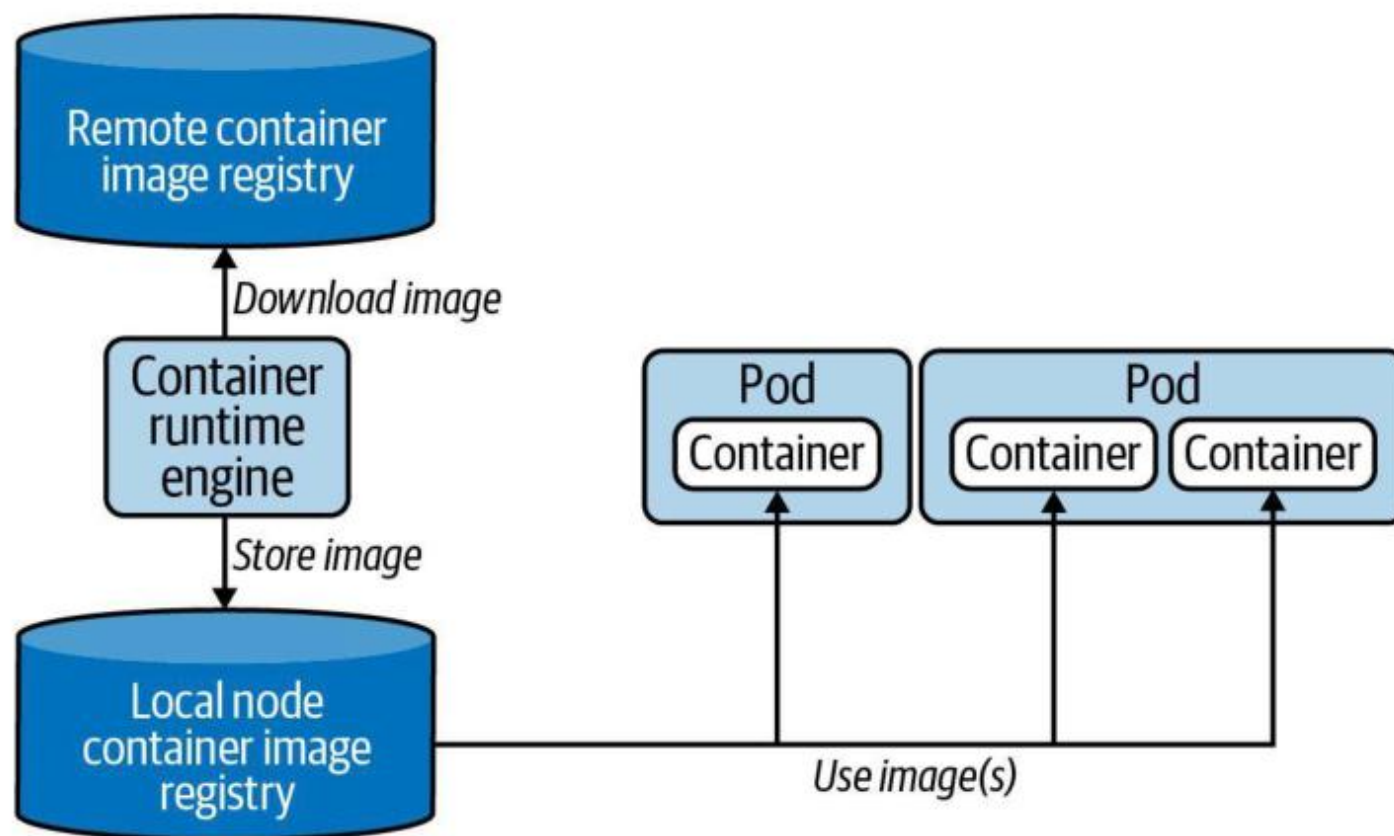
Imperative Commands

"Execute Hazelcast and configure it using parameters"

```
$ kubectl run hazelcast↵  
  --image=hazelcast/hazelcast↵  
  --restart=Never↵  
  --port=5701↵  
  --env="DNS_DOMAIN=cluster"↵  
  --labels="app=hazelcast,env=prod"  
pod/hazelcast created
```


Container Image Retrieval

Downloads container image from registry upon Pod creation



Creating a Namespace with Imperative Command

"Organize objects in namespace called `code-red`"

```
$ kubectl create namespace code-red
```

```
namespace/code-red created
```

```
$ kubectl run nginx --image=nginx --restart=Never -n code-red
```

```
pod/nginx created
```

```
$ kubectl get pods -n code-red
```

NAME	READY	STATUS	RESTARTS	AGE
nginx	1/1	Running	0	13s

Creating a Namespace with Imperative Command

```
kubectl run sample-pod-3 --image=nginx --namespace=dev
```

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    · name: sample-pod-3
5    · namespace: dev
6  spec:
7    · containers:
8      · - name: sample-pod-1
9        · image: nginx
```

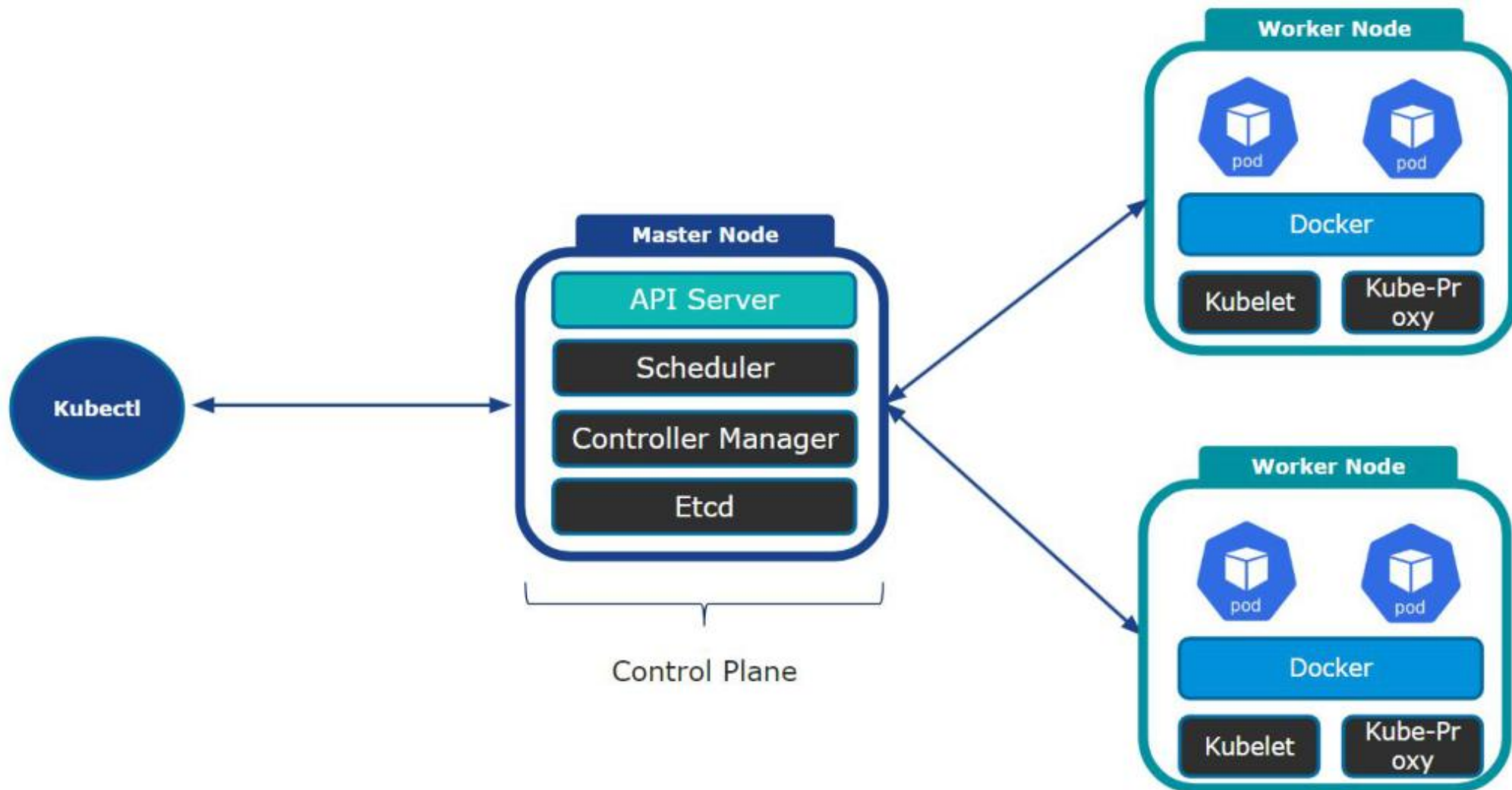
```
kubectl get pods --namespace=dev
```

Creating a CronJob with Imperative Command

"Render the current date on standard output every hour"

```
$ kubectl create
cronjob current-date
--schedule="* * * * *"
--image=nginx:1.24.0
-- /bin/sh -c 'echo "Current date: $(date)"'
cronjob.batch/current-date created
```

Kubernetes High-Level Architecture



Declarative Configurations

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: redis-pod1
5  spec:
6    containers:
7    - name: myapp-container
8      image: redis
```

- YAML
 - Lines and white spaces delimiters

Imperative Commands

"Execute Hazelcast and configure it using parameters"

```
$ kubectl run hazelcast↵  
  --image=hazelcast/hazelcast↵  
  --restart=Never↵  
  --port=5701↵  
  --env="DNS_DOMAIN=cluster"↵  
  --labels="app=hazelcast,env=prod"  
pod/hazelcast created
```


ONE INSTRUCTION = ONE LAYER

Anatomy of a Dockerfile

Base & setup

- FROM — pick a small, pinned base
- WORKDIR — set the working dir
- COPY / ADD — bring files in

Build steps

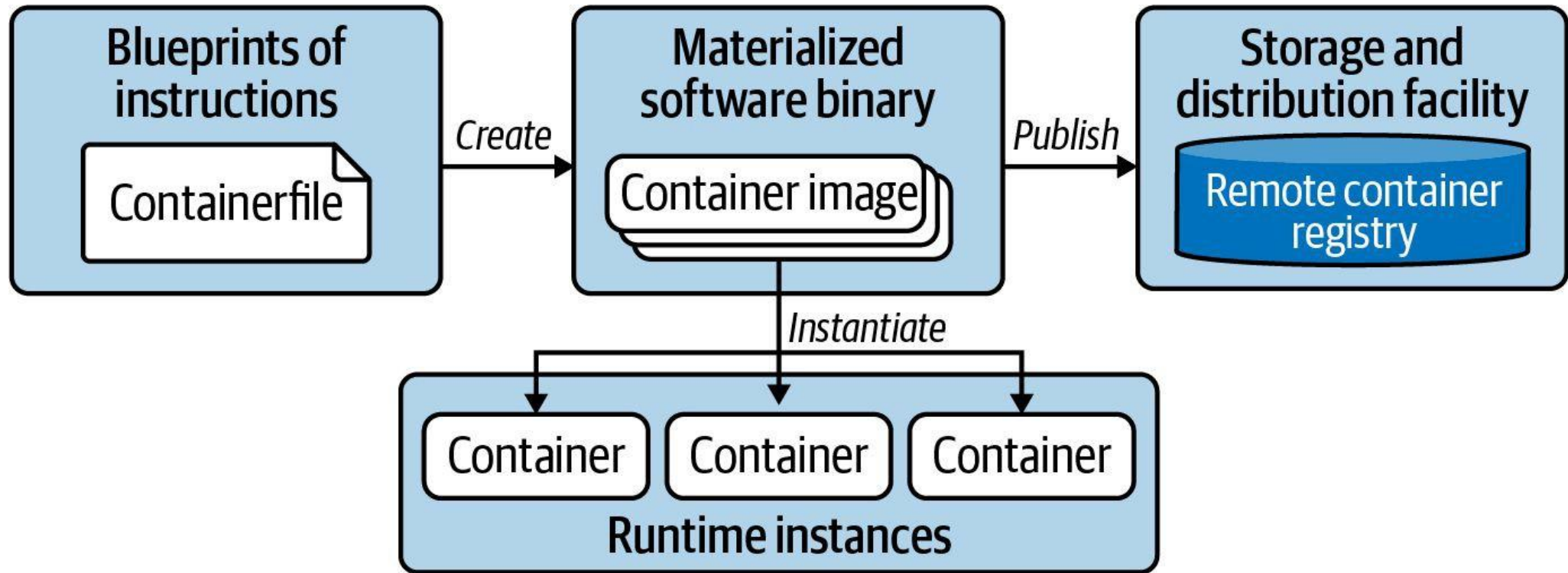
- RUN — install dependencies
- Each line = one cached layer
- Order steps for cache reuse

Runtime

- EXPOSE — document the port
- ENV — runtime configuration
- CMD / ENTRYPOINT — what runs

Containerization Process

Building and publishing an image, running image in container



Single-Stage vs Multi-Stage Builds

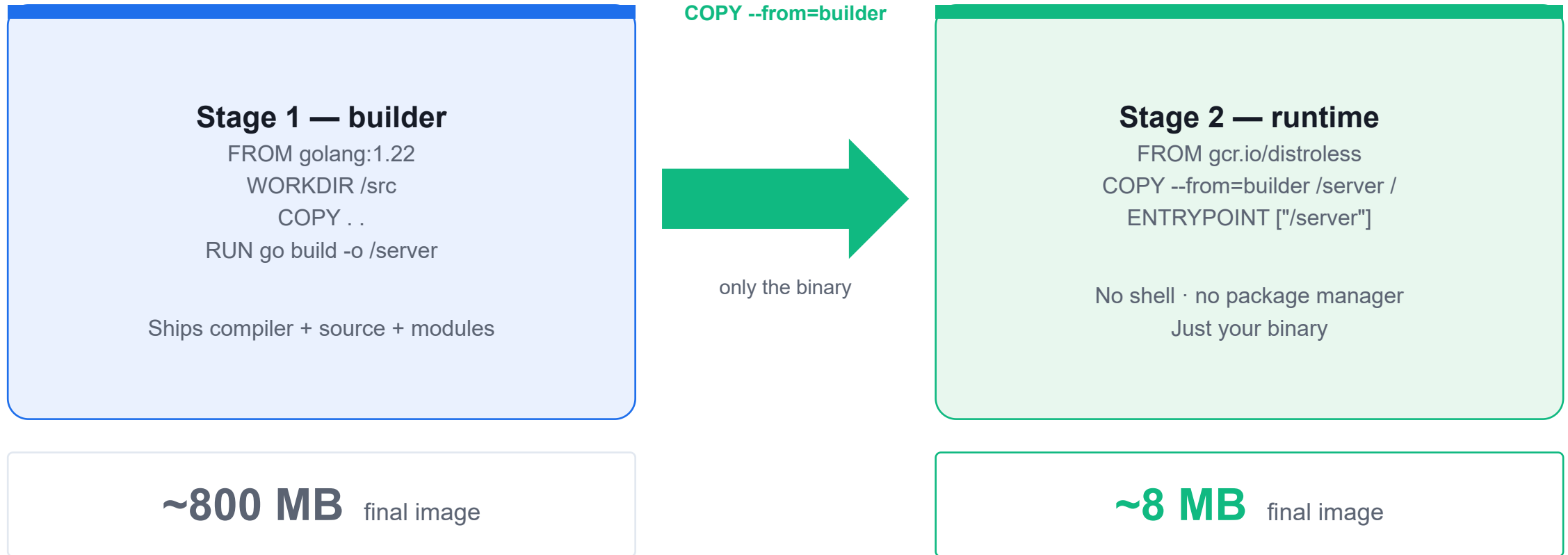
Single-stage

- One FROM — build + runtime together
- Ships the whole toolchain
- Large image (~800 MB for Go)
- Bigger attack surface

Multi-stage

- Multiple FROM stages with AS <name>
- COPY --from=builder copies only the binary
- Tiny image (~8 MB, distroless)
- No shell, no package manager

Multi-Stage Build — Shrink the Image



Build a Container Image with Docker

LAB 1

Container images

Write a Dockerfile from scratch, build a tagged image, run it locally and inspect its layers. CKAD expects you to read and write Dockerfiles confidently — you may be asked to fix a broken one or produce one under time pressure.

You'll build

Write a Dockerfile, build and tag an image, run it, then inspect its layers with docker history.

Key commands: `FROM python:3.12-slim` · `» `EXPOSE`` · `docker build` · `docker run` · `docker history`

Build a Container Image with Docker

Step 1 — Write the Dockerfile

```
FROM python:3.12-slim
WORKDIR /app
COPY app.py .
EXPOSE 8080
CMD ["python", "app.py"]
```

Note EXPOSE is documentation only — it does not publish the port. -p host:container is what actually opens it.

Step 2 — Build and tag the image

```
docker build -t ckad/hello:1.0 .
docker images ckad/hello
```

Build a Container Image with Docker (cont.)

Step 3 — Run and test the container

```
docker run -d --name hello -p 8080:8080 ckad/hello:1.0  
curl http://localhost:8080
```

Step 4 — Inspect the image layers

```
docker history ckad/hello:1.0  
docker rm -f hello && docker rmi ckad/hello:1.0
```

Build a Container Image with Docker

✓ Test it

curl http://localhost:8080 returns hello from CKAD 2026 lab 1, and docker history shows a distinct layer for each Dockerfile instruction.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Multi-Stage Dockerfile

LAB 2

Container images

Separate the build stage from the runtime stage to produce images that are 10x smaller and contain no compiler toolchain. Multi-stage builds are a CKAD staple — you must be able to write one from scratch and explain why it shrinks the image.

You'll build

Build a single-stage baseline, then a multi-stage image with COPY --from and distroless, and compare sizes.

Key commands: FROM golang:1.22 · docker build · » Only

Multi-Stage Dockerfile

Step 1 — Single-stage baseline (large image)

```
FROM golang:1.22
WORKDIR /src
COPY main.go .
RUN go mod init demo && go build -o app main.go
CMD ["/app"]
```

Step 2 — Multi-stage Dockerfile (small image)

```
# Stage 1: compile
FROM golang:1.22 AS builder
WORKDIR /src
COPY main.go .
RUN go mod init demo && CGO_ENABLED=0 go build -o /out/app main.go

# Stage 2: runtime only
FROM gcr.io/distroless/static-debian12
COPY --from=builder /out/app /app
ENTRYPOINT ["/app"]
```

Multi-Stage Dockerfile (cont.)

Step 3 — Build both and compare sizes

```
docker build -f Dockerfile.single -t demo:single .  
docker build -f Dockerfile.multi -t demo:multi .  
docker images | grep demo          # ~800MB vs ~8MB
```

Note Only the last FROM stage ends up in the final image. CGO_ENABLED=0 produces a static binary that runs in distroless/scratch — no shell, smaller attack surface.

Multi-Stage Dockerfile

✓ Test it

docker images | grep demo shows demo:multi roughly 100x smaller than demo:single, and demo:multi still serves hello from multi-stage build.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Tea Break

15 minutes



TOPIC 2

Pods

The smallest deployable unit & the exam-speed kubectl workflow

What is a Pod?

- A Pod is the smallest deployable unit — one or more containers sharing network & storage.
- Containers in a Pod share an IP and localhost; they can share volumes.
- Create Pods imperatively (`kubectl run`) or declaratively (`kubectl apply -f`).
- Pods are ephemeral — in production you run them via higher-level controllers (Day 2).

The CKAD kubectl Workflow

Generate

- kubectl run / create
- with --dry-run=client -o yaml
- Scaffold a manifest fast

Edit

- Redirect to a .yaml file
- Tweak labels, image, command
- Version-controllable

Apply

- kubectl apply -f file.yaml
- Idempotent & repeatable
- kubectl describe to verify

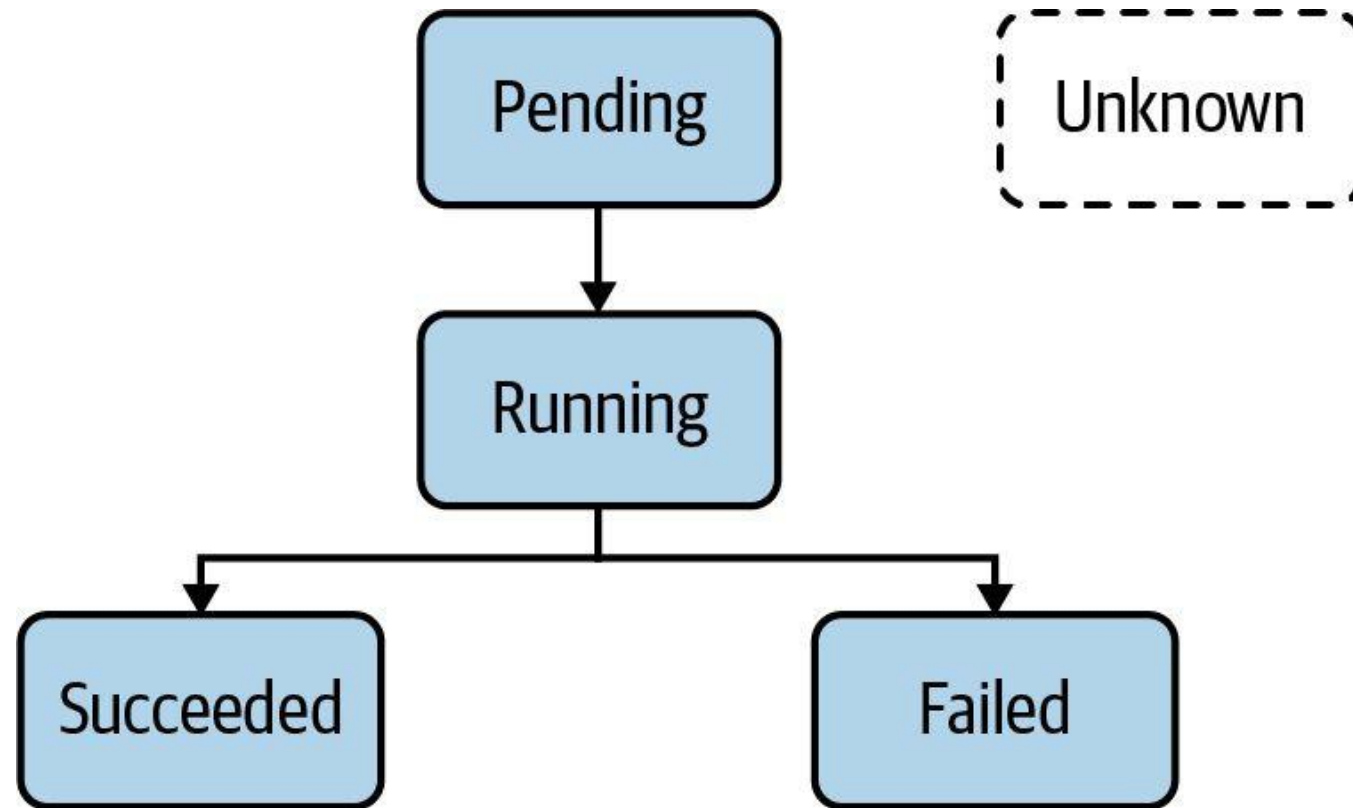
What is a Pod?

Runs a process or application in container(s)

- The smallest, most basic deployable objects in Kubernetes.
- It contains one or more containers, executed by the configured container runtime.
- Kubernetes enhances the basic functionality of a container with features such as security, storage, and more.

Pod Life Cycle Phases

Indicates operational status of Pod



SET THESE ON EXAM DAY

Exam-Speed kubectl (kubernetes.io quick reference)

- `alias k=kubectl` — set this first; saves keystrokes on every command.
- `export do="--dry-run=client -o yaml"` — scaffold any manifest without creating it.
- `source <(kubectl completion bash); complete -o default -F __start_kubectl k` — tab-completion.
- `kubectl get -o wide / -o yaml / -o jsonpath` · `kubectl describe` · `kubectl explain` — inspect & learn.

Create and Manage Pods

LAB 3

Pods

The Pod is the smallest schedulable unit in Kubernetes. Create Pods imperatively, generate YAML with `--dry-run`, edit live manifests, and use the exam-critical `$do` alias that saves 30+ seconds per question.

You'll build

Set exam-speed aliases, run Pods imperatively, scaffold YAML with `$do`, then apply and inspect.

Key commands: `alias k=kubectl · k run · k describe · » `--restart=Never``

Create and Manage Pods

Step 1 — Set exam-speed aliases (run these first on exam day)

```
alias k=kubectl  
export do="--dry-run=client -o yaml"
```

Step 2 — Create a Pod imperatively

```
k run web --image=nginx:1.25 --port=80  
k get pod web -o wide          # shows node and Pod IP
```

Step 3 — Generate a manifest without creating it

```
k run web2 --image=nginx:1.25 --port=80 $do > web2.yaml  
k apply -f web2.yaml  
k get pod web2 --show-labels
```

Create and Manage Pods (cont.)

Step 4 — Inspect a running Pod

```
k describe pod web
k get pod web -o jsonpath='{.status.podIP}'; echo
```

Step 5 — Override the container command

```
k run sleeper --image=busybox --restart=Never -- sh -c 'sleep 3600'
k delete pod web web2 sleeper --force --grace-period=0
```

Note `--restart=Never` creates a bare Pod (not a Deployment). Everything after `--` becomes the container command. `--force --grace-period=0` skips the 30s grace period.

Create and Manage Pods

✓ Test it

`k get pod web2 --show-labels` shows the `tier=frontend` label, and the `--dry-run=client -o yaml` workflow produces a valid manifest you can apply.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Pod Lifecycle Phases

Pending

Accepted by cluster; image pulling or scheduling

Running

At least one container is running

Succeeded

All containers exited with code 0

Failed

At least one container exited non-zero

Unknown

Pod state could not be obtained (node issue)

Check phase: `kubectl get pod <name> -o jsonpath='{.status.phase}'`

Common Pod Error Statuses

- **CrashLoopBackOff** — container keeps crashing; K8s backs off restart time
 - Causes: bad entrypoint, missing config, OOM kill, application crash
- **ImagePullBackOff / ErrImagePull** — cannot pull container image
 - Causes: wrong image name/tag, private registry, no imagePullSecret
- **OOMKilled** — container exceeded its memory limit; killed by kernel
- **Pending (forever)** — no node can satisfy scheduling requirements
 - Causes: insufficient CPU/memory, missing node label, taints without tolerations
- **RunContainerError** — container config error (bad entrypoint, missing volume mount)

Troubleshooting Pods & Containers

1 Check Pod status and events

```
kubectl get pod <name> -n <ns>
kubectl describe pod <name> -n <ns>    # shows Events section
```

2 View container logs

```
kubectl logs <pod-name>                # current logs
kubectl logs <pod-name> -p                # previous (crashed) container
kubectl logs <pod-name> -c <container>    # specific container
```

3 Shell and ephemeral debug containers

```
kubectl exec -it <pod-name> -- /bin/sh
kubectl debug <pod-name> -it --image=busybox    # ephemeral container
(use when main image has no shell, e.g. distroless images)
```

Environment Variables and Command/Args in Pods

Declaring environment variables

```
spec:
  containers:
  - name: app
    image: myapp:1.0
    env:
    - name: APP_ENV
      value: production
    - name: DB_URL
      value: postgres://db:5432/mydb
```

Overriding container ENTRYPOINT and CMD

```
spec:
  containers:
  - name: app
    image: myapp:1.0
    command: ["/bin/sh"] # overrides ENTRYPOINT
    args: ["-c", "echo hello"] # overrides CMD
```

What is a Namespace?

Kubernetes Cluster

default

General workloads
when no -n flag

Pod

Pod

kube-system

Internal K8s
components

Pod

Pod

my-app

Custom namespace
for your app

Pod

Pod

Namespace — Create, List, Delete

Create a namespace (imperative and declarative)

```
kubectl create namespace staging
```

```
apiVersion: v1
kind: Namespace
metadata:
  name: staging
```

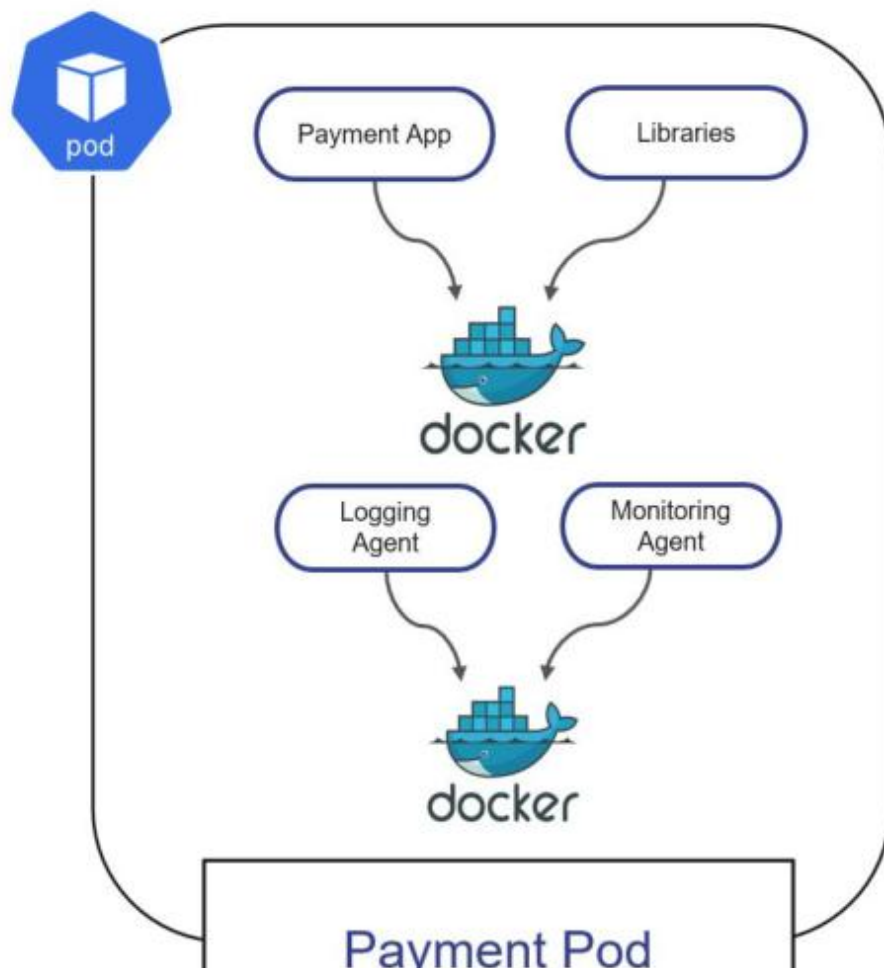
List, inspect, delete, and set default namespace

```
kubectl get namespaces          # list all
kubectl describe namespace staging # details
kubectl delete namespace staging # cascade-deletes all objects inside!

kubectl run nginx --image=nginx -n staging
kubectl config set-context --current --namespace=staging
```

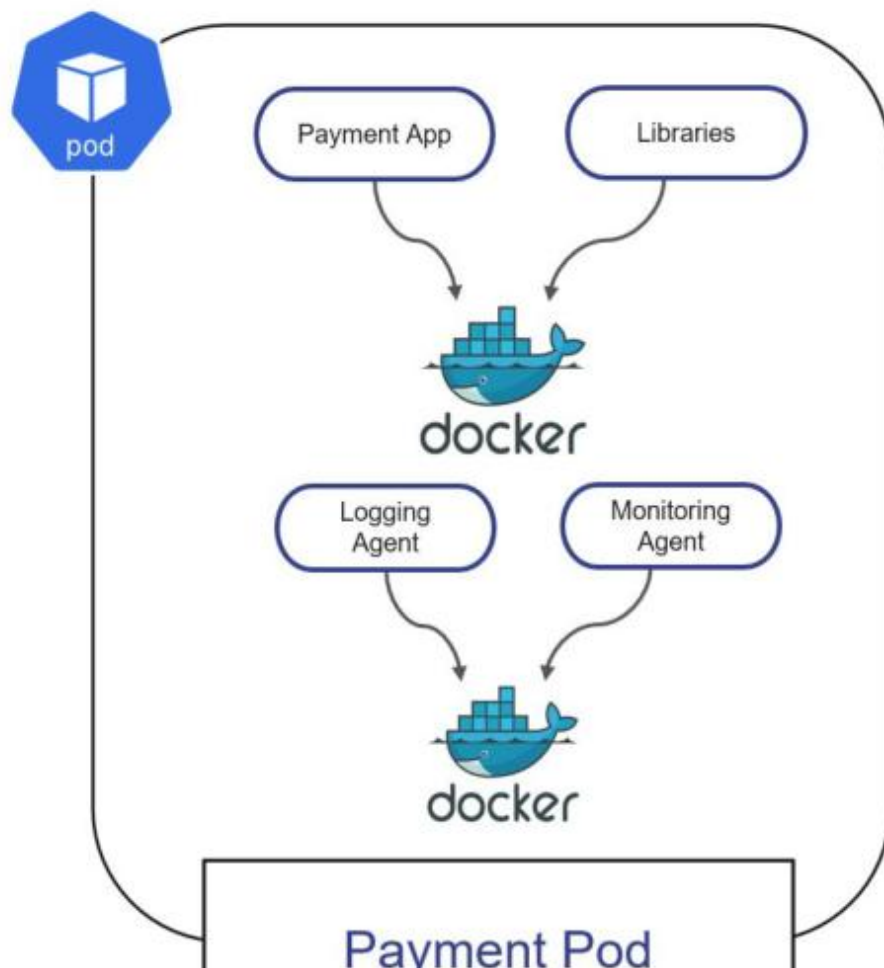
What is a Pod?

The smallest, most basic deployable objects in Kubernetes.



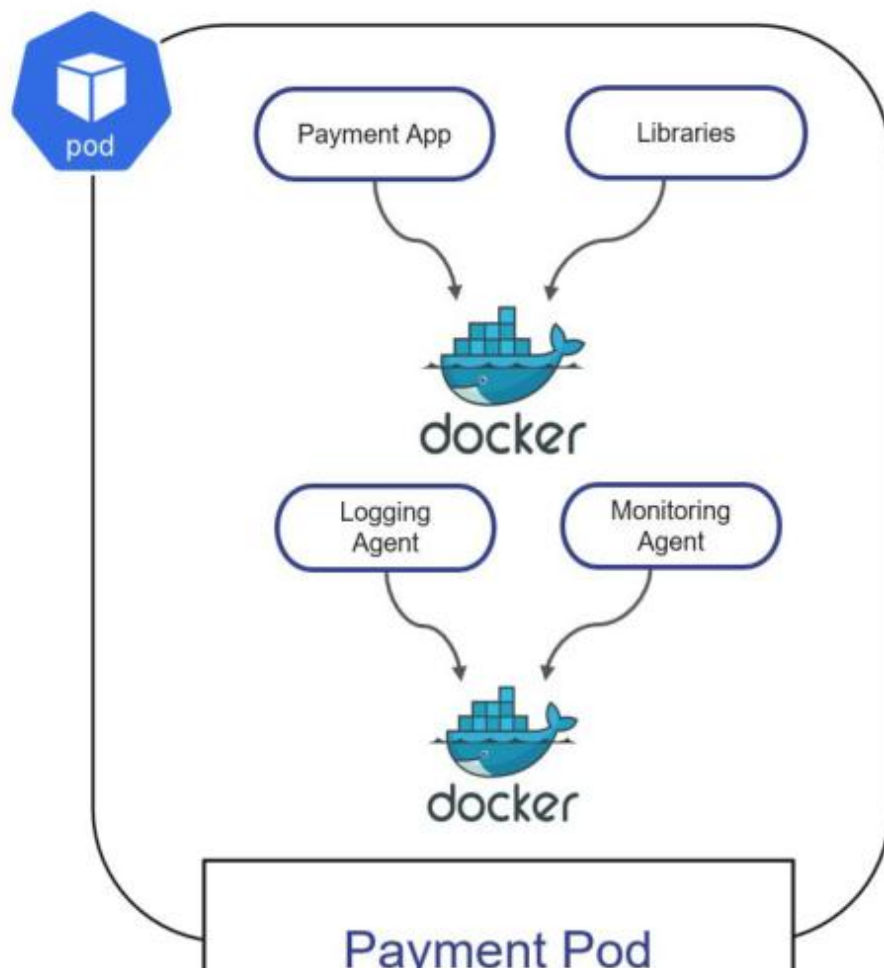
What is a Pod?

The smallest, most basic deployable objects in Kubernetes.



What is a Pod?

The smallest, most basic deployable objects in Kubernetes.



Retrieving High-Level Pod Information

Check the status of a Pod first

```
$ kubectl get pods -n t67
```

NAME	READY	STATUS	RESTARTS	AGE
misbehaving-pod	0/1	ErrImagePull	0	2s

```
$ kubectl get pods -n k31
```

NAME	READY	STATUS	RESTARTS	AGE
successful-pod	1/1	Running	0	5s

Failed to pull
container image

Looks successful on
the surface-level

Inspecting Events Across all Pods

Lists events for all Pods in the given namespace

```
$ kubectl get events
```

LAST SEEN	TYPE	REASON	OBJECT	MESSAGE
-----------	------	--------	--------	---------

3m14s	Warning	BackOff	pod/custom-cmd	Back-off restarting
-------	---------	---------	----------------	---------------------

2s	Warning	FailedNeedsStart	cronjob/google-ping	↵failed container
----	---------	------------------	---------------------	-------------------

Opening an Interactive Shell to a Container

Allows for exploring container file system and processes

```
$ kubectl logs failing-pod
```

```
/bin/sh: can't create /root/tmp/curr-date.txt: nonexistent directory
```

```
$ kubectl exec failing-pod -it -- /bin/sh
```

```
# mkdir -p
```

```
~/tmp # cd
```

```
~/tmp
```

```
# ls
```

```
-l
```

```
total
```

```
4
```

```
-rw-r--r-- 1 root root 112 May 9 23:52
```

Copying a File from the Container

The `tar` binary needs to exist in container

```
$ kubectl exec -it nginx -n code-red -- /bin/sh
# tar cvf error-log.tar
/var/log/nginx/error.log # exit

$ kubectl cp code-red/nginx:error-log.tar error-log.tar

$ ls
error-log.tar
```

Namespace

File to be copied

Accessing Container Logs

Renders logs produced by application or process running in a container

```
$ kubectl logs hazelcast
```

```
...
```

```
May 25, 2020 3:36:26 PM com.hazelcast.core.LifecycleService
```

```
INFO: [10.1.0.46]:5701 [dev] [4.0.1] [10.1.0.46]:5701 is STARTED
```

- You can stream the logs with the command line option `-f`. This option is helpful if you want to see logs in real time.
- You can still get back to the logs of the previous container by adding the `-p` command line option in case the container crashes or is restarted.

Executing a Command in a Container

Interactive exploration of a container, or seeing the direct effect of a command

```
$ kubectl exec -it hazelcast -- /bin/sh
```

```
# ...
```

```
$ kubectl exec hazelcast -- env
```

```
...
```

```
DNS_DOMAIN=cluster
```

- The command line option `-it` opens an interactive shell. This is useful if you want to inspect the configuration of your application or debug the current state of your application.
- For direct execution of a command in the container simply append to `--`

Declaring Command and Args

Assign or overwrite container entrypoint and arguments

```
apiVersion: v1
kind: Pod
metadata:
  name: spring-boot-app
spec:
  containers:
  - image: bmuschko/spring-boot-app:1.5.3
    name: spring-boot-app
    command: ["/bin/sh"]
    args: ["-c", "while true; do date; sleep 10; done"]
```


Declaring Command and Args

Assign or overwrite container entrypoint and arguments

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: sample-pod-1
5  spec:
6    containers:
7      - name: sample-pod-1
8        image: python:3.6-alpine
9        command: ["sleep"]
10       args: ["5"]
```

Namespace YAML Manifest

"Organize objects in namespace called `code-red`"

```
apiVersion: v1
kind: Namespace
metadata:
  name: code-red
```


Deleting a Namespace

Cascade-deletes containing objects

```
$ kubectl delete namespace code-red
```

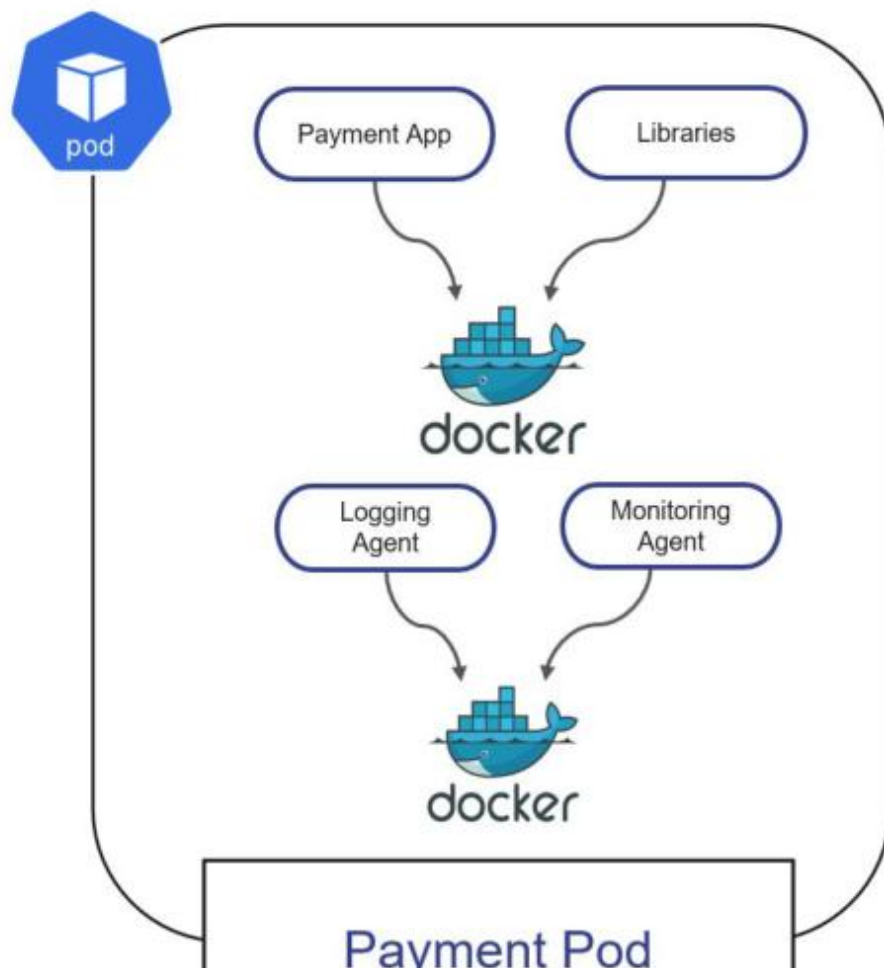
```
namespace "code-red" deleted
```

```
$ kubectl get pods -n code-red
```

```
No resources found in code-red namespace.
```

What is a Pod?

The smallest, most basic deployable objects in Kubernetes.



Lunch Break

1 hour



TOPIC 3

Jobs & CronJobs

Run-to-completion & scheduled batch workloads

Batch Workloads in Kubernetes

- A Job runs Pods until a set number of successful completions, then stops — it does not restart on success.
- A CronJob creates Jobs on a schedule using standard cron syntax (*/* * * * * = every minute).
- Job Pod templates must use restartPolicy: Never or OnFailure — never Always.
- Trigger a CronJob now with: `kubectl create job --from=cronjob/<name> <job-name>`.

Key Fields to Memorise

Job

- completions — successes needed
- parallelism — Pods at once
- backoffLimit — failures allowed

Job (timing)

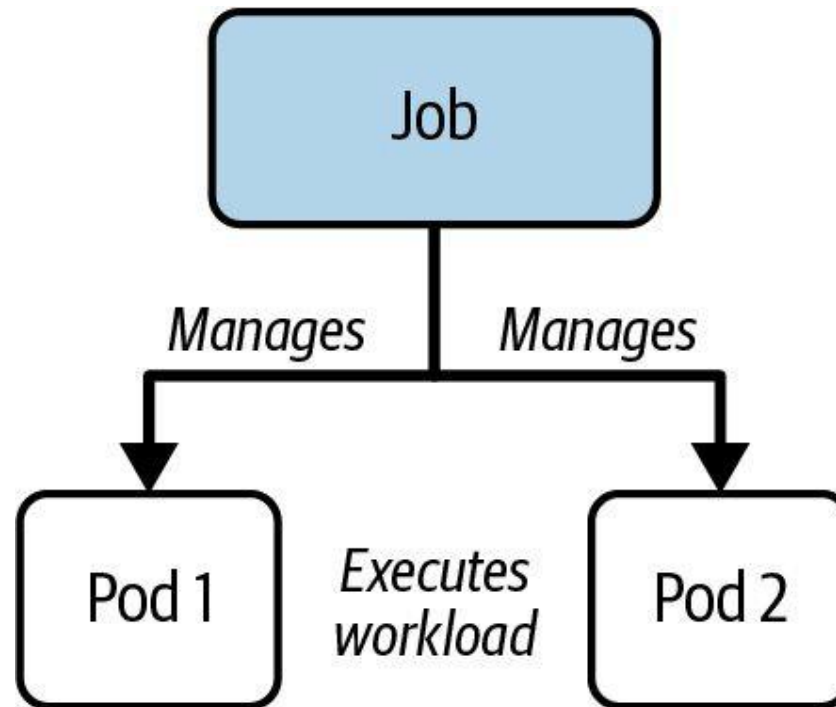
- activeDeadlineSeconds
- Hard wall-clock ceiling
- Overrides backoffLimit

CronJob

- schedule + timeZone
- concurrencyPolicy:
Allow/Forbid/Replace
- successful/failedJobsHistoryLimit

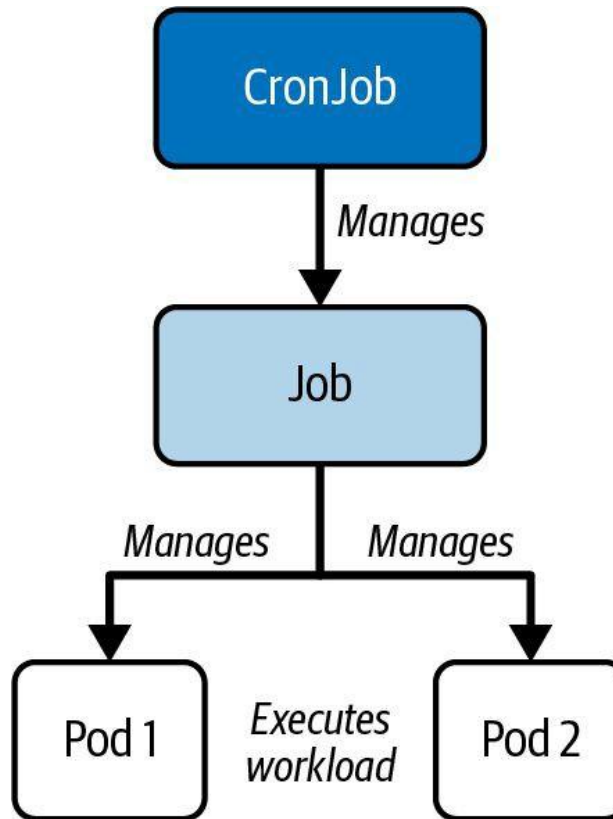
Job Object Hierarchy

A Job manages Pods to run the workload



CronJob Object Hierarchy

A CronJob manages Jobs which delegate the workload to Pods



Jobs — Run-to-Completion Workloads

LAB 4

Batch workloads

A Job runs Pods until a required number of successful completions is reached, then stops. CKAD tests completions, parallelism, backoffLimit and activeDeadlineSeconds in almost every sitting — you must write a Job manifest from memory.

You'll build

Create a one-shot Job, a parallel Job with completions, and explore backoffLimit and activeDeadlineSeconds.

Key commands: `k create · apiVersion: batch/v1 · k apply · » `restartPolicy:`

Jobs — Run-to-Completion Workloads

Step 1 — One-shot Job (imperative)

```
k create job hello --image=busybox -- echo "hello CKAD 2026"  
k logs -l job-name=hello
```

Jobs — Run-to-Completion Workloads (cont.)

Step 2 — Parallel Job with multiple completions

```
apiVersion: batch/v1
kind: Job
metadata:
  name: pi-parallel
spec:
  completions: 5
  parallelism: 2
  backoffLimit: 4
  template:
    spec:
      restartPolicy: Never
      containers:
      - name: pi
        image: perl:5.34
        command: ["perl", "-Mbignum=bpi", "-wle", "print bpi(50)"]
```

Jobs — Run-to-Completion Workloads (cont.)

Step 3 — Apply and watch completions climb

```
k apply -f parallel.yaml  
k get job pi-parallel -w      # COMPLETIONS ticks 0/5 → 5/5
```

Note restartPolicy: Never is mandatory in Job Pod templates. activeDeadlineSeconds is a hard wall-clock ceiling — it overrides backoffLimit.

Jobs — Run-to-Completion Workloads

✓ Test it

k get job pi-parallel reaches 5/5 completions with no more than two Pods running simultaneously during the run.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

What is a Job?

Kubernetes primitive for executing a one-time operation

- It runs a specific task until a specified number of completions has been reached.
- The actual work managed by a Job is still running inside of a Pod. Therefore, you can think of a Job as a higher-level coordination instance for Pods executing the workload.
- Upon completion of a Job and its Pods, Kubernetes does not automatically delete the objects - they will stay until they're explicitly deleted. Keeping those objects helps with debugging the command run inside of the Pod and gives you a chance to inspect the logs.

Job Operation Types

Execution runtime behavior of workload can be fine-tuned

- The default behavior of a Job is to run the workload in a single Pod and expect one successful completion (non-parallel Job).
- The attribute `spec.completions` controls the number of required successful completion.
- The attribute `spec.parallelism` allows for executing the workload by multiple pods in parallel.

Job Operation Types

Pick combination of completions and parallelism based on workload use case

Type	Completions	Parallelism	Description
Non-parallel with one completion count	1	1	Completes as soon as its Pod terminates successfully.
Parallel with a fixed completion count	≥ 1	≥ 1	Completes when specified number of tasks finish successfully.
Parallel with worker queue	unset	≥ 1	Completes when at least one Pod has terminated successfully and all Pods are terminated.

Restart Behavior

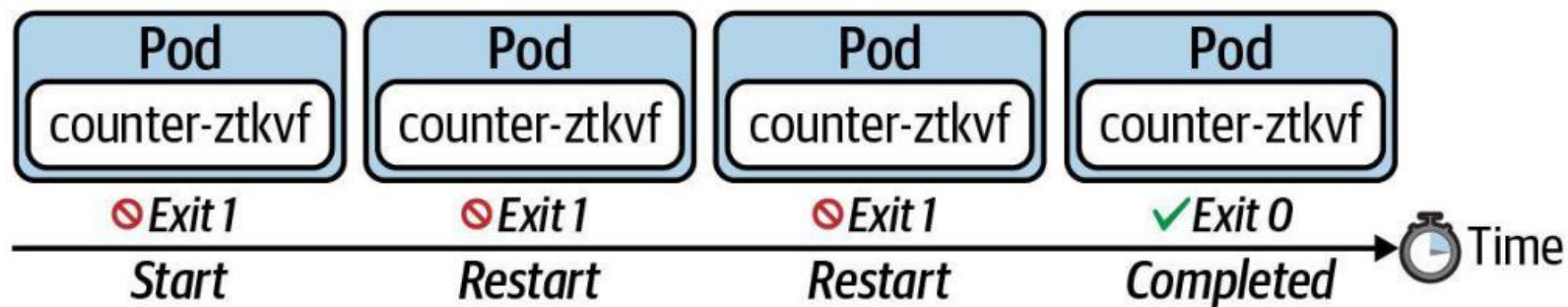
Re-running workload if execution is unsuccessful

- The attribute `spec.backoffLimit` determines the number of retries a Job attempts to successfully complete the workload until the executed command finishes with an exit code 0. The default is 6, which means it will execute the workload 6 times before the Job is considered unsuccessful.
- The attribute `spec.template.spec.restartPolicy` needs to be declared explicitly. The restart policy of a Job can only be `OnFailure` or `Never`.

Restarting a Container on Failure

Restart container if it exits with a non-zero exit code

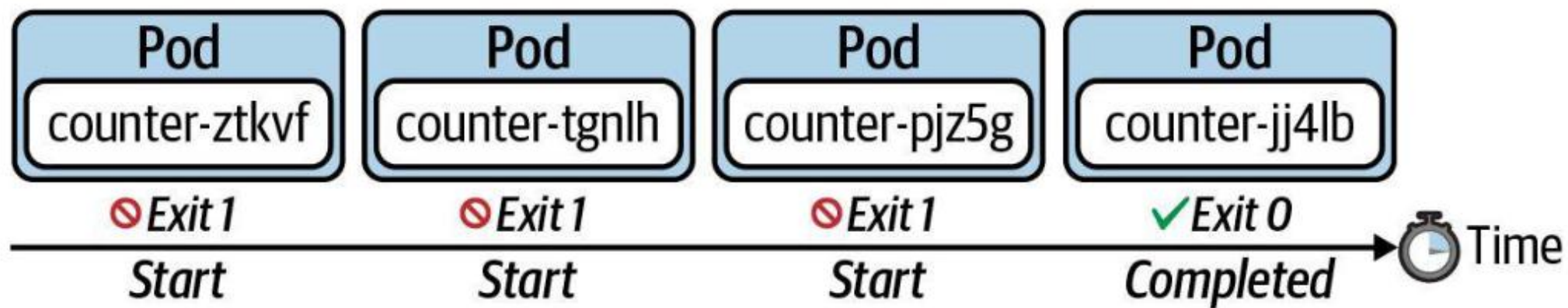
```
spec.template.spec.restartPolicy: OnFailure
```



Starting a New Pod on Failure

The policy does not restart the container upon a failure, it starts a new Pod.

```
spec.template.spec.restartPolicy: Never
```



CronJob YAML Manifest

"Increment a counter and render its value on the terminal"

```
apiVersion: batch/v1
kind: CronJob
metadata:
  name: counter
spec:
  schedule: "*/1 * * * *"
  jobTemplate:
    spec:
      template:
        spec:
          restartPolicy: Never
          containers:
            - args:
              - /bin/sh
              - -c
              - ...
              image: nginx:1.24.0
```

The crontab expression used to run CronJob periodically

Run in a new Pod

CronJob YAML Manifest

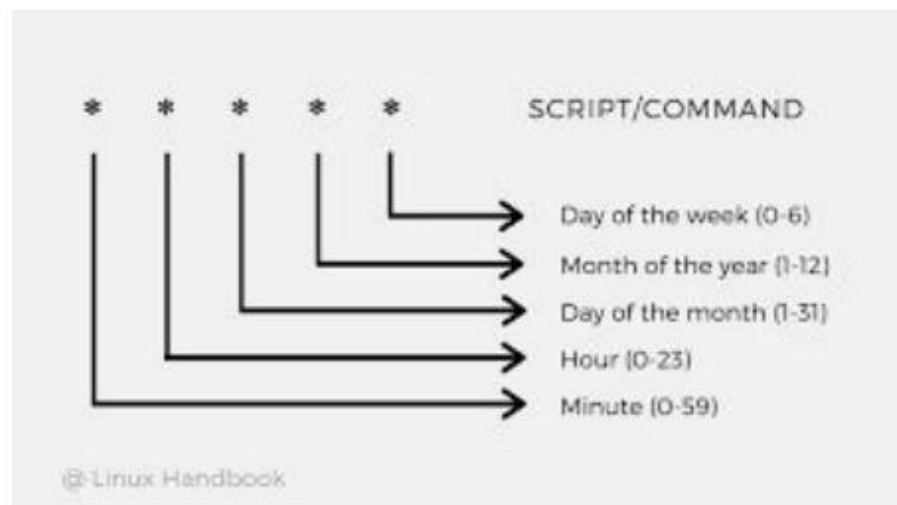
job-definition.yaml

```
apiVersion: batch/v1
kind: Job
metadata:
  name: calculate-pod-job
spec:
  completions: 3
  parallelism: 3
  template:
    spec:
      containers:
      - image: ubuntu
        name: calculate-pod
        command: ['expr','2','+', '1']
  restartPolicy: Never
```

cron-job-definition.yaml

```
apiVersion: batch/v1beta1
kind: CronJob
metadata:
  name: calculate-pod-cron-job
spec:
  schedule: "*/1 * * * *"
  jobTemplate:
    spec:
      spec:
        completions: 3
        parallelism: 3
        template:
          spec:
            containers:
            - image: ubuntu
              name: calculate-pod
              command: ['expr','2','+', '1']
      restartPolicy: Never
```


CronJob YAML Manifest



cron-job-definition.yaml

```
apiVersion: batch/v1beta1
kind: CronJob
metadata:
  name: calculate-pod-cron-job
spec:
  schedule: "*/1 * * * *"
  jobTemplate:
    spec:
      spec:
        completions: 3
        parallelism: 3
      template:
        spec:
          containers:
            - image: ubuntu
              name: calculate-pod
              command: ['expr', '2', '+', '1']
          restartPolicy: Never
```

What is a Job?

Kubernetes primitive for executing a one-time operation

- It runs a specific task until a specified number of completions has been reached.
- The actual work managed by a Job is still running inside of a Pod. Therefore, you can think of a Job as a higher-level coordination instance for Pods executing the workload.
- Upon completion of a Job and its Pods, Kubernetes does not automatically delete the objects - they will stay until they're explicitly deleted. Keeping those objects helps with debugging the command run inside of the Pod and gives you a chance to inspect the logs.

CronJobs — Scheduled Workloads

LAB 5

Batch workloads

A CronJob creates Jobs on a time-based schedule using standard cron syntax. CKAD tests concurrencyPolicy, timeZone, startingDeadlineSeconds, history limits, manual triggers and suspend/resume — these fields appear verbatim in exam questions.

You'll build

Create a CronJob imperatively and declaratively, suspend/resume it, then trigger it manually.

Key commands: `k create · apiVersion: batch/v1 · k patch · » `concurrencyPolicy`:`

CronJobs — Scheduled Workloads

Step 1 — Create a CronJob imperatively

```
k create cronjob date-printer --image=busybox \  
  --schedule="*/1 * * * *" -- sh -c "date; echo from cronjob"  
k get cronjob date-printer
```

CronJobs — Scheduled Workloads (cont.)

Step 2 — Declarative CronJob with all key fields

```
apiVersion: batch/v1
kind: CronJob
metadata:
  name: report
spec:
  schedule: "*/2 * * * *"
  timeZone: "Asia/Singapore"
  concurrencyPolicy: Forbid
  successfulJobsHistoryLimit: 2
  startingDeadlineSeconds: 30
  jobTemplate:
    spec:
      template:
        spec:
          restartPolicy: OnFailure
          containers:
            - name: report
              image: busybox
              command: ["sh", "-c", "echo report at $(date)"]
```

CronJobs — Scheduled Workloads (cont.)

Step 3 — Suspend, resume and trigger manually

```
k patch cronjob report -p '{"spec":{"suspend":true}}'      # pause
k patch cronjob report -p '{"spec":{"suspend":false}}'     # resume
k create job --from=cronjob/report report-manual           # run now
```

Note concurrencyPolicy: Allow (default) / Forbid / Replace. timeZone is a newer field — always set it. create job --from=cronjob/<name> runs the workload immediately.

CronJobs — Scheduled Workloads

✓ Test it

k get cronjob report shows SUSPEND True after the patch, and k create job --from=cronjob/report report-manual produces an immediate one-off run.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Configuring Retained History

Being able to know what happened during workload execution retroactively

- Even after completing a task in a Pod controlled by a CronJob, it will not be deleted automatically. Keeping a historical record of Pods can be tremendously helpful for troubleshooting failed workloads or inspecting the logs.
- The attribute `spec.successfulJobsHistoryLimit` configures the number of successful executions. Defaults to 3 successful Jobs.
- The attribute `spec.failedJobsHistoryLimit` configures the number of failed executions. Defaults to 1 failed Job.

Tea Break

15 minutes



TOPIC 4

Multi-Container Pods

Sidecar & init-container design patterns

Why More Than One Container?

- Containers in a Pod share the network namespace and can share volumes (emptyDir is the usual glue).
- Sidecar — a helper alongside the main app (log shipper, proxy) sharing a volume.
- Init container — runs to completion before main containers (seed a volume, wait for a service).
- Always target a specific container with `kubectl logs -c <name>` and `kubectl exec -c <name>`.

Container Patterns

Sidecar

- Runs beside the main app
- Shares an emptyDir volume
- Log shipping / proxy / sync

Init container

- Runs first, to completion
- Pod shows Init:N/M
- Seed volume · wait for DNS

Native sidecar

- initContainer + restartPolicy: Always
- Kubernetes 1.29+
- Runs for the Pod's lifetime

Design Patterns

Emerged from Pod requirements



Init Container

- Provide initialization logic concerns to be run before the main application even starts.
- Example: Downloading a configuration files required by application.



Adapter

- Transforms the output produced by the application to make it consumable in the format needed by another part of the system.
- Example: Massaging log data.



Sidecar

- The sidecars are not part of the main traffic or API of the primary application and operate asynchronously.
- Example: Watcher capabilities.

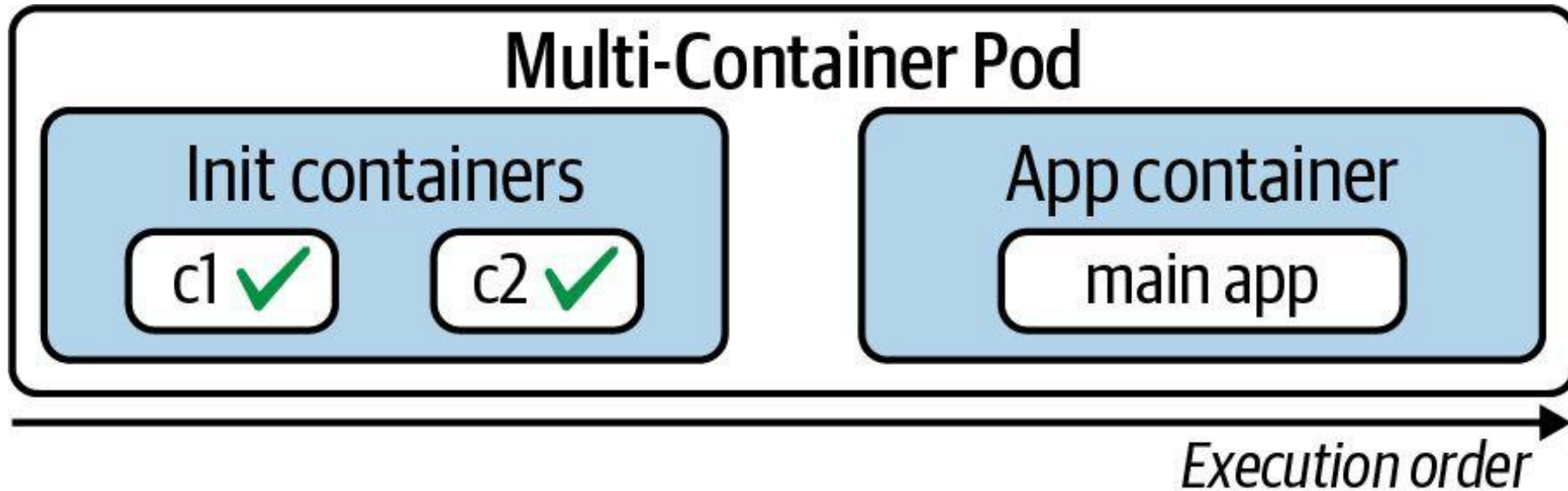


Ambassador

- Provides a proxy for communicating with external services to hide and/or abstract the complexity.
- Example: Rate-limiting functionality for HTTP(S) calls to an external service.

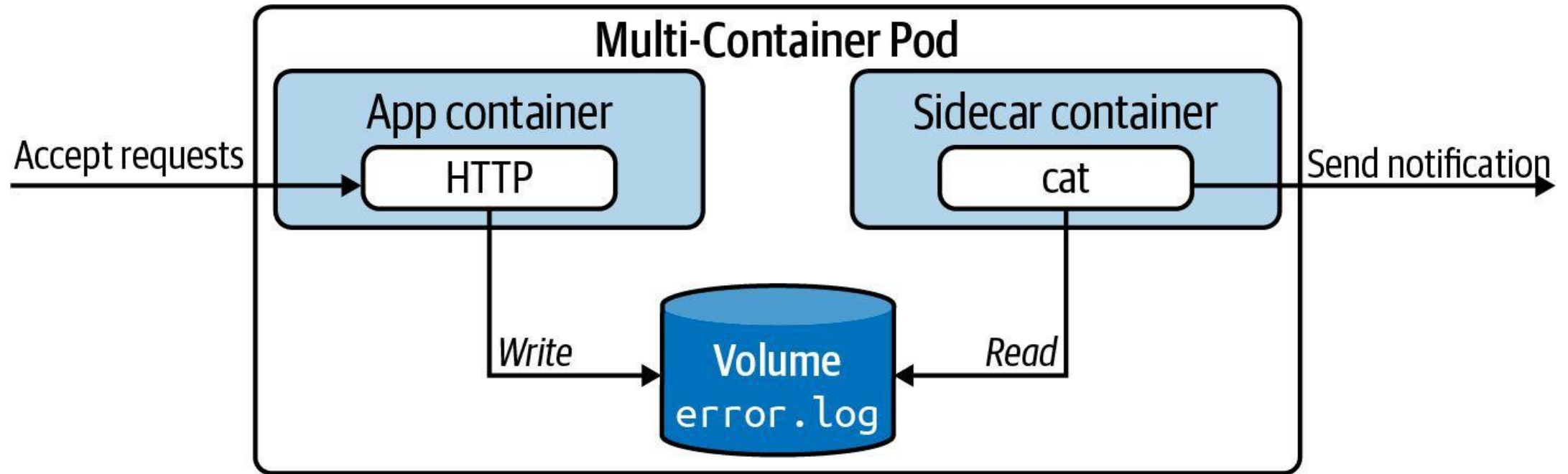
Init Container Life Cycle

Initialization logic before main application containers



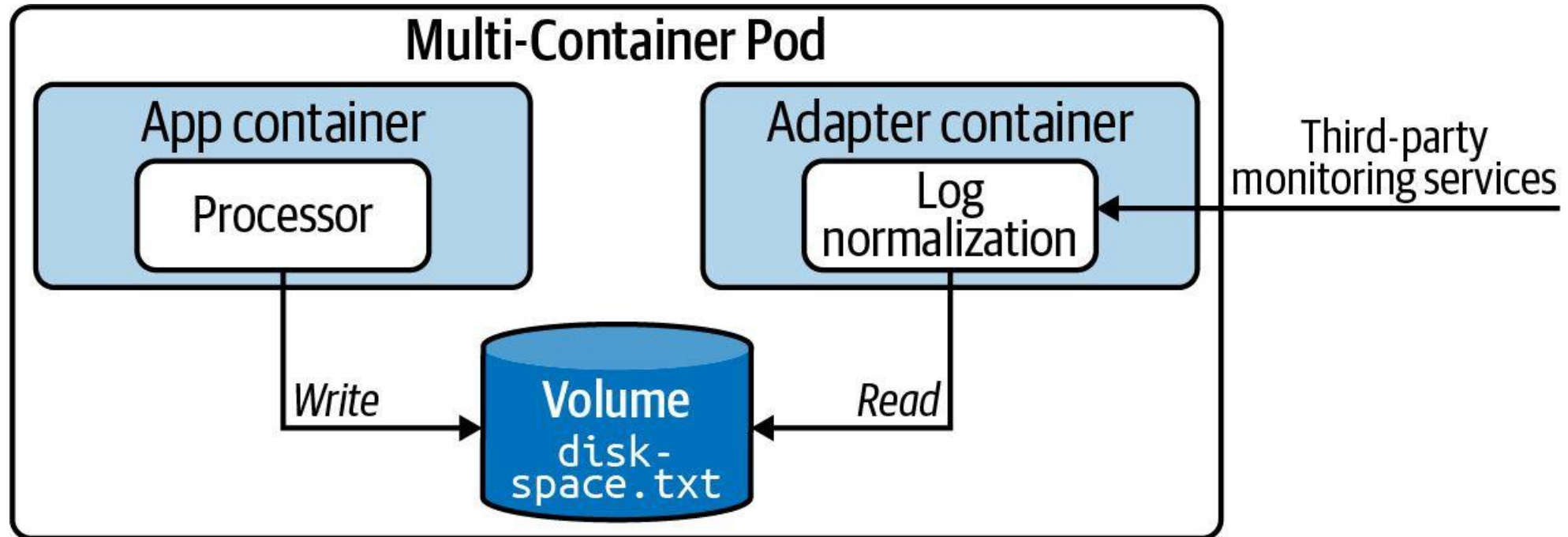
Sidcar Pattern

"Send a notification of error log contains entries"



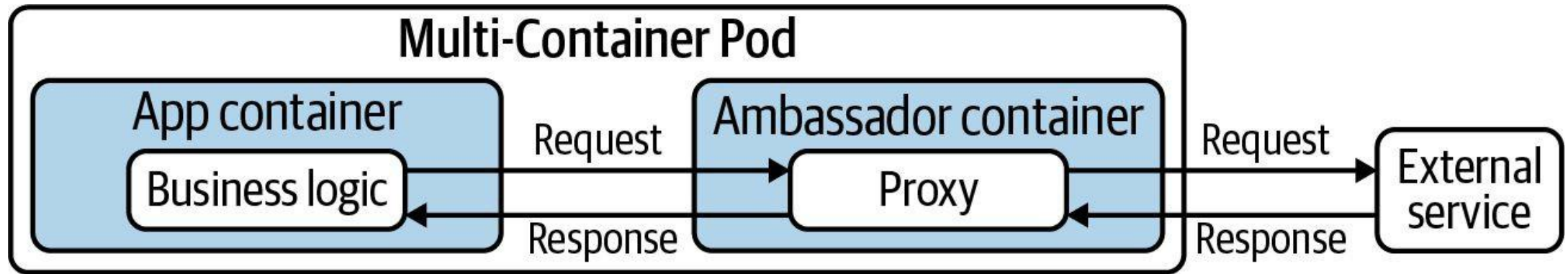
Adapter Pattern

"Transform log data to JSON to make it consumable by external system"



Ambassador Pattern

"OAuth authentication handled by network communication proxy"



Multi-Container Pods — Sidecar Pattern

LAB 6

Multi-container Pods

Containers in the same Pod share a network namespace and can share volumes. CKAD tests the sidecar pattern (a helper reads/writes a shared volume) and the native sidecar feature (Kubernetes 1.29+). You must exec into and read logs from each container.

You'll build

Run a sidecar that tails a shared emptyDir log, read each container's logs, then build a native sidecar.

Key commands: `apiVersion: v1 · k logs · initContainers: · »` With

Multi-Container Pods — Sidecar Pattern

Step 1 — Sidecar sharing a log file via emptyDir

```
apiVersion: v1
kind: Pod
metadata:
  name: app-with-sidecar
spec:
  volumes:
    - name: logs
      emptyDir: {}
  containers:
    - name: app
      image: busybox
      command: ["sh", "-c", "while true; do date >> /var/log/app.log; sleep 2; done"]
      volumeMounts:
        - {name: logs, mountPath: /var/log}
    - name: log-shipper
      image: busybox
      command: ["sh", "-c", "tail -F /var/log/app.log"]
      volumeMounts:
        - {name: logs, mountPath: /var/log}
```


Multi-Container Pods — Sidecar Pattern (cont.)

Step 2 — Read logs from each container separately

```
k logs app-with-sidecar -c app | head -5
k logs app-with-sidecar -c log-shipper | head -5
k exec -it app-with-sidecar -c app -- sh -c 'wc -l /var/log/app.log'
```

Step 3 — Native sidecar container (Kubernetes 1.29+)

```
initContainers:
- name: log-collector
  image: busybox
  restartPolicy: Always
  command: ["sh", "-c", "while true; do echo alive; sleep 5; done"]
```

Note With multiple containers in a Pod, always specify `-c <name>` for logs and exec. `emptyDir` is the standard glue volume between sidecar and main containers.

Multi-Container Pods — Sidecar Pattern

✓ Test it

k logs app-with-sidecar -c log-shipper shows the same lines app is writing, proving both containers share the emptyDir volume.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

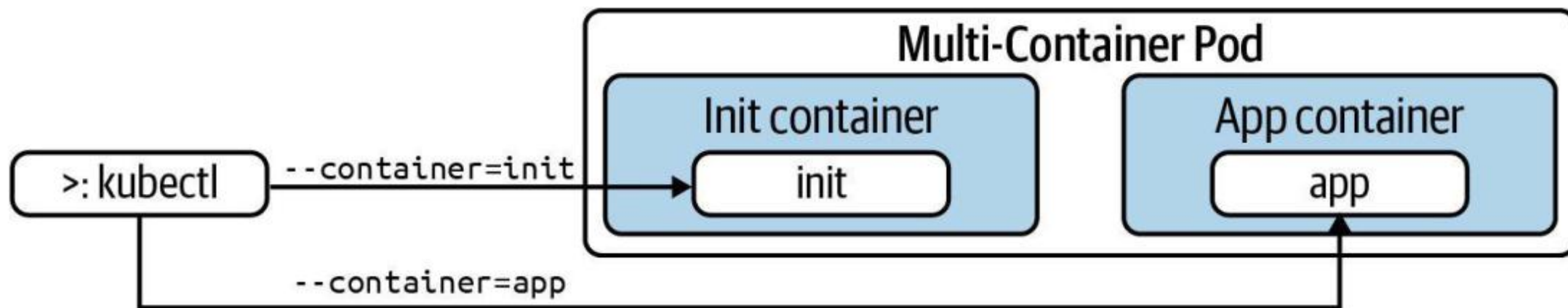


Multi-Container Pods

Defining and interacting with multiple containers, design patterns

Interacting with a Container

Run `kubectl` with `--container` or `-c` CLI option



Shelling into a Container

Defaults to main application container if not explicit

```
$ kubectl exec -it -c web-server business-app -- /bin/sh
# exit
```

```
$ kubectl exec -it business-app -- /bin/sh
Defaulted container "web-server" out of: web-server, ↵
app-initializer
(init) # exit
```

Init Containers

LAB 7

Multi-container Pods

Init containers run in order, to completion, before any main container starts. Use them to seed shared volumes, wait for upstream services, or do one-time setup. CKAD asks you to write init containers and read Init:N/M Pod status correctly.

You'll build

Seed an nginx web root with an init container, then add an init container that waits for a Service via DNS.

Key commands: `apiVersion: v1 · k exec · initContainers: · »` The

Init Containers

Step 1 — Init container that seeds the web root

```
apiVersion: v1
kind: Pod
metadata:
  name: web-init
spec:
  volumes:
    - name: html
      emptyDir: {}
  initContainers:
    - name: seed
      image: busybox
      command: ["sh", "-c", "echo '<h1>Seeded by init</h1>' > /work/index.html"]
      volumeMounts:
        - {name: html, mountPath: /work}
  containers:
    - name: web
      image: nginx:1.25
      volumeMounts:
        - {name: html, mountPath: /usr/share/nginx/html}
```

Init Containers (cont.)

Step 2 — Verify the seeded content

```
k exec web-init -- cat /usr/share/nginx/html/index.html
```

Step 3 — Init container that waits for a Service

```
initContainers:  
- name: wait-for-db  
  image: busybox  
  command: ["sh", "-c", "until nslookup db.default.svc.cluster.local; do sleep 2; done"]
```

Note The Pod stays in Init:0/1 until DNS resolves. Creating the db Service unblocks it. If an init container fails it is restarted per the Pod's restartPolicy.

Init Containers

✓ Test it

`k exec web-init -- cat /usr/share/nginx/html/index.html` returns the seeded heading, proving the init container populated the volume before nginx started.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>



TOPIC 5

Volumes

Persist & share Pod data: emptyDir, tmpfs & hostPath

Pod Storage Basics

- Anything written to the container filesystem is lost when the container restarts.
- Volumes are declared under `spec.volumes` and mounted via `spec.containers[].volumeMounts`.
- A volume can be shared by every container in the Pod — that is how sidecars exchange data.
- Pod-scoped volumes live and die with the Pod; durable storage (PV/PVC) comes later in the course.

Volume Types in Domain 1

emptyDir: {}

- Created with the Pod
- Shared by all containers
- Deleted with the Pod

emptyDir: Memory

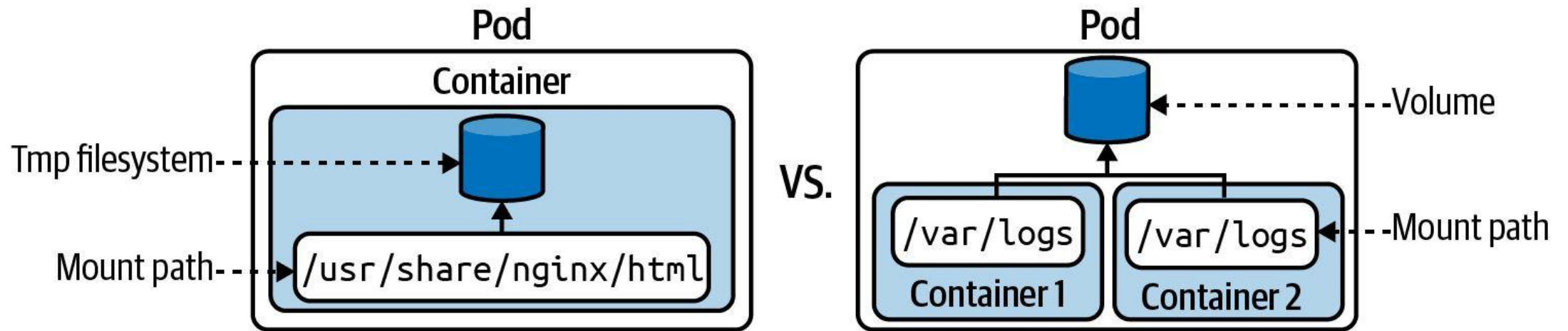
- tmpfs — backed by RAM
- Fast scratch space
- Counts to memory limit

hostPath

- Mounts a node directory
- DaemonSets / node tools only
- Security risk in multi-tenant

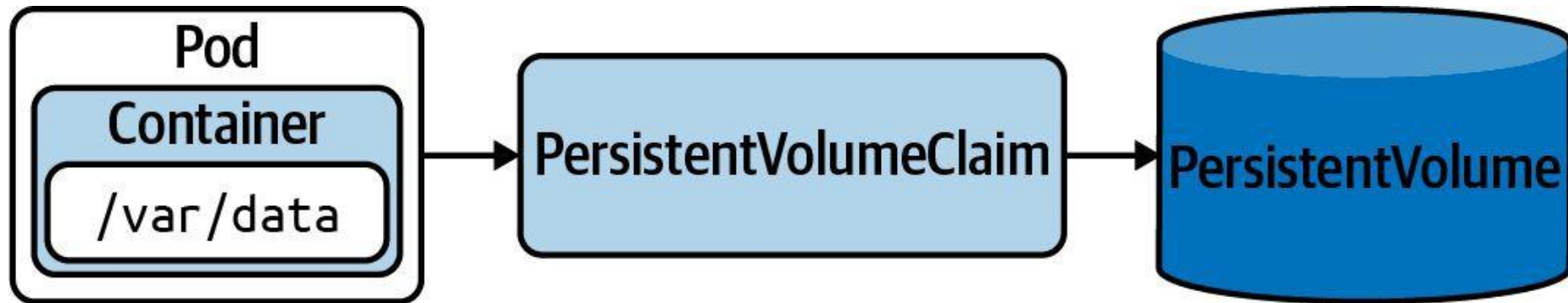
Ephemeral Volume

Useful for sharing data between containers, or for caching data



Persistent Volume

Persist data that outlives a Pod, node, or cluster restart



Volumes — emptyDir and hostPath

LAB 8

Volumes

Containers are ephemeral — data written to the container filesystem is lost on restart. Volumes survive restarts and can be shared between containers. CKAD tests emptyDir, emptyDir.medium: Memory and hostPath, plus mounting strategies.

You'll build

Share an emptyDir between two containers, mount a RAM-backed tmpfs volume, then mount a node directory with hostPath.

Key commands: `volumes: · » Volumes`

Volumes — emptyDir and hostPath

Step 1 — emptyDir shared between two containers

```
volumes:
- name: shared
  emptyDir: {}
containers:
- name: writer
  image: busybox
  command: ["sh", "-c", "echo hello-shared > /data/msg.txt; sleep 3600"]
  volumeMounts: [{name: shared, mountPath: /data}]
- name: reader
  image: busybox
  command: ["sh", "-c", "sleep 5; cat /data/msg.txt; sleep 3600"]
  volumeMounts: [{name: shared, mountPath: /data}]
```


Volumes — emptyDir and hostPath (cont.)

Step 2 — emptyDir backed by RAM (tmpfs)

```
volumes:  
- name: fast  
  emptyDir:  
    medium: Memory  
    sizeLimit: 64Mi
```

Step 3 — hostPath: access files on the node

```
volumes:  
- name: etc  
  hostPath:  
    path: /etc  
    type: Directory
```

Note Volumes are declared under `spec.volumes` and consumed via `spec.containers[].volumeMounts`. Avoid `hostPath` in multi-tenant clusters — it is a security risk; use it only for DaemonSets and node-level tools.

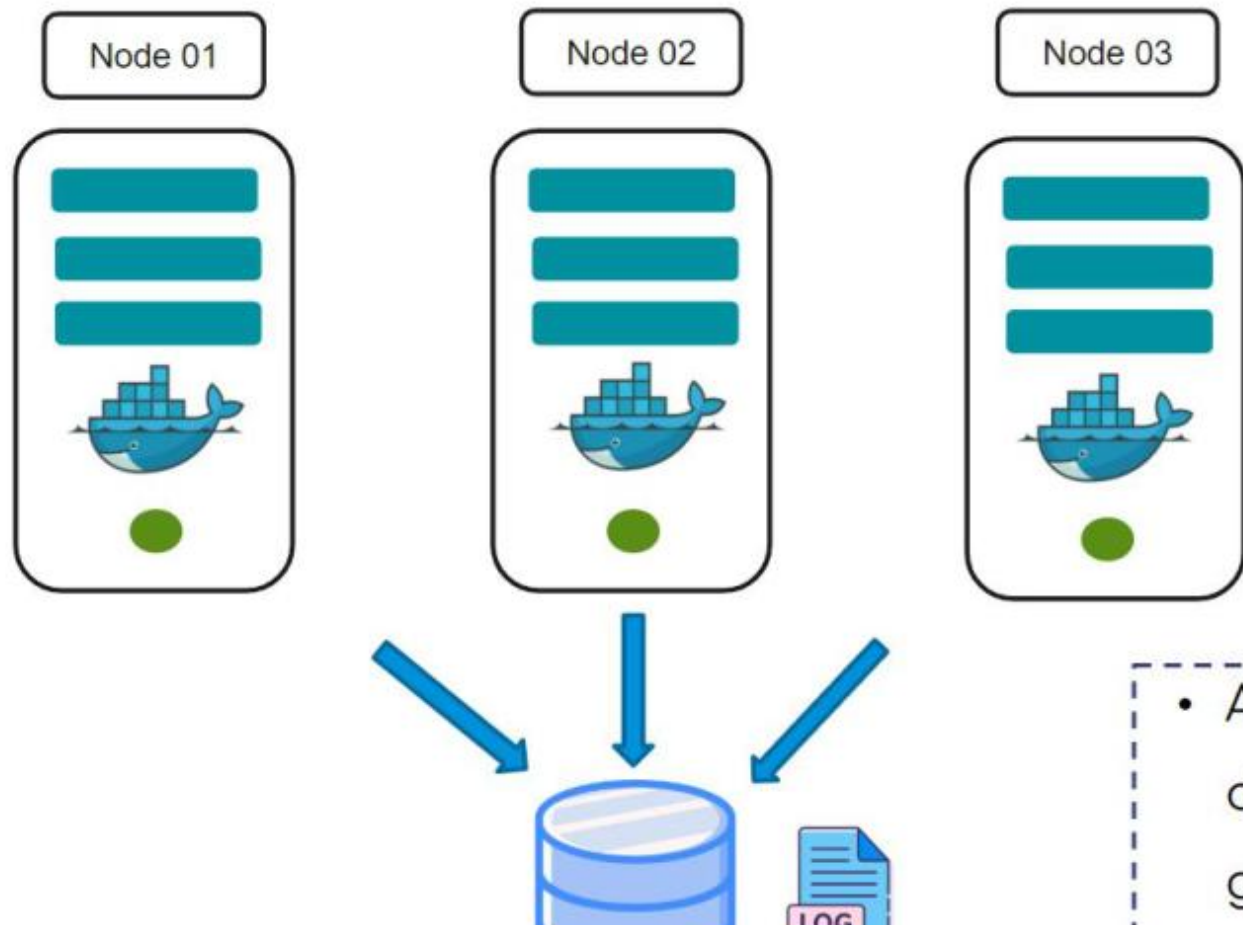
Volumes — emptyDir and hostPath

✓ Test it

k logs scratch -c reader prints hello-shared, proving the writer and reader containers share the same emptyDir volume.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

How do you store the data?



- Allows you to store data even if the pod gets destroyed

Static vs. Dynamic Provisioning

PersistentVolume object is created manually or automatically

- *Static Provisioning:* Storage device needs to be created first. The PersistentVolume object references the storage device and needs to be created manually.
- *Dynamic Provisioning:* The PersistentVolumeClaim object references a storage class. The PersistentVolume object is created automatically.
- *Storage Class:* Object that knows how to provision a storage device with specific performance requirements.

Defining a PersistentVolume

Define storage capacity, access mode, and host path

```
1  apiVersion: v1
2  kind: PersistentVolume
3  metadata:
4    name: pv0001
5  spec:
6    capacity:
7      storage: 5Gi
8    accessModes:
9      - ReadWriteOnce
10   hostPath:
11     path: /tmp/data
```



apiVersion, kind of object

Defining a PersistentVolume

Define storage capacity, access mode, and host path

```
1  apiVersion: v1
2  kind: PersistentVolume
3  metadata:
4    name: pv0001
5  spec:
6    capacity:
7      storage: 5Gi
8    accessModes:
9      - ReadWriteOnce
10   hostPath:
11     path: /tmp/data
```



Name of persistent volume

Defining a PersistentVolume

Define storage capacity, access mode, and host path

```
1  apiVersion: v1
2  kind: PersistentVolume
3  metadata:
4    name: pv0001
5  spec:
6    capacity:
7      storage: 5Gi
8    accessModes:
9      - ReadWriteOnce
10   hostPath:
11     path: /tmp/data
```



Capacity of the storage

Defining a PersistentVolume

Define storage capacity, access mode, and host path

```
1  apiVersion: v1
2  kind: PersistentVolume
3  metadata:
4    name: pv0001
5  spec:
6    capacity:
7      storage: 5Gi
8    accessModes:
9      - ReadWriteOnce
10   hostPath:
11     path: /tmp/data
```

- 
- ReadWriteOnce – mounted as read write by a single node
 - ReadWriteMany – mounted as read write by multiple nodes
 - ReadOnlyMany – mounted as read only by many nodes

Defining a PersistentVolume

Define storage capacity, access mode, and host path

```
1  apiVersion: v1~
2  kind: PersistentVolume
3  metadata:~
4    name: pv0001~
5  spec:~
6    capacity:~
7    storage: 5Gi~
8    accessModes:~
9    - ReadWriteOnce~
10   hostPath:~
11     path: /tmp/data~
```



Volume plugin for the type of storage

Access Mode

Defines read/write capabilities of Volume

Type	Description
<code>ReadWriteOnce</code>	Read-write access by a single node.
<code>ReadOnlyMany</code>	Read-only access by many nodes.
<code>ReadWriteMany</code>	Read-write access by many nodes.
<code>ReadWriteOncePod</code>	Read/write access mounted by a single Pod.

Creating a PersistentVolume

Summarized most important configuration and status

```
$ kubectl create -f db-pv.yaml
```

```
persistentvolume/db-pv
```

```
created
```

```
$ kubectl get pv db-pv
```

NAME	CAPACITY	ACCESS MODES	RECLAIM POLICY	STATUS	...
db-pv	1Gi	RWO	Retain	Available	...



The object hasn't been
consumed by a claim yet

Reclaim Policy

What should happen to PersistentVolume when Claim is deleted?

Type	Description
Retain Delete	Default. When PVC is deleted, PV is “released” and can be reclaimed.
Recycle	Deletion removes PV and associated storage.
	<i>Deprecated.</i> Use dynamic binding instead.

Mounting a PersistentVolumeClaim in a Pod

Use Volume type `persistentVolumeClaim`

```
apiVersion: v1
kind: Pod
metadata:
  name: app-consuming-pvc
spec:
  volumes:
    - name: app-storage
      persistentVolumeClaim:
        claimName: db-pvc
  containers:
    - image: alpine
      ...
      volumeMounts:
        - mountPath:
            "/mnt/data" name:
              app-storage
```

← Reference the Volume by claim name

Assigning a StorageClass for Dynamic Provisioning

Creates PersistentVolume object automatically via storage class

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: aws-ebs
provisioner: kubernetes.io/aws-ebs
```

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: pvc
spec:
  accessModes:
    - ReadWriteMany
  resources:
    requests:
      storage: 256Mi
  storageClassName: aws-ebs
```

Bundling the Template Files into a Chart Archive File

Archive file is usually published to a chart repository for consumption

```
$ helm package .
```

```
Successfully packaged chart and saved it to:  
/Users/bmuschko/helm/web-app-2.5.4.tgz
```

```
$ ls
```

```
web-app-2.5.4.tgz
```

WRAP-UP

Day 1 done — you can design and build applications.

Tomorrow: Application Deployment — Deployments, rollouts, Helm & Kustomize.

COURSE SLIDES

Certified Kubernetes Application Developer (CKAD)

WSQ Course Code: TGS-2025053212

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

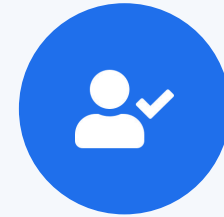
DAY 2

Application Deployment

Deployments · Rollouts · Blue/Green & Canary · Helm · Kustomize · Daemon/StatefulSet (CKAD Domain 2 — 20%)

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your phone camera and submit your attendance.
- A minimum of 75% attendance is required for assessment and funding.



**Minimum 75%
attendance required**

Recap & Today

- Yesterday (Domain 1): images, Pods, Jobs/CronJobs, multi-container Pods and volumes.
- Today (Domain 2 — 20% of the exam): how applications are deployed, scaled and upgraded.
- From a single Pod to controllers that keep replicas running and roll out new versions safely.
- Packaging with Helm and Kustomize, plus DaemonSets and StatefulSets for specialised workloads.

KUBERNETES PRIMITIVES

Labels and Annotations

Label assignment and selection, providing meta information to objects

Labels and Annotations

Labels

- Key-value pairs assigned to an object
- Must follow naming conventions
- Used to query and filter objects from the CLI or from other objects

Annotations

- Represents human-readable metadata for an object
- Not meant for querying purposes
- Reserved annotations can add runtime behaviour to objects

What is a Label?

- Key-value pairs assigned to Kubernetes objects with the imperative label command or using the `metadata.labels[]` attribute.
- Not meant for elaborate, multi-word descriptions of its functionality.
- Kubernetes limits the length of a label to a maximum of 63 characters and a range of allowed alphanumeric and separator characters.

Example Labels

- Three different Pods, each with multiple label key-value pairs:
 - Pod 1: tier=backend, env=prod, app=miracle
 - Pod 2: tier=frontend, env=dev, app=crawler
 - Pod 3: tier=frontend, env=staging, version=v2.1
- Labels are set at creation time or added later with `kubectl label`.
- Multiple labels on the same object create a unique identity for filtering.

Assigning Labels with Imperative Approach

Assign multiple labels to a Pod

```
kubectl label pod nginx tier=backend env=prod app=miracle  
pod/nginx labeled
```

List Pods with their labels

```
kubectl get pods --show-labels  
NAME      ...LABELS  
nginx     ...tier=backend,env=prod,app=miracle
```

Remove a label by appending '-' to its key

```
kubectl label pod nginx tier-  
pod/nginx labeled
```

Assigning Labels in a YAML Manifest

Pod manifest with three label key-value pairs

```
apiVersion: v1
kind: Pod
metadata:
  name: pod1
  labels:
    tier: backend
    env: prod
    app: miracle
spec:
  ...
```

Recommended Labels

Well-known label keys with the app.kubernetes.io prefix

```
metadata:  
  labels:  
    app.kubernetes.io/name: wordpress  
    app.kubernetes.io/instance: wordpress-abcxyz  
    app.kubernetes.io/version: "4.9.4"  
    app.kubernetes.io/managed-by: helm  
    app.kubernetes.io/component: server  
    app.kubernetes.io/part-of: wordpress
```

Selecting Labels

- A label selector uses a set of criteria to query for Kubernetes objects.
- Equality-based requirements — match objects with a specific key or key=value:
 - tier=frontend (equal) • tier!=frontend (not equal)
- Set-based requirements — match objects where a key's value is in a set:
 - 'tier in (frontend, backend)' • 'version notin (v1)'
- Multiple selectors are ANDed together.

Label Selection from the CLI

Equality-based: Pods with tier=frontend AND env=dev

```
kubectl get pods -l tier=frontend,env=dev --show-labels
NAME      ...LABELS
pod2      ...app=crawler,env=dev,tier=frontend
```

Key-only: Pods that have a 'version' label (any value)

```
kubectl get pods -l version --show-labels
NAME      ...LABELS
pod3      ...app=crawler,env=staging,version=v2.1
```

Set-based with Boolean OR

```
kubectl get pods -l 'tier in (frontend,backend),env=dev' --show-labels
```

Label Selection in YAML

A NetworkPolicy selects Pods via podSelector.matchLabels

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: frontend-network-policy
spec:
  podSelector:
    matchLabels:
      tier: frontend
  ...
```

What is an Annotation?

- Key-value pairs for providing descriptive, human-readable metadata to Kubernetes objects via the `metadata.annotations[]` attribute.
- Cannot be used for querying or selecting objects.
- Make sure to put the value of an annotation into single- or double-quotes if it contains special characters or spaces.

Example Annotations

- A Pod can carry any number of annotation key-value pairs:
 - commit: 866a8dc
 - author: 'Benjamin Muschko'
 - branch: 'bm/bugfix'
- Unlike labels, annotation values are unrestricted in length.
- Annotations are visible via `kubectl describe` and `kubectl get -o yaml`.

Assigning Annotations with Imperative Approach

Assign annotations to a Pod

```
kubectl annotate pod nginx commit='866a8dc' branch='bm/bugfix'  
pod/nginx annotated
```

Inspect annotations via describe

```
kubectl describe pods nginx  
Annotations:  branch: bm/bugfix  
              commit: 866a8dc
```

Remove an annotation by appending '-' to its key

```
kubectl annotate pod nginx commit-  
pod/nginx annotated
```

Assigning Annotations in a YAML Manifest

Pod manifest with three annotation key-value pairs

```
apiVersion: v1
kind: Pod
metadata:
  name: pod1
  annotations:
    commit: 866a8dc
    author: 'Benjamin Muschko'
    branch: 'bm/bugfix'
spec:
  ...
```

Well-Known Annotations

Reserved annotations may influence runtime behaviour (e.g. Pod Security Standards)

```
apiVersion: v1
kind: Namespace
metadata:
  name: secured
  annotations:
    kubernetes.io/description: "Manages objects for business app."
    pod-security.kubernetes.io/warn: "baseline"
```



EXERCISE

Using Labels and Annotations

Exam Essentials

- Labels are key-value pairs assigned to Kubernetes objects. Values must follow naming and length restrictions (max 63 chars, alphanumeric + separators).
- Primitives such as Deployments, Services, and NetworkPolicies use label selection for filtering, querying, and routing.
- Use annotations to provide human-readable metadata that cannot be queried. Some reserved annotations influence runtime behaviour (e.g. Pod Security Standards at namespace level).

Well-Known Annotations

Reserved annotations that may influence runtime treatment

```
apiVersion: v1
kind: Namespace
metadata:
  name: secured
  annotations:
    kubernetes.io/description: "Manages objects for business app."
                              "baseline"
```



Enforces Pod
Security Standards
at the namespace



TOPIC 1

Deployments & ReplicaSets

The controller that keeps your Pods running

Why Deployments?

- A bare Pod is not rescheduled if it dies — you need a controller to keep the desired count.
- A Deployment manages a ReplicaSet, which manages the Pods — a three-tier ownership chain.
- `kubectl scale` changes the replica count; the ReplicaSet controller reconciles immediately.
- Old ReplicaSets are kept for rollback (`revisionHistoryLimit`, default 10).

The Ownership Chain

Deployment

- You declare desired state
- replicas + Pod template
- Owns one active ReplicaSet

ReplicaSet

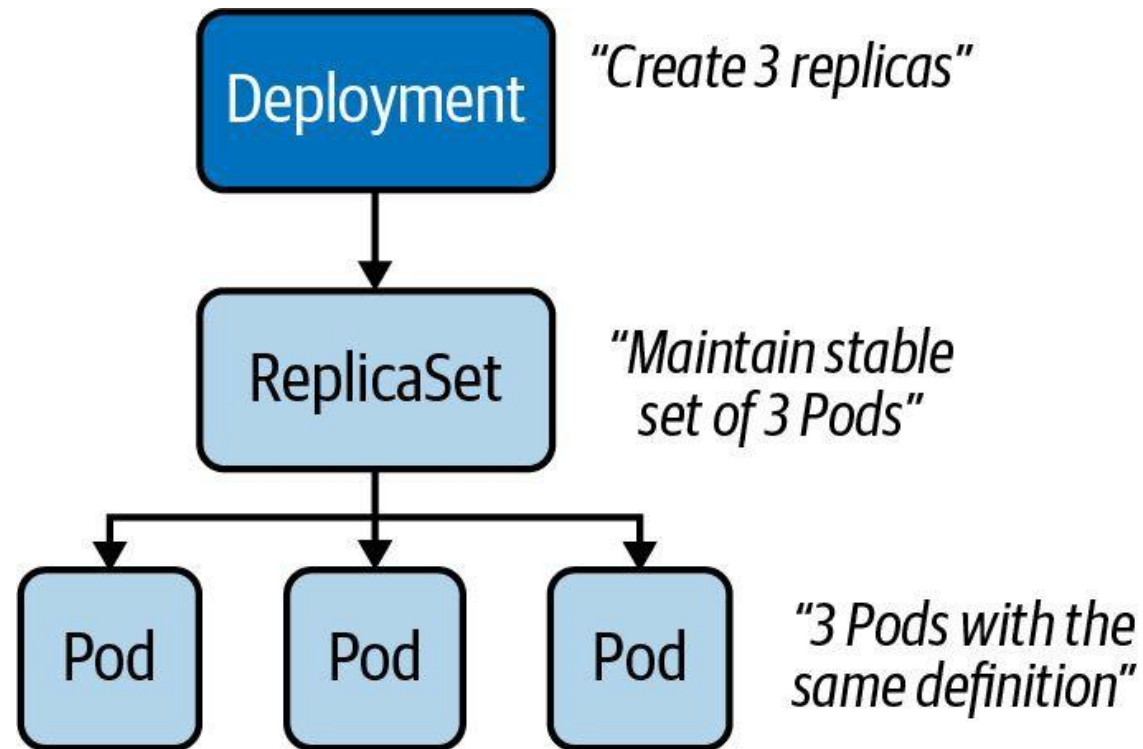
- Keeps N Pods running
- Hash-suffixed name
- Self-heals deleted Pods

Pod

- The running workload
- Recreated if it dies
- Never edited directly

Example Deployment

A Deployment that controls three replicas



Deployments and ReplicaSets

LAB 9

Deployments & ReplicaSets

A Deployment manages a ReplicaSet, which manages Pods. CKAD tests the full ownership chain, scaling, environment-variable injection and reading ReplicaSet history — required to debug failed rollouts.

You'll build

Create a Deployment imperatively, scale it, generate YAML, inject env vars, then read its ReplicaSet history.

Key commands: `k create · k scale · k set · » `kubectl`

Deployments and ReplicaSets

Step 1 — Create a Deployment imperatively

```
k create deployment web --image=nginx:1.25 --replicas=3  
k get deploy,rs,pod -l app=web
```

Step 2 — Scale up and down

```
k scale deployment web --replicas=5  
k scale deployment web --replicas=2  
k get pods -l app=web
```

Step 3 — Generate Deployment YAML and apply

```
k create deployment api --image=httpd:2.4 --replicas=2 $do > api.yaml  
k apply -f api.yaml
```

Deployments and ReplicaSets (cont.)

Step 4 — Inject an env var and observe ReplicaSet history

```
k set env deployment/api APP_COLOR=blue  
k get rs -l app=api      # old RS 0/0/0, new RS 2/2/2
```

Note kubectl set env / set image mutate the Pod template and trigger a new ReplicaSet (a rolling update). Old ReplicaSets are kept for rollback — revisionHistoryLimit (default 10) controls how many.

Deployments and ReplicaSets

✓ Test it

k get deploy,rs,pod -l app=web shows one Deployment owning one ReplicaSet owning the Pods, and k get rs -l app=api shows two ReplicaSets after the env change.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Creating a Deployment with Imperative Approach

Creates objects for the Deployment, ReplicaSet and Pod(s)

```
kubectl create deployment deployment-1 --image=nginx:1.18.0
```

Create command

Deployment name

Image

Deployment Template Attributes

Replica configuration is defined under `spec.template`

Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: my-deploy
    name: my-deploy
spec:
  replicas: 3
  selector:
    matchLabels:
      tier: backend
  template:
    metadata:
      labels:
        tier: backend
    spec:
      containers:
        - image: nginx:1.14.2
          name: nginx
```

Template spec attributes
are the same as for a
Pod spec

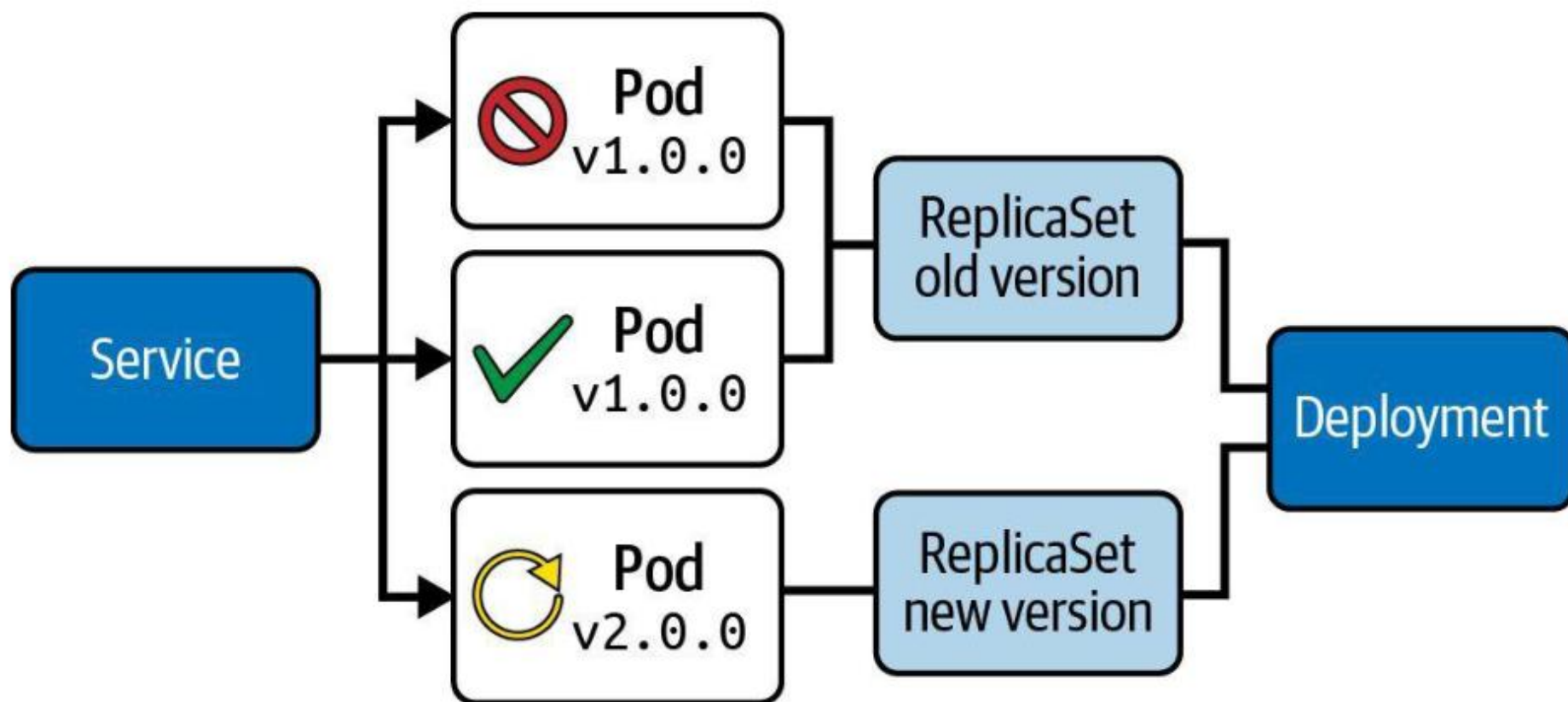


Pod

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
    - image: nginx:1.14.2
      name: nginx
```


Rolling Update Deployment Strategy

Secondary ReplicaSet is created with new version of application



Rolling Update Deployment Strategy

Runtime behavior can be fine-tuned

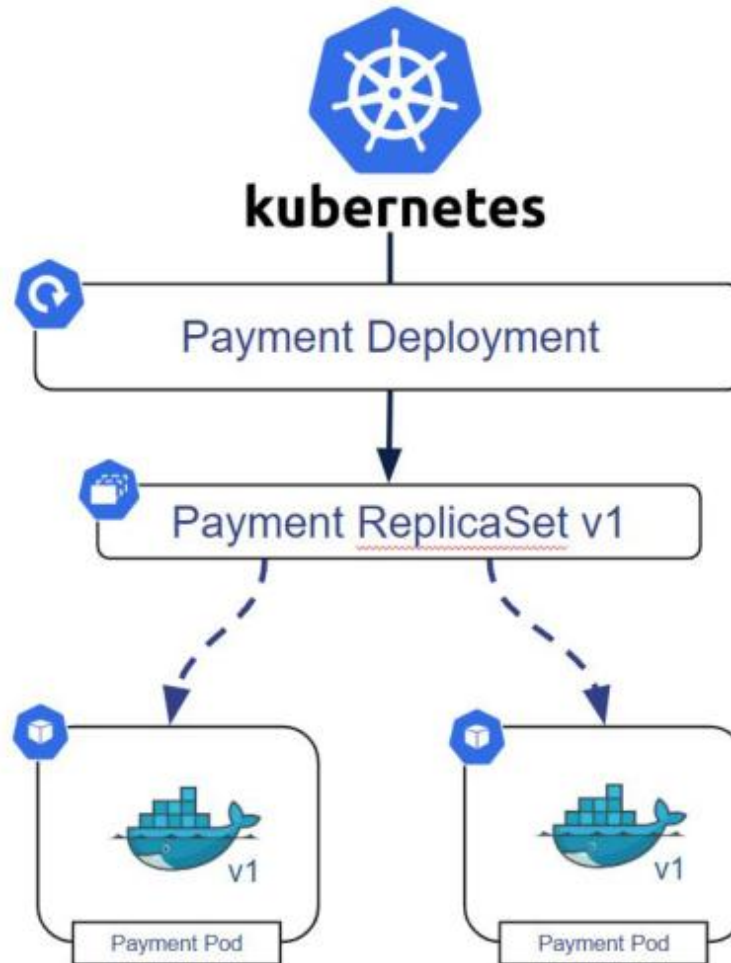
```
spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 2
      maxUnavailable: 0
```

← Determines how update
should be performed

- *Pros:* New version is slowly distributed over time, no downtime
- *Cons:* Breaking API changes can make consumers incompatible

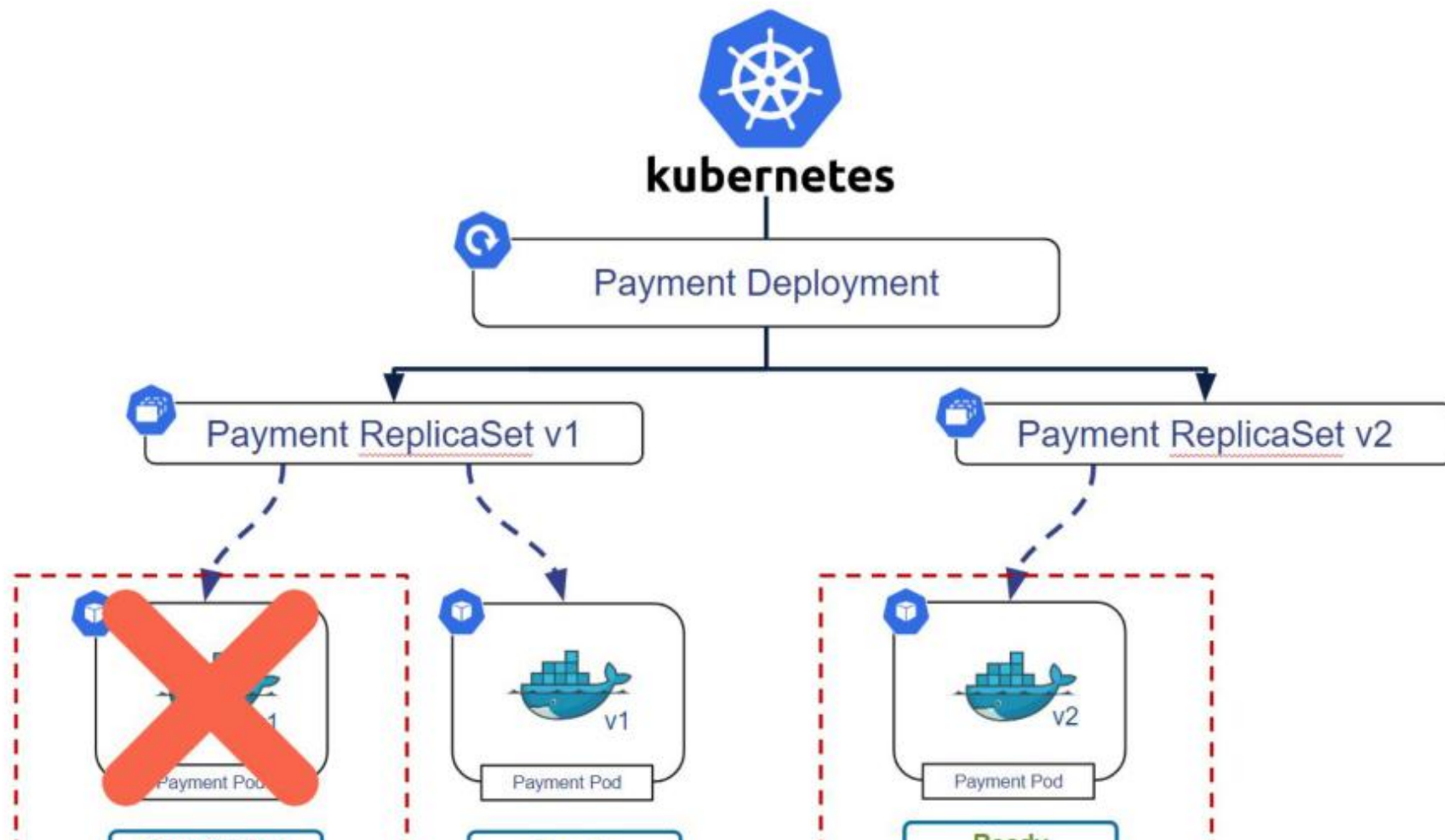
Rolling Update Deployment Strategy

Secondary ReplicaSet is created with new version of application



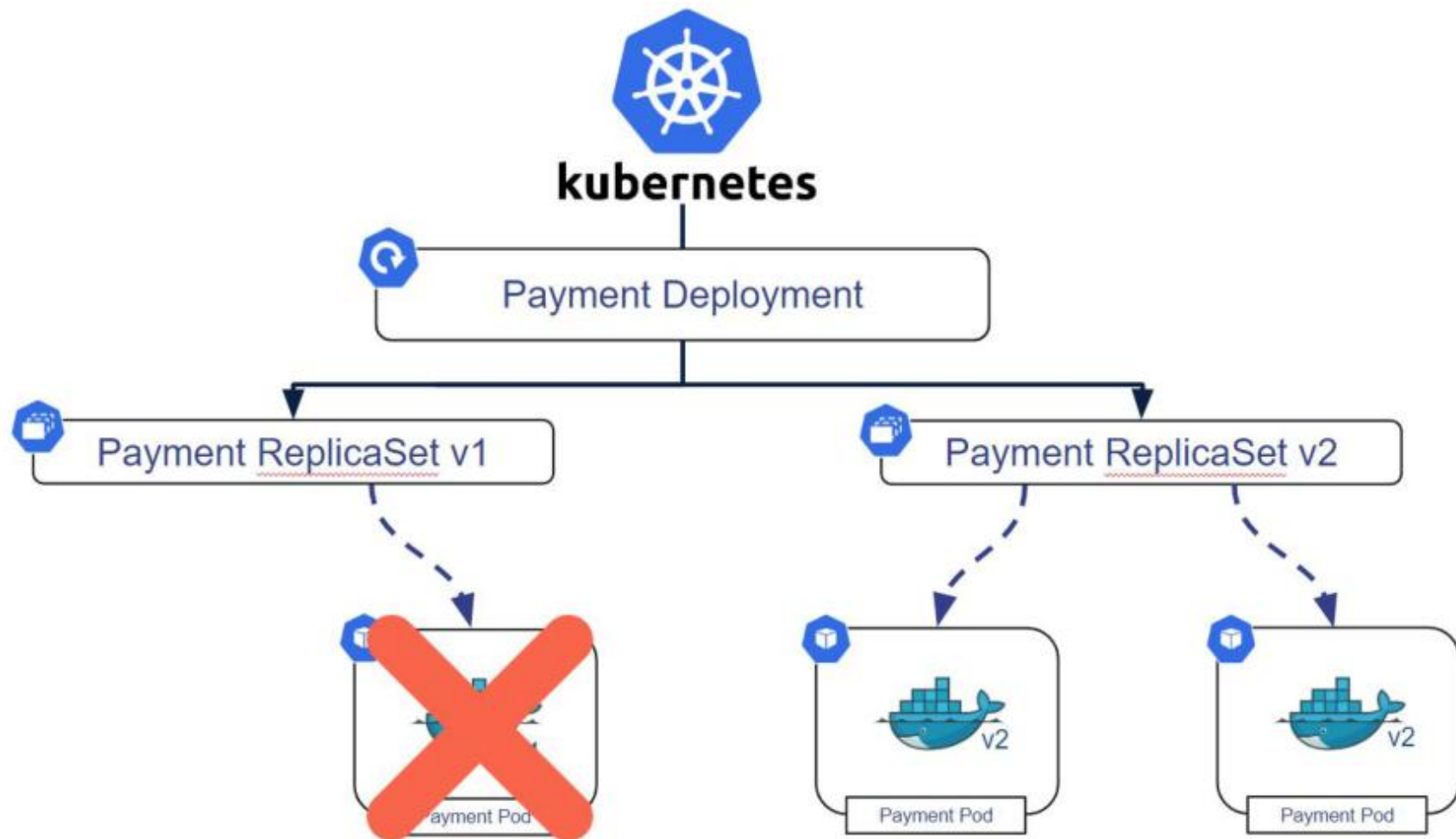
Rolling Update Deployment Strategy

Secondary ReplicaSet is created with new version of application



Rolling Update Deployment Strategy

Secondary ReplicaSet is created with new version of application



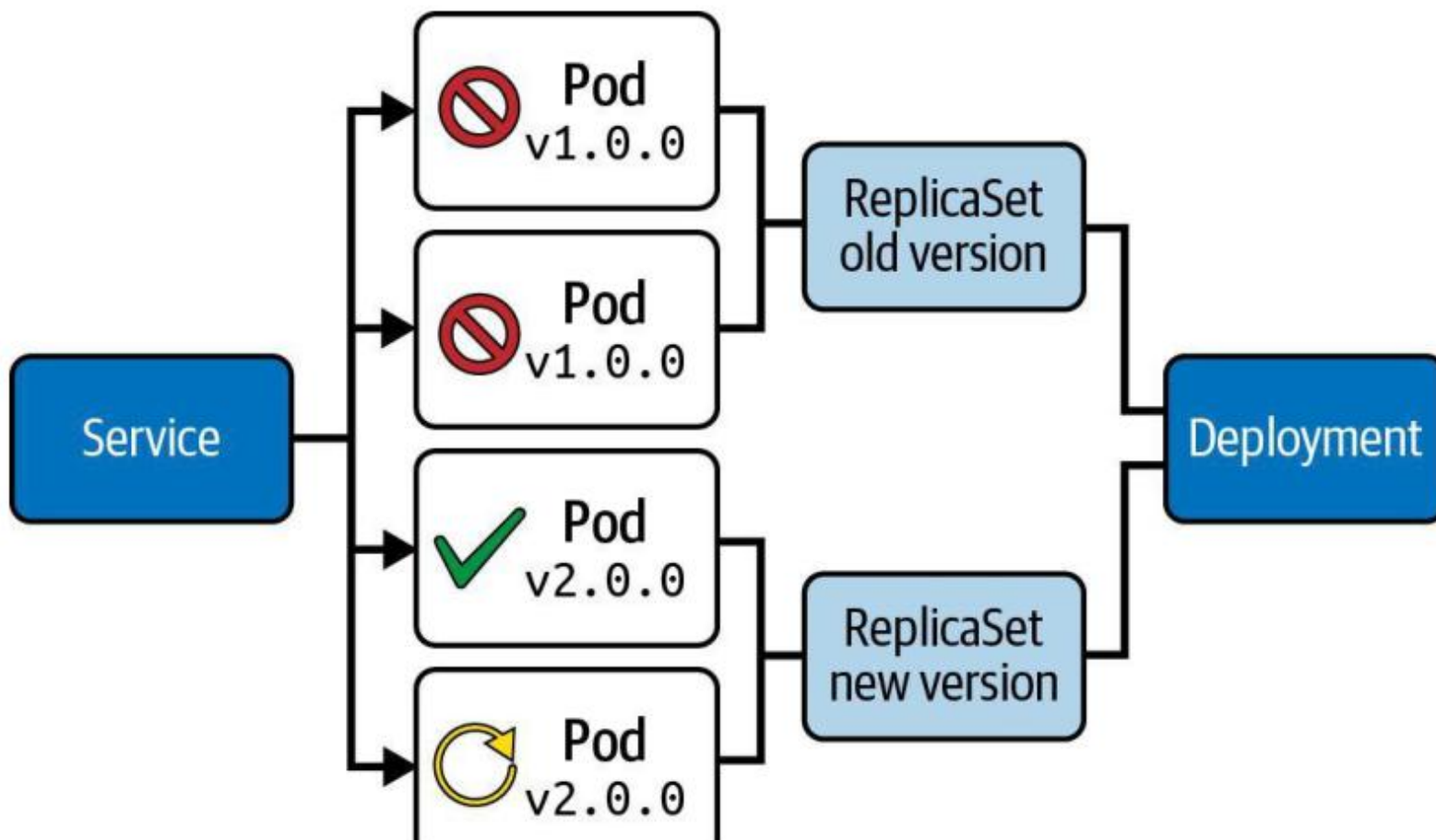


Secondary ReplicaSet is created with new version of application



Recreate Deployment Strategy

Terminates all the running instances then recreate them



Recreate Deployment Strategy

No configuration options available

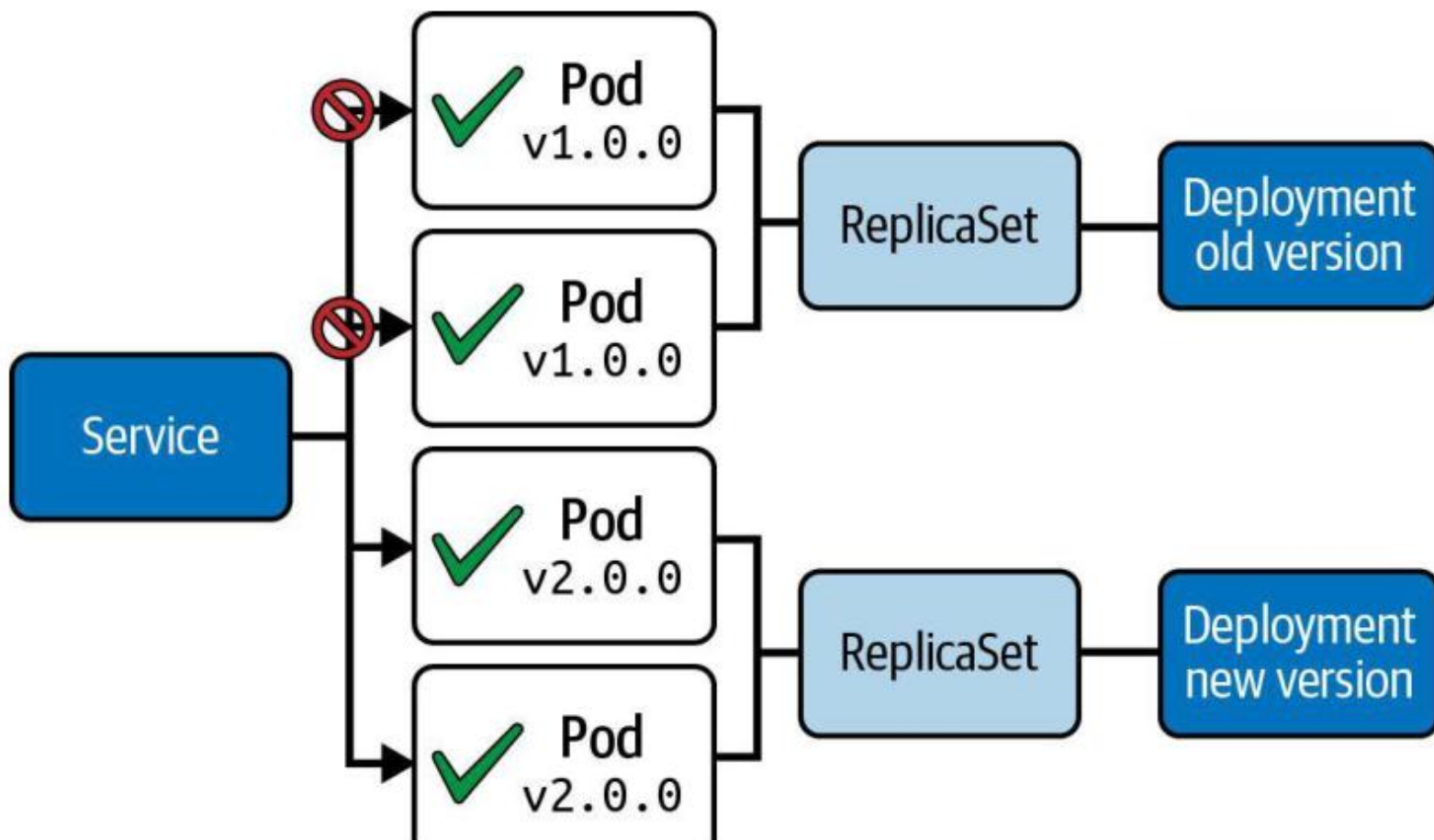
```
spec:  
  replicas: 3  
  strategy:  
    type: Recreate
```

← Set the strategy explicitly

- *Pros:* Good option for development environments as everything is renewed at once
- *Cons:* Can cause downtime while applications are starting up

Blue/Green Deployment Strategy

Deployment object with new version is created alongside



Blue/Green Deployment Strategy

Requires a duplicate set of objects

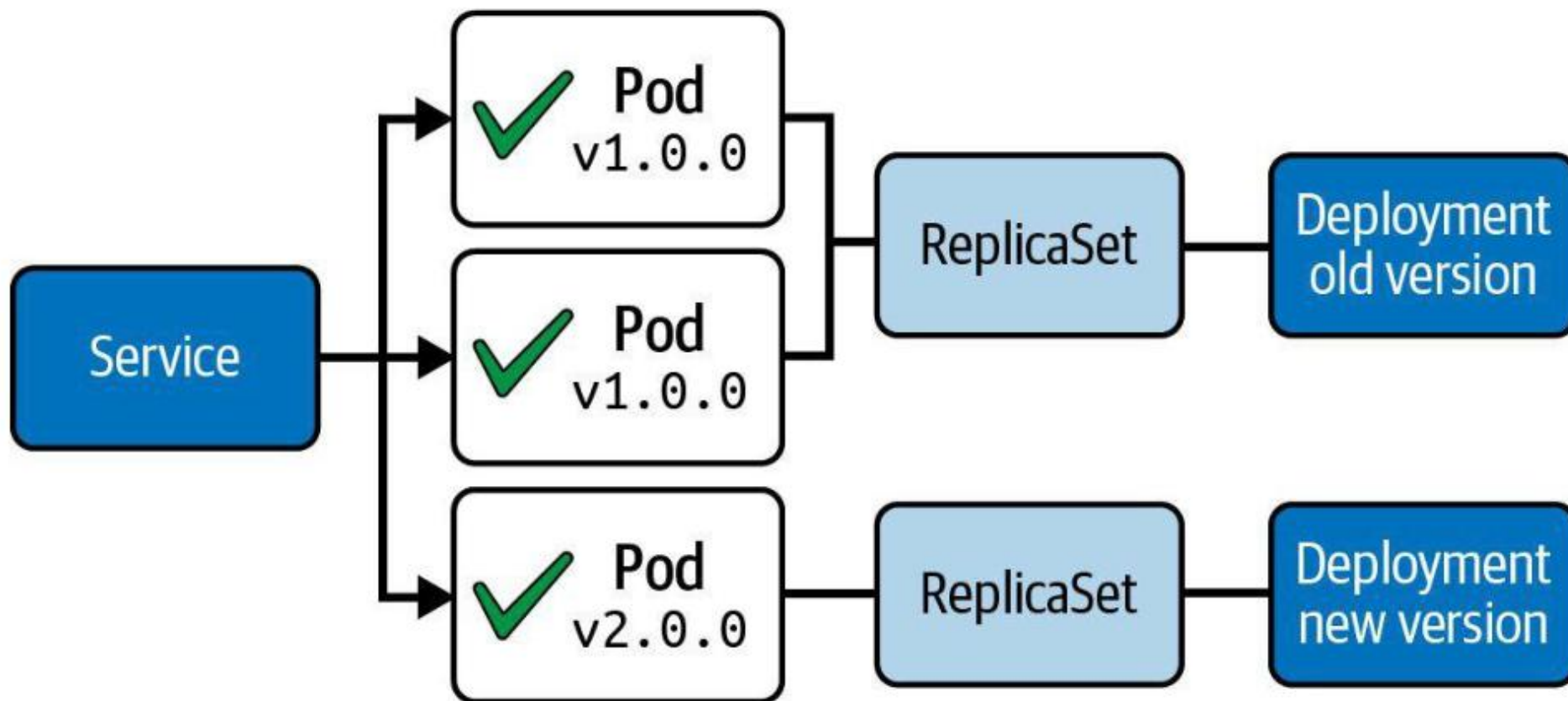
```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: blue
spec:
  selector:
    matchLabels:
      app: my-deploy
      version: 1.0.0
```

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: green
spec:
  selector:
    matchLabels:
      app: my-deploy
      version: 2.0.0
```

- *Pros:* No downtime, traffic can be routed when ready
- *Cons:* Resource duplication, configuration of network routing

Canary Deployment Strategy

Two Deployment objects with different traffic distribution



Canary Deployment Strategy

The new Deployment initially starts with a smaller number of replicas

```
spec:  
  replicas: 3
```

```
spec:  
  replicas: 1
```

- *Pros:* New version released to subset of users, A/B testing of features and performance
- *Cons:* May require a load balancer for fine-grained distribution

Tea Break

15 minutes



TOPIC 2

Rolling Updates & Rollback

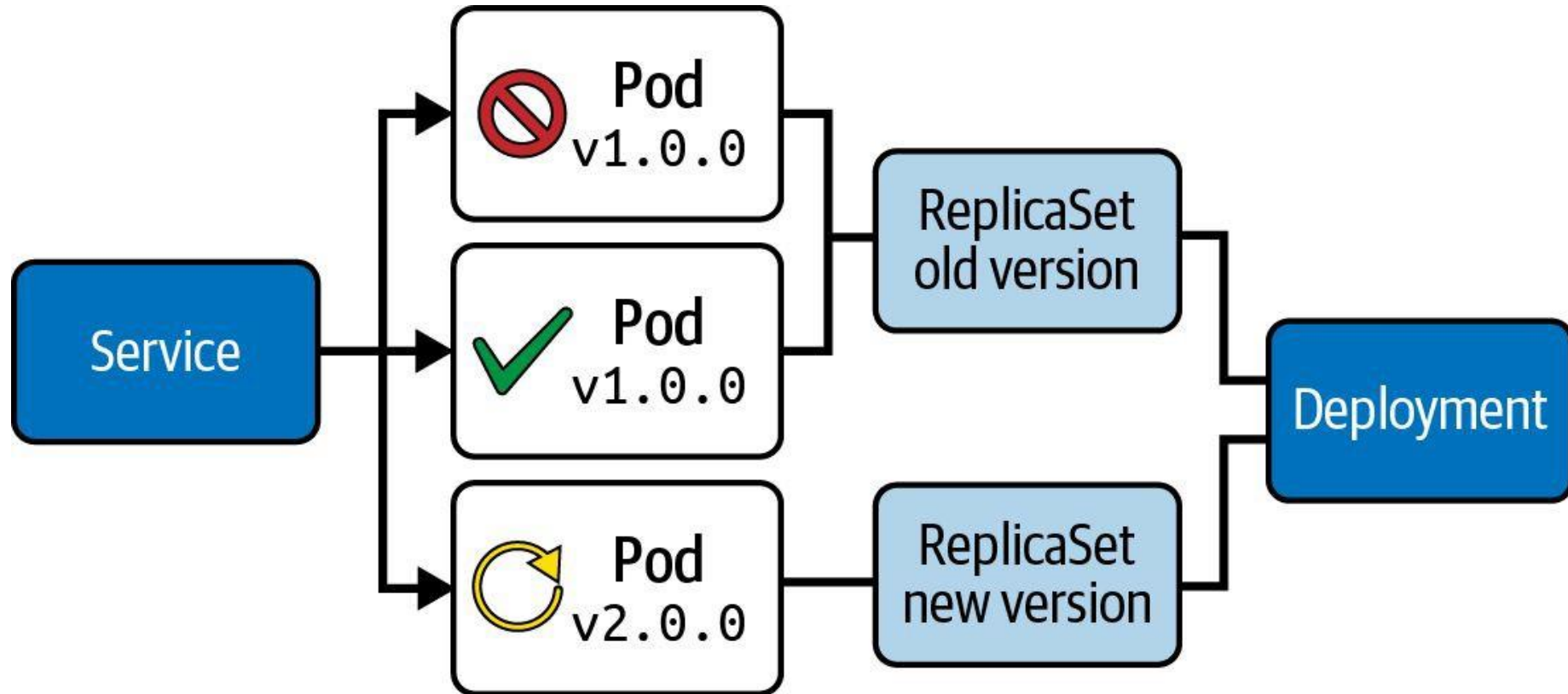
Zero-downtime upgrades, one ReplicaSet at a time

How a Rolling Update Works

- `kubectl set image` mutates the Pod template — this creates a new ReplicaSet and starts a rollout.
- `maxSurge` = extra Pods allowed above desired; `maxUnavailable` = Pods allowed down — together they tune speed vs availability.
- `kubectl rollout status` watches progress; rollout history lists every revision.
- rollout pause batches several changes into one ReplicaSet; rollout undo reverts instantly.

Rolling Update Deployment Strategy

Secondary ReplicaSet is created with new version of application



Rolling Update Deployment Strategy

Runtime behavior can be fine-tuned

```
spec:  
  replicas: 3  
  strategy:  
    type: RollingUpdate
```

```
    rollingUpdate:  
      maxSurge: 2  
      maxUnavailable: 0
```

—————
Determines how update should
be performed

- *Pros:* New version is slowly distributed over time, no downtime
- *Cons:* Breaking API changes can make consumers incompatible

Rolling Updates and Rollback

LAB 10

Rolling Updates & Rollback

A Deployment performs zero-downtime upgrades by gradually replacing Pods one ReplicaSet at a time. CKAD tests maxSurge, maxUnavailable, rollout pause/resume, history inspection and rollback — often as a timed multi-step question.

You'll build

Tune the rolling-update strategy, trigger an image update, inspect history, pause/resume, then roll back.

Key commands: `k create` · `k patch` · `k set` · `k rollout` · » Pausing

Rolling Updates and Rollback

Step 1 — Create the initial Deployment

```
k create deployment web --image=nginx:1.24 --replicas=4
k rollout status deployment/web
```

Step 2 — Tune the rolling-update strategy

```
k patch deployment web -p \
'{"spec":{"strategy":{"rollingUpdate":{"maxSurge":1,"maxUnavailable":1}}}}'
```

Step 3 — Trigger an image update and watch the rollout

```
k set image deployment/web nginx=nginx:1.25
k rollout status deployment/web
k get rs -l app=web          # old RS drains to 0, new RS ramps to 4
```

Rolling Updates and Rollback (cont.)

Step 4 — View history, pause, batch changes, resume

```
k rollout history deployment/web
k rollout pause deployment/web
k set image deployment/web nginx=nginx:1.26
k set env deployment/web APP_ENV=production
k rollout resume deployment/web
```

Step 5 — Roll back

```
k rollout undo deployment/web
k rollout undo deployment/web --to-revision=1
```

Note Pausing batches multiple mutations into a single new ReplicaSet — one rollout, not two. rollout undo reverts to the previous (or a specified) revision instantly.

Rolling Updates and Rollback

✓ Test it

k rollout history deployment/web lists each revision, and k rollout undo returns the Deployment to the prior image with zero downtime.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Listing Deployment with Imperative Approach

Listing objects for the Deployment, ReplicaSet and Pod(s)

Kubectl get deployments

Get all Deployments

Kubectl describe deployment deployment-1



Setting a Change Cause for a Revision

Tracked by annotation with key `kubernetes.io/change-cause`

```
$ kubectl rollout history deployment my-deploy
```

```
deployment.apps/my-deploy
```

```
REVISION  CHANGE-CAUSE
```

```
2          <none>
```

```
3          <none>
```

```
$ kubectl annotate deployment
```

```
my-deploy kubernetes.io/change-cause="image updated to 1.16.1"
```

```
deployment.apps/my-deploy annotated
```

```
$ kubectl rollout history deployment my-deploy
```

```
deployment.apps/my-deploy
```

```
REVISION  CHANGE-CAUSE
```

```
2          <none>
```

```
3          image updated to 1.16.1
```

← What changed
with this revision?

Rolling Update Declaratively

```
kubectl edit deployment deployment-1
```

Edit deployment command

Deployment name

```
spec:  
  containers:  
  - image: nginx:1.18.0
```


Rolling Update Declaratively

```
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: deployment-1
5  spec:
6    replicas: 3
7    selector:
8      matchLabels:
9        env: dev
10   template:
11     metadata:
12       labels:
13         env: dev
14     spec:
15       containers:
16       - name: nginx-deployment
17         image: nginx:1.18.0
```

NAME	READY	STATUS	RESTARTS	AGE
deployment-1-5c8bbc875-8tsc7	0/1	ContainerCreating	0	12s
deployment-1-b9d67f686-rp6kd	1/1	Running	0	3m7s
deployment-1-b9d67f686-w56vw	1/1	Running	0	3m7s
deployment-1-b9d67f686-wz5tg	1/1	Running	0	3m7s

NAME	READY	STATUS	RESTARTS	AGE
deployment-1-5c8bbc875-8tsc7	1/1	Running	0	26s
deployment-1-5c8bbc875-8tvdv	1/1	Running	0	12s
deployment-1-5c8bbc875-gp5cc	1/1	Running	0	14s

Rolling Update Declaratively

```
kubectl set image deployment deployment-1 nginx=nginx:1.19.1
```

Edit deployment image

Deployment name

Container name=version

Rolling Update Declaratively

NAME	READY	STATUS
deployment-1-68ff495bcc-4nvvgg	1/1	Running
deployment-1-68ff495bcc-vd8lx	0/1	ContainerCreating
deployment-1-75fb796bf8-h542s	1/1	Running
deployment-1-75fb796bf8-j7747	1/1	Terminating

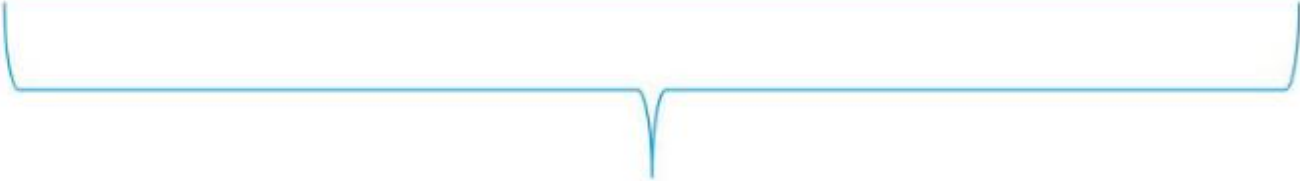
```
Containers:
  deployment-1:
    Image:      nginx:1.19.1
```



Rollout Status



```
kubectl rollout status deployment payment-deployment
```



Rollout command



Deployment name



What happens if there is an error?

Lunch Break

1 hour



TOPIC 3

Blue/Green & Canary

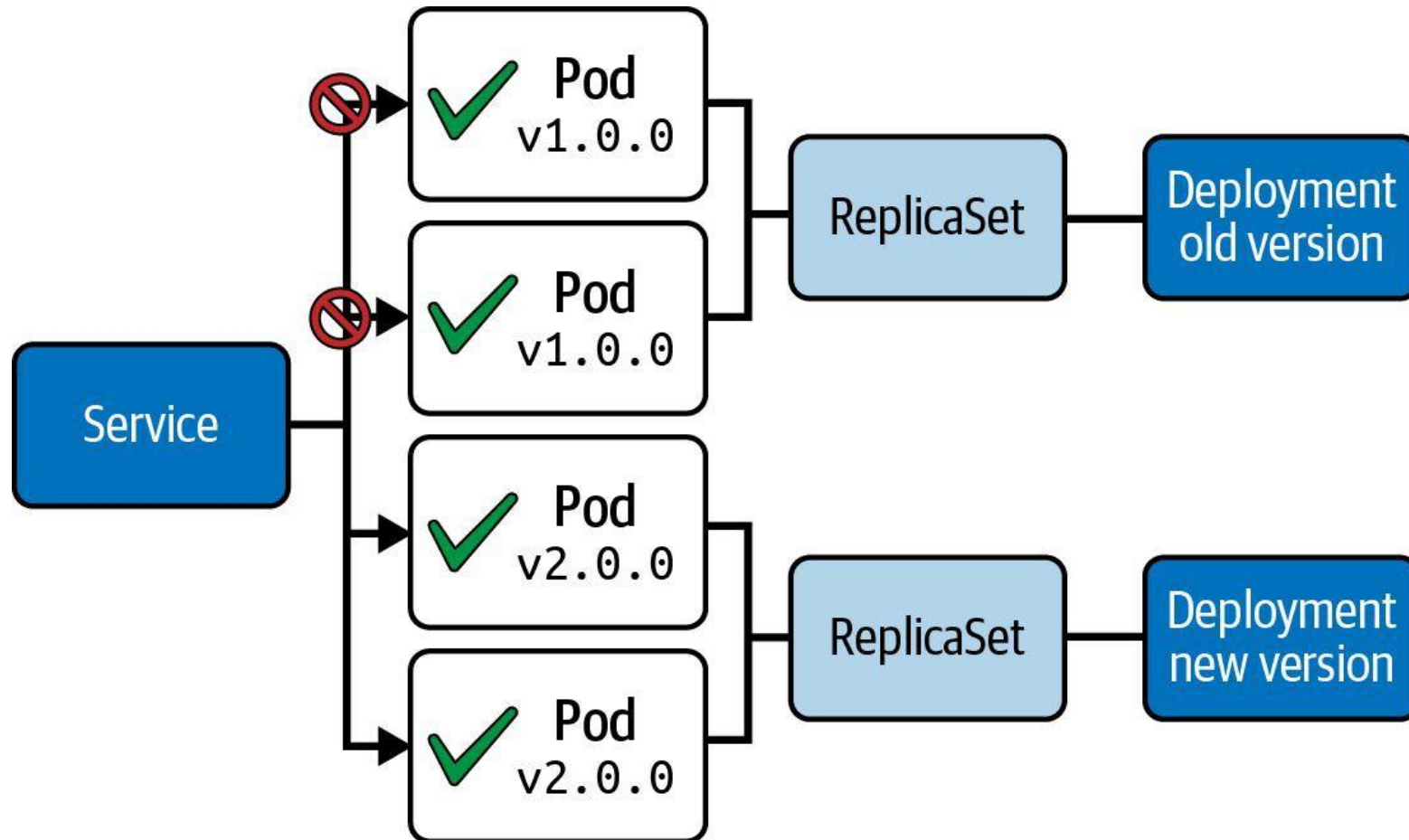
Release strategies built from Deployments + Services

Beyond the Default Rollout

- Blue/Green runs two complete versions; a Service selector switch flips 100% of traffic atomically.
- Canary sends a small slice of traffic to the new version by running it at a low replica ratio.
- Both are built from plain Deployments and Services using labels — no extra tooling for CKAD.
- Rollback is instant: re-patch the selector (blue/green) or scale the canary back down.

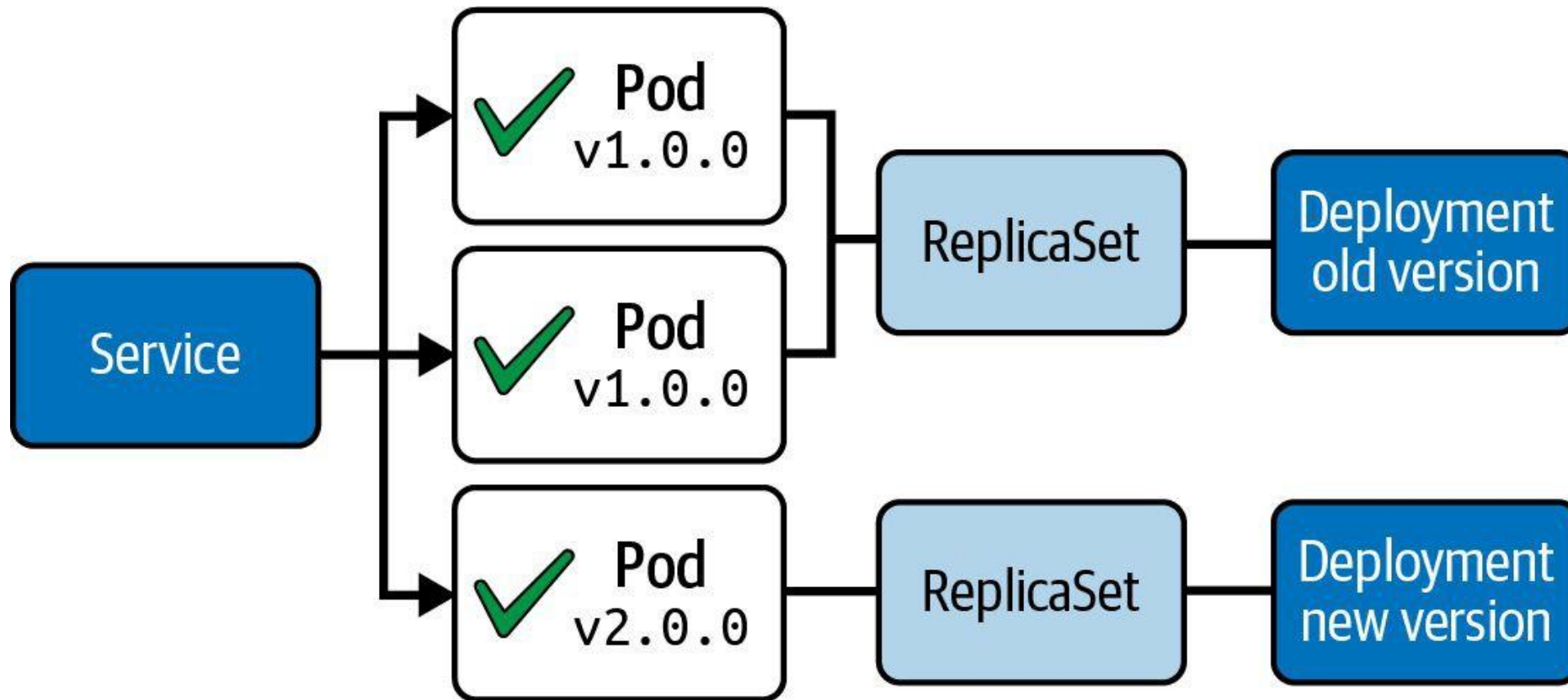
Blue/Green Deployment Strategy

Deployment object with new version is created alongside



Canary Deployment Strategy

Two Deployment objects with different traffic distribution



Blue/Green Deployment

LAB 11

Blue/Green Deployment

Blue/Green keeps two complete copies of the app alive at once. A Service selector switch flips 100% of traffic from blue to green in one atomic operation — with instant rollback if anything breaks.

You'll build

Deploy blue (live) and green (idle), smoke-test green, flip the Service selector, then roll back with one patch.

Key commands: `k apply · GREEN_POD=$(k get · k patch · » Cutover`

Blue/Green Deployment

Step 1 — Deploy blue (current production) behind a Service

```
# Deployment web-blue (labels app=web, version=blue) + Service web
# selector: {app: web, version: blue}
k apply -f blue.yaml
k get pods -l app=web --show-labels
```

Step 2 — Deploy green (next version, no live traffic yet)

```
# Deployment web-green (labels app=web, version=green), nginx:1.25
k apply -f green.yaml
k get pods -l app=web --show-labels
```

Step 3 — Smoke-test green directly before cutover

```
GREEN_POD=$(k get pod -l version=green -o jsonpath='{.items[0].metadata.name}')
k exec $GREEN_POD -- curl -sI localhost:80 | head -2
```

Blue/Green Deployment (cont.)

Step 4 — Flip the Service to green (atomic cutover)

```
k patch service web -p '{"spec":{"selector":{"app":"web","version":"green"}}}'  
k describe svc web | grep Selector
```

Step 5 — Roll back instantly if needed

```
k patch service web -p '{"spec":{"selector":{"app":"web","version":"blue"}}}'
```

Note Cutover and rollback are both a single kubectl patch service selector update — atomic, milliseconds, no Pod churn. Blue stays running as an instant fallback.

Blue/Green Deployment

✓ Test it

After the patch, `k describe svc web | grep Selector` shows `version=green`, and patching it back to `version=blue` restores the old version instantly.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Canary Deployment

LAB 12

Canary Deployment

A canary release sends a small fraction of live traffic to a new version while the bulk stays on stable. In Kubernetes the split is approximated by replica ratios behind one broad Service selector — no special tooling required.

You'll build

Run a 9-replica stable and 1-replica canary behind one Service, verify the ~10% split, then promote.

Key commands: `k apply · k run · k scale · » Traffic`

Canary Deployment

Step 1 — Stable Deployment (90% of traffic)

```
# Deployment web-stable: replicas 9, labels app=web,track=stable
# Service web: selector {app: web}, port 5678
k apply -f stable.yaml
```

Step 2 — Canary Deployment (10% of traffic)

```
# Deployment web-canary: replicas 1, labels app=web,track=canary
k apply -f canary.yaml
k get pods -l app=web --show-labels
```

Step 3 — Verify the traffic split

```
k run probe --image=busybox --restart=Never -it --rm -- sh -c \
  'for i in $(seq 1 30); do wget -qO- web:5678; done | sort | uniq -c'
# ~27 stable, ~3 canary
```


Canary Deployment (cont.)

Step 4 — Promote: scale up canary, retire stable

```
k scale deployment web-canary --replicas=9  
k scale deployment web-stable --replicas=0
```

Note Traffic split is approximated by replica ratio (9:1 $\approx$ 90/10). Promotion is just `kubectl scale` on both Deployments — no Service change. For precise percentage splits use a service mesh (Istio/Linkerd) — beyond CKAD scope.

Canary Deployment

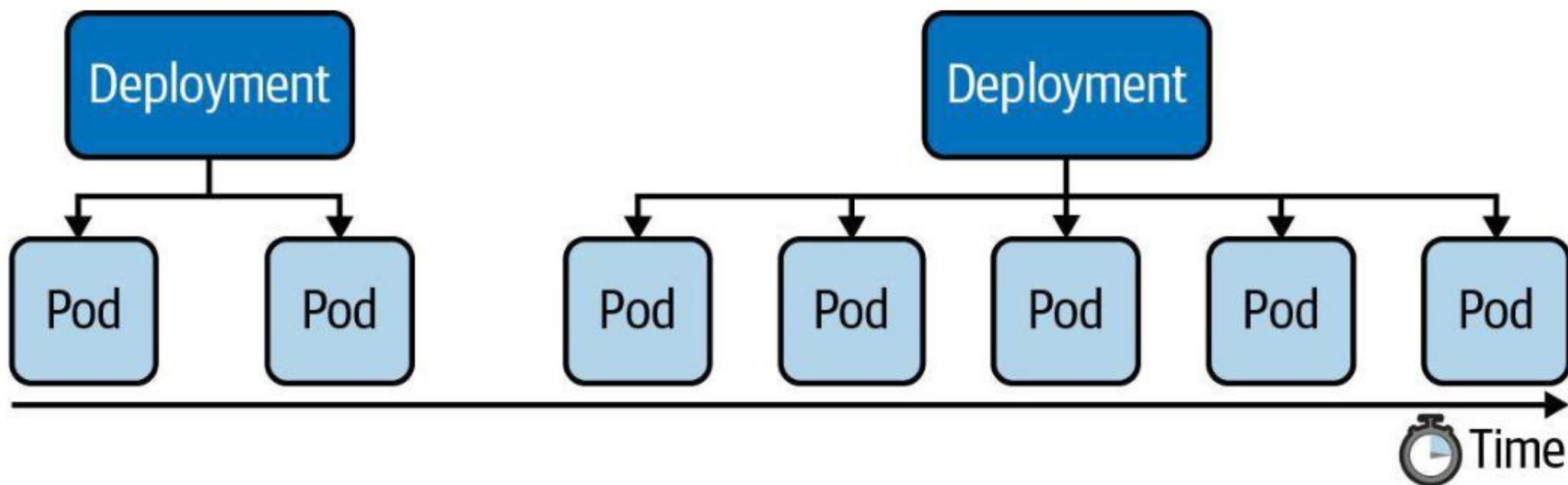
✓ Test it

The probe loop returns roughly 90% stable and 10% canary, and after scaling, all traffic shifts to the canary version with no Service edit.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Manually Scaling a Deployment

"We measured performance under load. Scale up to exactly x number of Pods."



Common Deployment Strategies

Choosing the right deployment procedure depends on the needs

- *Rolling Update*: Release a new version on a rolling update fashion, one after the other.
- *Recreate*: Terminate the old version and release the new one.
- *Blue/green*: Release a new version alongside the old version then switch traffic.
- *Canary*: Release a new version to a subset of users, then proceed to a full rollout.

Common Deployment Strategies

Choosing the right deployment procedure depends on the needs

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- *Recreate*: Terminate the old version and release the new one.
- *Blue/green*: Release a new version alongside the old version then switch traffic.
- *Canary*: Release a new version to a subset of users, then proceed to a full rollout.

Types of Autoscalers

HPA — Horizontal

- Horizontal Pod Autoscaler (HPA)
- Scales the number of Pod replicas based on CPU/memory thresholds
- Standard Kubernetes feature
- Uses the metrics-server component

VPA — Vertical

- Vertical Pod Autoscaler (VPA)
- Scales CPU/memory allocation for existing Pods from historic metrics
- Provided by cloud providers as an add-on or installed manually
- Both autoscalers require metrics-server and resource requests/limits on Pods

Horizontal Pod Autoscaler (HPA)

- The HPA is a Kubernetes primitive that queries the metrics server at regular intervals.
- It compares current resource utilisation against the defined threshold.
- If utilisation exceeds the threshold → HPA increases the replica count (scale out).
- If utilisation drops below the threshold → HPA decreases the replica count (scale in).
- The HPA targets a Deployment (or ReplicaSet) via `scaleTargetRef`.

Autoscaling a Deployment by CPU

- The HPA monitors average CPU utilisation across all running replicas.
- Example: threshold = 80% CPU, current = 15% → no scale-out needed.
- Example: threshold = 80% CPU, current = 95% → HPA adds replicas.
- Scaling decisions respect minReplicas and maxReplicas bounds.
- `kubectl autoscale` is the imperative shortcut to create an HPA.

Horizontal Pod Autoscaler YAML Manifest

HPA targeting a Deployment, scaling on CPU utilisation

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: app-cache
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: app-cache
  minReplicas: 3
  maxReplicas: 5
  metrics:
  - type: Resource
    resource:
      name: cpu
      target:
        type: Utilization
        averageUtilization: 80
```

Inspecting the Horizontal Pod Autoscaler

List HPAs — shows current utilisation vs threshold

```
kubectl get hpa
```

NAME	REFERENCE	TARGETS	MINPODS	MAXPODS	REPLICAS	AGE
app-cache	Deployment/app-cache	15%/80%	3	5	3	2m14s

Describe for detailed events and scaling history

```
kubectl describe hpa app-cache
```



EXERCISE

Creating a Horizontal Pod Autoscaler for a Deployment

Tea Break

15 minutes



TOPIC 4

Helm

The Kubernetes package manager

Charts, Releases & Revisions

- A chart is a bundle of templated YAML; a release is one deployed instance of a chart.
- Workflow: repo add → install → upgrade → rollback → uninstall.
- --set overrides values at install/upgrade; --reuse-values keeps prior overrides.
- helm history tracks every revision; helm get manifest shows what was actually applied.

What is Helm?

Package manager and templating engine for a set of manifests

- The artifact produced by the Helm executable is a so-called *chart file* bundling the manifests that comprise the API resources of an application.
- Chart files can be distributed to a chart repository e.g. [central chart repository](#) and consumed from there (e.g., for running Grafana or PostgreSQL).
- At runtime, it replaces placeholders in YAML template files with actual, end-user defined values.



Standard Chart Structure

Predefined directory structure with mandatory `Chart.yaml` and `values.yaml`

```
$ tree
```

```
.
├──
├── templates
│   ├── web-app-pod-template.yaml
│   └── web-app-service-template.yaml
└──
```

```
values.yaml
```


Helm — Install, Upgrade, Rollback

LAB 13

Helm

Helm is the Kubernetes package manager. A chart is a bundle of templated YAML; a release is a deployed instance. CKAD tests install, upgrade, rollback, list and value overrides — all under exam time pressure.

You'll build

Add a chart repo, install a release with overrides, inspect it, upgrade, roll back, then uninstall.

Key commands: `helm repo · helm install · helm get · helm upgrade · » --reuse-values`

Helm — Install, Upgrade, Rollback

Step 1 — Add a chart repository

```
helm repo add bitnami https://charts.bitnami.com/bitnami
helm repo update
helm search repo bitnami/nginx | head -5
```

Step 2 — Install a release with value overrides

```
helm install web bitnami/nginx \
  --set service.type=ClusterIP --set replicaCount=2
helm list
```

Step 3 — Inspect the generated manifests and values

```
helm get manifest web | head -40
helm get values web
```

Helm — Install, Upgrade, Rollback (cont.)

Step 4 — Upgrade, then roll back

```
helm upgrade web bitnami/nginx --set replicaCount=4 --reuse-values
helm history web
helm rollback web 1
helm uninstall web
```

Note `--reuse-values` preserves prior overrides while changing only the new one. `helm history <release>` tracks every revision; `rollback` creates a new revision rather than deleting history.

Helm — Install, Upgrade, Rollback

✓ Test it

helm history web lists each revision with its status, and helm rollback web 1 returns the release to its first revision as a new history entry.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Finding a Chart

Searches the Artifact Hub for a chart with a matching name

```
$ helm search hub jenkins
```

```
URL
```

```
https://artifacthub.io/packages/helm/bitnami/jenkins
```

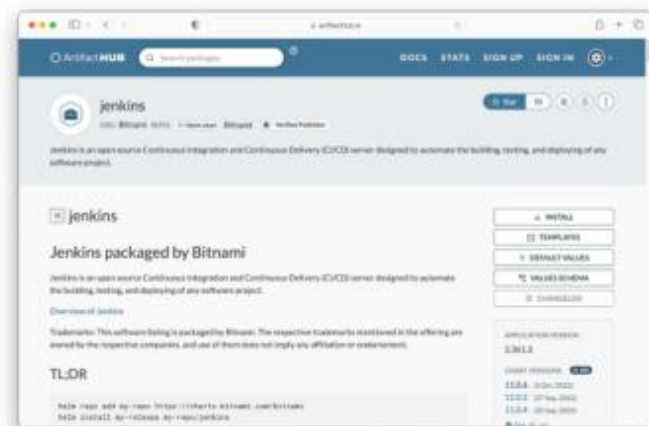
```
...
```

```
CHART
```

```
11.0.6
```

```
VERSION APP
```

```
2.361.2
```



Installing a Chart

You may have to add an external repository that hosts chart file

```
$ helm repo add bitnami https://charts.bitnami.com/bitnami
"bitnami" has been added to your repositories
```

```
$ helm install jenkins bitnami/jenkins
```

```
...
```

```
CHART NAME: jenkins
```

```
CHART VERSION: 11.0.6
```

```
APP VERSION: 2.361.2
```

```
** Please be patient while the chart is being deployed **
```

Listing Installed Charts

Information about chart details

```
$ helm list
```

NAME	NAMESPACE	REVISION	...	STATUS	CHART	APP VERSION
jenkins	default	1		deployed	jenkins-11.0.6	2.361.2

```
$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
pod/bitnami-jenkins-764b44cc99-tw6f6f	1/1	Running	0	7m34s
...				

Uninstalling a Chart

Deletes Kubernetes objects controlled by chart

```
$ helm uninstall jenkins  
release "jenkins" uninstalled
```

```
$ kubectl get pods  
No resources found in default namespace.
```


Values File

Key-value pairs used at runtime to replace placeholders in the YAML manifests

values.yaml

```
db_host: mysql-service
db_user: root
db_password: password
service_port: 3000
```

Additional [built-in objects](#) are available for use.

Previewing Final Templates

Replaces placeholders with actual values from `values.yaml`

```
$ helm template .  
---  
# Source: Web Application/templates/web-app-pod-template.yaml  
apiVersion: v1  
kind: Pod  
metadata:  
  name: web-app  
spec:  
  containers:  
  - image: bmuschko/web-app:1.0.1  
    name: web-app  
    env:  
  
...
```

Installing a Chart from Local Files

Helpful for trying out the deployment before publishing the chart file

```
$ helm install web-app .
```

```
NAME: web-app
```

```
LAST DEPLOYED: Wed Apr 19 11:58:34  
2023
```

```
NAMESPACE:
```

```
default STATUS:
```

```
deployed
```

```
REVISION: 1
```

```
$ kubectl get pods,services
```

NAME	READY	STATUS	RESTARTS	AGE
pod/web-app	1/1	Running	0	3m24s

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/web-app-service	NodePort	10.109.44.239	<none>	3000:30456/TCP	3m24s

Overriding Values

End user can tweak runtime behavior to personal requirements

```
$ helm install --namespace custom --create-namespace --set service_port=9090 web-app .
```

```
NAME: web-app  
LAST DEPLOYED: Wed Apr 19 11:58:34  
2023  
NAMESPACE:  
custom STATUS:  
deployed  
REVISION: 1  
TEST SUITE: None
```

Providing a custom
namespace (defaults to
default namespace)

Override values in
values.yaml



TOPIC 5

Kustomize

Environment overlays without templating

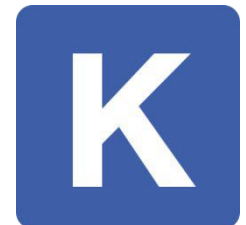
Base + Overlays

- Kustomize reuses one base set of manifests and layers environment-specific changes on top.
- `namePrefix` and `commonLabels` are applied to every resource in the overlay.
- The `images` block overrides image tags without editing the base files; patches do strategic merges.
- Built into `kubectl`: `kubectl apply -k` and `kubectl kustomize` — no extra binary required.

What is Kustomize?

Customizing application manifests without templates

- Writing manifests for application stacks is straightforward, however, deploying them to different Kubernetes runtime environments with varying configuration gets tricky.
- Instead of copying and modifying manifests per environment, you can use kustomize to merge configuration change. Kustomize does not require using a template engine or changing the original manifests.
- Kustomize can be used as a subcommand/flag of `kubectl` or as a separate [executable](#).



Environment Kustomize Setup

Subdirectories contain environment-specific configuration

```
.  
├── base  
│   ├── deployment.yaml  
│   ├── kustomization.yaml  
│   └── service.yaml  
└── dev
```

```
├── config.properties  
└── kustomization.yaml  
└── prod  
    ├── config.properties  
    └── kustomization.yaml
```

Create a dedicated directory +
configuration per environment

Kustomize Overlays

LAB 14

Kustomize

Kustomize reuses a single base set of manifests and applies environment-specific overlays without templating. It is built into kubectl (-k). CKAD tests namePrefix, commonLabels, images and strategic-merge patches.

You'll build

Build a base, add dev and prod overlays (prefix, labels, image tag, replica patch), preview, then apply.

Key commands: resources: · kubectl kustomize · » `namePrefix`

Kustomize Overlays

Step 1 — Base kustomization (shared by all environments)

```
# base/kustomization.yaml
resources:
- deployment.yaml
- service.yaml
```

Step 2 — Dev overlay (name prefix + common label)

```
# overlays/dev/kustomization.yaml
resources:
- ../../base
namePrefix: dev-
commonLabels:
  env: dev
```

Kustomize Overlays (cont.)

Step 3 — Prod overlay (image bump + replica patch)

```
# overlays/prod/kustomization.yaml
resources:
- ../../base
namePrefix: prod-
commonLabels: {env: prod}
images:
- name: nginx
  newTag: "1.26"
patches:
- path: replica-patch.yaml
```

Step 4 — Preview, then apply

```
kubectl kustomize overlays/prod | grep -E "name:|replicas:|image:"
kubectl apply -k overlays/dev
kubectl apply -k overlays/prod
```

Kustomize Overlays (cont.)

Note namePrefix and commonLabels apply to every resource in the overlay. The images block overrides tags without editing base files. apply -k / kubectl kustomize are built into kubectl — no extra binary.

Kustomize Overlays

✓ Test it

`kubectl kustomize overlays/prod` shows name `prod-web`, 5 replicas and `nginx:1.26`, while the dev overlay stays at 1 replica and `nginx:1.25`.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>



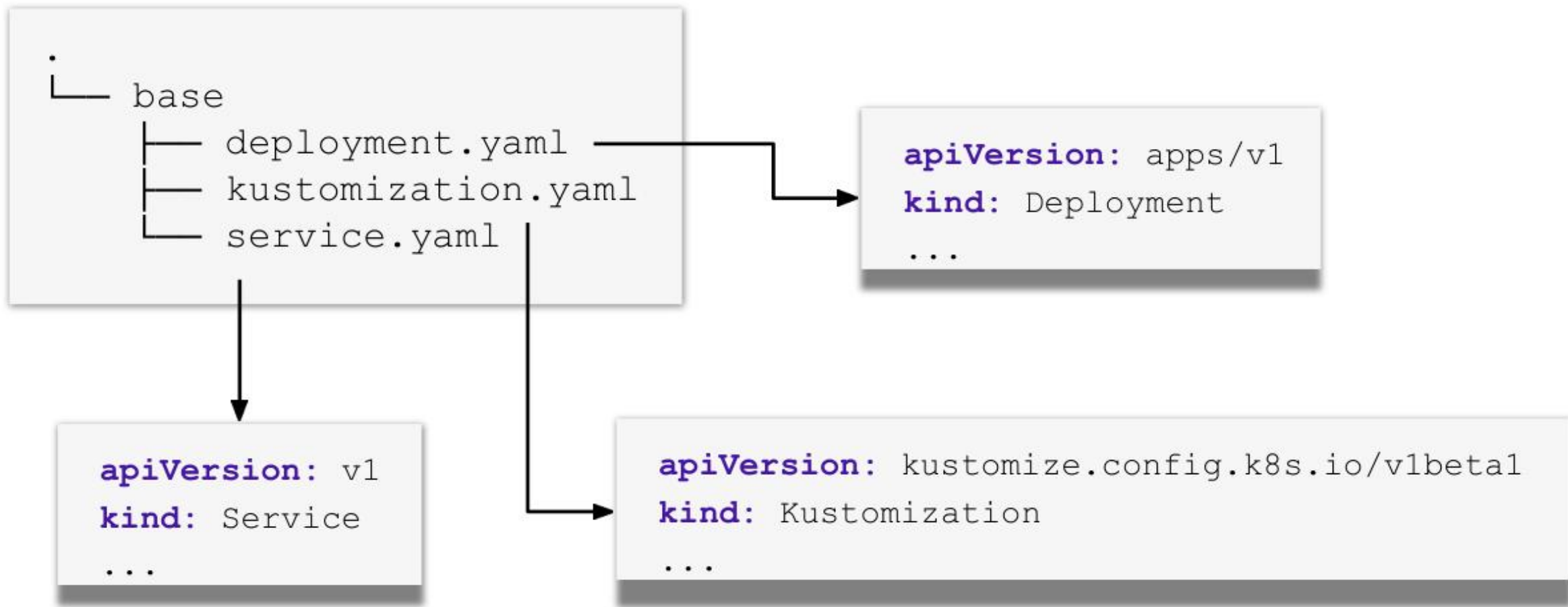
Kustomize Use Cases

Convenient manifest management

- Generating manifests from other sources. For example, creating a ConfigMap and populating its key-value pairs from a properties file.
- Adding common configuration across multiple manifests. For example, adding a namespace and a set of labels for a Deployment and a Service.
- Composing and customizing a collection of manifests. For example, setting resource boundaries for multiple Deployments.

Basic Kustomize Setup

Regular YAML manifests + a `kustomization.yaml` file



Kustomization File Structure

Generate or transform other Kubernetes Resource Model (KRM) objects

```
apiVersion: kustomize.config.k8s.io/v1beta1
```

```
kind: Kustomization
```

```
resources:
```

```
- {pathOrUrl}  
- ...
```

```
generators:
```

```
- {pathOrUrl}  
- ...
```

```
transformers:
```

```
- {pathOrUrl}  
- ...
```

```
validators:
```

```
- {pathOrUrl}  
- ...
```

What *existing* resources
are to be customized

What *new* resources
should be created

What to *do* to the
aforementioned resources

A transformer that doesn't
transform but can (just like a

Example Kustomization File

Merges labels into a set of resources

kustomization.yaml

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
```

```
labels:
  layer: backend
```

```
resources:
- deployment.yaml
- service.yaml
```

Defines labels for all
processed resources

Declares resources
to be modified

Generating a Kustomize File

Imperative command to create `kustomization.yaml` file

```
$ cd base
$ ls .
deployment.yaml service.yaml

$ kustomize create --resources=deployment.yaml,service.yaml ↵
  --labels=layer:backend

$ ls .
deployment.yaml kustomization.yaml service.yaml
```

Building and Previewing a Set of Resources

Applies the rules in `kustomization.yaml` and renders the result to the output

```
$ kustomize build base
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  labels:
```

```
    layer: backend
```

```
...
```

```
---
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  labels:
```

```
    layer: backend
```

```
...
```

Alternatively, you can
also use `kubectl`
`kustomize base`

Applied labels
from rules

Example Production Environment

Base configuration is merged with the configuration of an environment

prod/kustomization.yaml

```
apiVersion: kustomize.config.k8s.io/v1beta1
kind: Kustomization
resources:
- ../base/ namespace:
  prod namePrefix:
  prod-
  configMapGenerator:
- name: backend-configmap
  env: config.properties
replicas:
- name: backend-deployment
  count: 10
```

Merging Environment Configuration

Base configuration + environment configuration

```
$ kustomize build prod
apiVersion: v1
data:
  DB_USERNAME: prod-user
kind: ConfigMap
metadata:
  name: prod-backend-configmap-7gbb88gbhc
  namespace: prod
---
apiVersion: v1
kind: Service
metadata:
  labels:
    layer: backend
  name: prod-backend-service
  namespace: prod
...
```

Kustomize Benefits

A good option for operators managing their own manifests

- *Native to `kubectl`*: Kustomize does not necessarily require you to install additional tooling.
- *DRY approach*: Define a base configuration just once with the ability to reuse it. Environment-specific configuration will be added as an overlay.
- *Easy to learn*: Does not require learning another template engine. You'll use standard YAML manifests to define resources.
- *Scalable variants*: New configuration e.g. environments can be easily added.

Tea Break

15 minutes



TOPIC 6

DaemonSets & StatefulSets

Node-wide agents & stable-identity workloads

Specialised Controllers

- A DaemonSet runs exactly one Pod per matching node — log collectors, CNI agents, node exporters.
- A StatefulSet gives Pods stable names (db-0, db-1) and ordered, graceful start/stop.
- A StatefulSet needs a headless Service (clusterIP: None) for per-Pod DNS.
- Scale-down always removes the highest ordinal first; scale-up adds the next in order.

Pick the Right Controller

Deployment

- Stateless, interchangeable
- Any number of replicas
- Rolling updates

DaemonSet

- One Pod per node
- Node-level agents
- Scales with the cluster

StatefulSet

- Stable identity + storage
- Ordered start/stop
- Databases, queues

DaemonSet vs StatefulSet

DaemonSet — one Pod per node

node-1

agent Pod

node-2

agent Pod

node-3

agent Pod

StatefulSet — ordered, stable names

headless Service (db)

clusterIP: None

db-0

db-0.db.svc.cluster.local

db-1

db-1.db.svc.cluster.local

db-2

db-2.db.svc.cluster.local

created db-0→db-2 · deleted db-2→db-0

DaemonSet = **one Pod per matching node**; StatefulSet = **stable identity + ordered start/stop**.

DaemonSets and StatefulSets

LAB 15

DaemonSets & StatefulSets

A DaemonSet runs exactly one Pod per matching node (log collectors, CNI agents). A StatefulSet gives Pods stable identities and ordered start/stop (databases). CKAD expects you to know when to use each and write the YAML.

You'll build

Run a DaemonSet on every node, then a StatefulSet with a headless Service, stable DNS and ordered scaling.

Key commands: `k apply · nslookup db-0.db.default.svc.cluster.local · » DaemonSet`

DaemonSets and StatefulSets

Step 1 — DaemonSet on every node

```
# DaemonSet node-agent, image busybox
k apply -f ds.yaml
k get ds,pods -l app=node-agent -o wide    # DESIRED = node count
```

Step 2 — Headless Service for the StatefulSet

```
# Service db: clusterIP: None, selector app=db, port 5432
k apply -f headless.yaml
```

Step 3 — StatefulSet with stable identities

```
# StatefulSet db: serviceName db, replicas 3
k apply -f sts.yaml
k rollout status sts/db
k get pods -l app=db          # db-0 → db-1 → db-2 in order
```

DaemonSets and StatefulSets (cont.)

Step 4 — Verify stable per-Pod DNS and scale

```
nslookup db-0.db.default.svc.cluster.local
k scale sts db --replicas=4      # adds db-3
k scale sts db --replicas=2      # removes db-3 then db-2
```

Note DaemonSet = one Pod per matching node. StatefulSet = stable names (db-0, db-1) and ordered start/stop; a headless Service (clusterIP: None) gives each Pod its own DNS. Scale-down always removes the highest ordinal first.

DaemonSets and StatefulSets

✓ Test it

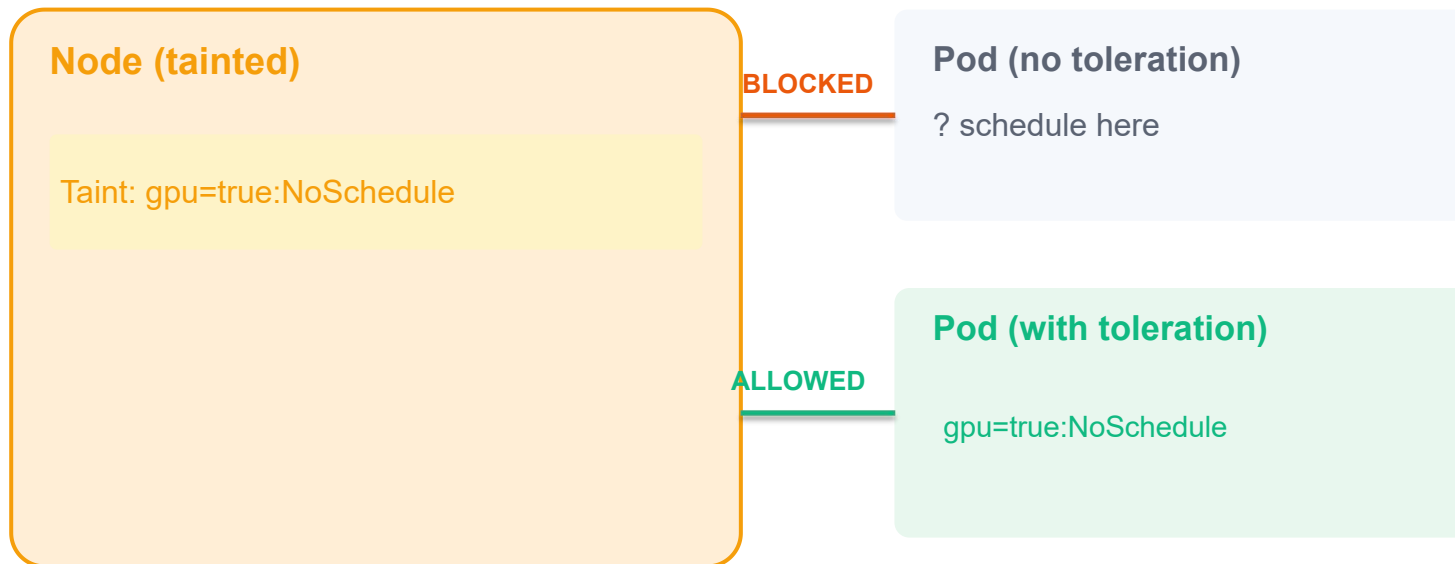
k get pods -l app=db shows db-0, db-1, db-2 created in order, and nslookup db-0.db.default.svc.cluster.local resolves to a single Pod IP.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Taints & Tolerations

Taint — mark a Node to repel Pods that do not tolerate it.

Toleration — added to a Pod to allow it to schedule onto tainted Nodes.



Exam Tip

- Taints repel Pods from Nodes
- Tolerations let a Pod tolerate a taint
- A Pod with a toleration **MAY** go to a tainted node — it's not forced

Taint Effects

NoSchedule

New Pods without matching toleration are NOT scheduled here.
Already-running Pods are unaffected.

PreferNoSchedule

Scheduler TRIES to avoid placing Pods here, but will if no other option exists.
Soft constraint.

NoExecute

New Pods are not scheduled AND existing Pods without toleration are EVICTED.
Use tolerationSeconds for grace period.

Tolerations in a Pod Spec

Adding a taint to a Node

```
kubectl taint nodes node1 gpu=true:NoSchedule
kubectl taint nodes node1 gpu=true:NoSchedule- # remove
```

Toleration in the Pod spec

```
spec:
  tolerations:
    - key: "gpu"
      operator: "Equal"
      value: "true"
      effect: "NoSchedule"
  containers:
    - name: gpu-job
      image: cuda:11.0
```

Node Affinity

Node Affinity constrains which Nodes a Pod can be scheduled on, based on Node labels.

More expressive than nodeSelector — supports operators: In, NotIn, Exists, DoesNotExist, Gt, Lt

nodeSelector (simple)

```
spec:
  nodeSelector:
    disktype: ssd
```

- Key=value match only
- No operator support
- Cannot express 'not in' or 'exists'

Node Affinity (expressive)

```
spec:
  affinity:
    nodeAffinity:
      required...:
        nodeSelectorTerms:
          - matchExpressions:
            - key: disktype
              operator: In
              values: [ssd, nvme]
```

Node Affinity Types

- `requiredDuringSchedulingIgnoredDuringExecution`
 - Hard constraint — Pod WILL NOT schedule if no matching node
 - Use when workload requires specific hardware (GPU, SSD)
- `preferredDuringSchedulingIgnoredDuringExecution`
 - Soft constraint — scheduler PREFERS matching nodes but places elsewhere if needed
 - Use when you want to spread workloads across zones if possible
- `IgnoredDuringExecution`
 - Both types ignore affinity rules for already-running Pods
- Combine multiple `matchExpressions` for fine-grained control
 - Expressions within a `nodeSelectorTerm` are ANDed; multiple terms are ORed

What is a DaemonSet?

- Resource that ensures exactly one (or more) copy of a Pod runs on every (or every matching) Node in the cluster
- Contrast with a Deployment, which runs N replicas spread across nodes as the scheduler sees fit

Yaml Manifest

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: fluentd
  namespace: logging
spec:
  selector:
    matchLabels:
      app: fluentd
  template:
    metadata:
      labels:
        app: fluentd
    spec:
      containers:
        - name: fluentd
          image: fluent/fluentd:latest
          volumeMounts:
            - name: varlog
              mountPath: /var/log
      volumes:
        - name: varlog
          hostPath:
```

selector: must match the Pod template's labels

template: same as for Deployments—a Pod spec.

hostPath (common): mount host filesystems into each

Deploy & Verify

- `kubectl apply -f daemonset.yaml`
- `kubectl get daemonset -n logging`
- `kubectl get pods -n logging -o wide`
- `kubectl rollout status daemonset/fluentd -n logging`

If you add a new node, the DaemonSet controller automatically launches a new Pod on it; when you remove a node, its Pod is cleaned up.

What is a StatefulSet?

- Set of pods with stable unique identities and optional stable storage
- Pods are created in ordinal order, from 0 up to N-1
- Controller waits for Pod 0 to be running and ready before creating pod 1 etc.
- Like Deployments, each Pod is scheduled by the kube-scheduler based on capacity, taints, selectors, affinity, etc.

Yaml Manifest

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: web
  namespace: default
spec:
  serviceName: web
  replicas: 3
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
        - name: nginx
          image: nginx:1.25.5
          ports:
            - containerPort: 80
          volumeMounts:
            - name: www-storage
              mountPath: /usr/share/nginx/html
  volumeClaimTemplates:
    - metadata:
        name: www-storage
      spec:
        accessModes: ["ReadWriteOnce"]
        resources:
```

must match headless service

template: same as for
Deployments—a Pod spec.

will yield PVCs:
www-storage-web-0
www-storage-web-1

Deploy & Verify

- `kubectl apply -f statefulset.yaml`
- `kubectl get statefulset`

StatefulSet creates the amount of replica pods you specified, and **names them in order**.



What happens if we want to stop a pod from getting deployed on certain nodes?



What happens if we want a pod to only go to a specific node?

WRAP-UP

Day 2 done — you can deploy, scale and upgrade applications.

Tomorrow: Observability and Configuration & Security — probes, logging, ConfigMaps, Secrets, RBAC.

COURSE SLIDES

Certified Kubernetes Application Developer (CKAD)

WSQ Course Code: TGS-2025053212

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

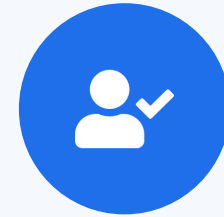
DAY 3

Observability + Config & Security

Probes · Logging · Metrics · Debug · ConfigMaps · Secrets · SecurityContext · RBAC (CKAD Domains 3 & 4 — 40%)

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your phone camera and submit your attendance.
- A minimum of 75% attendance is required for assessment and funding.



**Minimum 75%
attendance required**

Recap & Today

- Yesterday (Domain 2): Deployments, rollouts, release strategies, Helm, Kustomize, Daemon/StatefulSet.
- Morning — Domain 3 Observability (15%): probes, logging, metrics, debugging, API deprecations.
- Afternoon — Domain 4 Configuration & Security (25%): ConfigMaps, Secrets, securityContext, RBAC, quotas.
- Together these two domains are 40% of the CKAD exam — the largest single day.



DOMAIN 3

Observability & Maintenance

Probes · logging · metrics · debugging · API versions (15%)



TOPIC 1

Health Probes

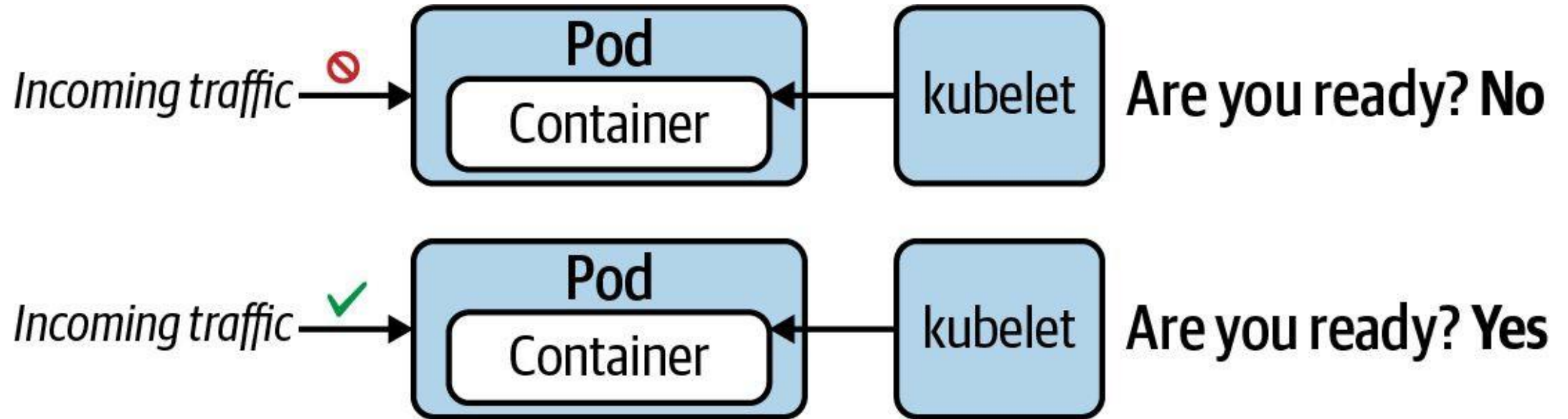
liveness · readiness · startup

Three Probes, Three Jobs

- livenessProbe — restarts the container when it has hung or deadlocked.
- readinessProbe — removes the Pod from Service endpoints until it can serve traffic (no restart).
- startupProbe — gives slow apps time to boot; it blocks the other two until it succeeds.
- Each probe uses one handler: httpGet, tcpSocket or exec — pick by what the container exposes.

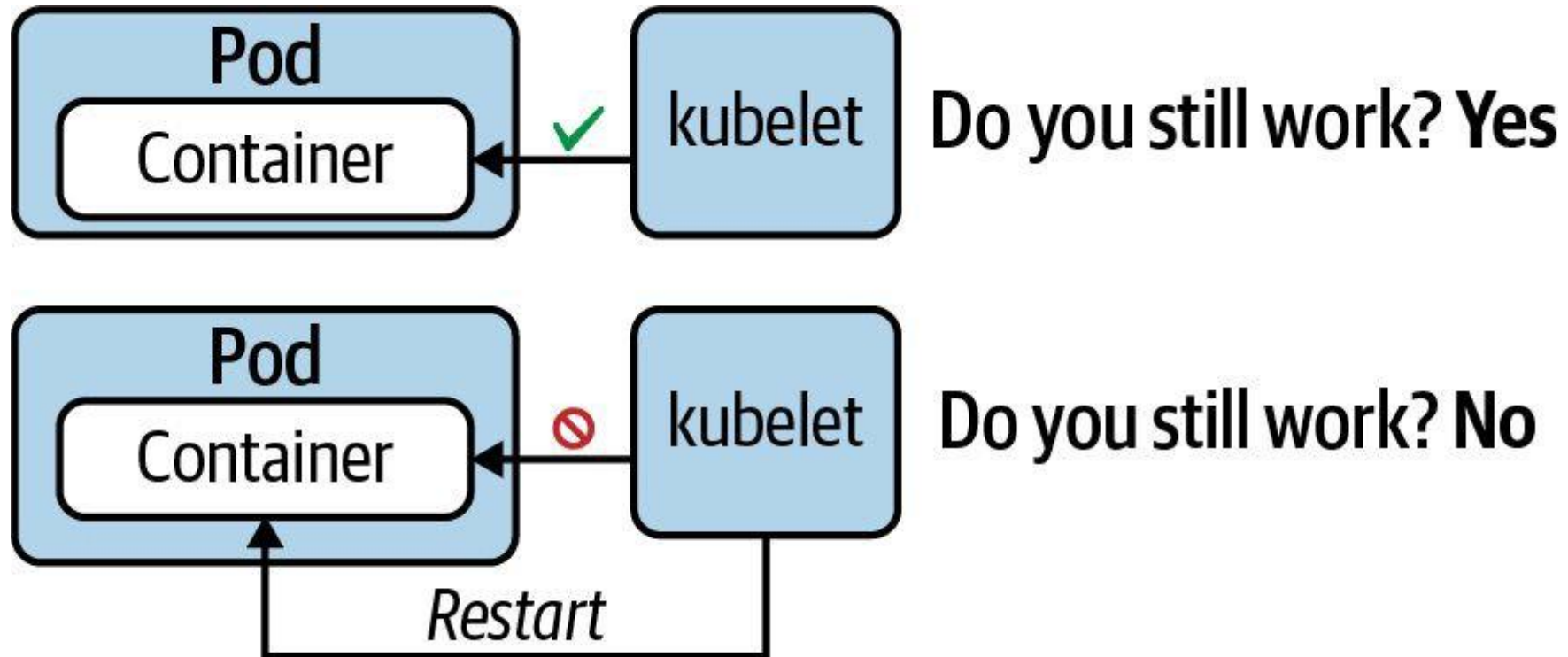
Readiness Probe

"Is application ready to handle requests?"



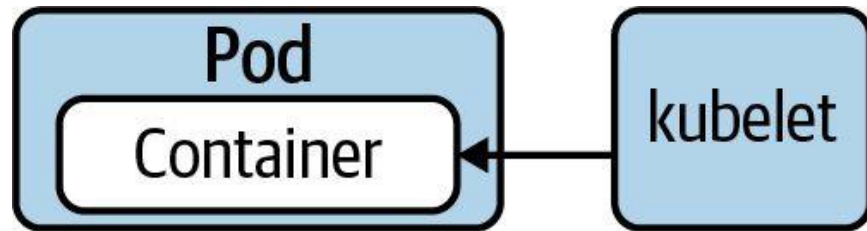
Liveness Probe

"Does the application still function without errors?"

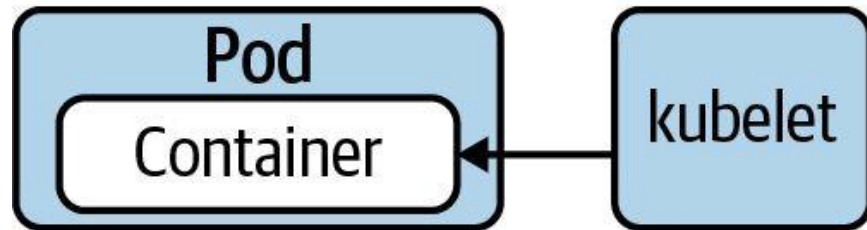


Startup Probe

"Legacy application may need longer to start. Hold off on liveness probing."



Are you started? **No**  *Start liveness probe*



Are you started? **Yes**  *Start liveness probe*

Liveness, Readiness & Startup Probes

LAB 16

Probes

Kubernetes uses three probes to manage health: livenessProbe restarts a failed container, readinessProbe removes it from Service endpoints, startupProbe gives slow apps time to boot. CKAD tests all three handlers (httpGet, tcpSocket, exec) and timing.

You'll build

Add HTTP readiness+liveness probes, trigger a liveness restart, add a TCP probe, then a startupProbe with exec.

Key commands: readinessProbe: · k exec · startupProbe: · » Three

Liveness, Readiness & Startup Probes

Step 1 — HTTP liveness + readiness probe

```
readinessProbe:
  httpGet: {path: /, port: 80}
  initialDelaySeconds: 2
  periodSeconds: 5
livenessProbe:
  httpGet: {path: /, port: 80}
  initialDelaySeconds: 10
  periodSeconds: 10
  failureThreshold: 3
```

Step 2 — Trigger a liveness failure (watch RESTARTS climb)

```
k exec web-probes -- rm /usr/share/nginx/html/index.html
sleep 35
k get pod web-probes
```

Liveness, Readiness & Startup Probes (cont.)

Step 3 — TCP socket probe

```
readinessProbe:
  tcpSocket: {port: 6379}
  periodSeconds: 5
```

Step 4 — startupProbe for slow-starting apps

```
startupProbe:
  exec: {command: ["cat", "/tmp/ready"]}
  failureThreshold: 30
  periodSeconds: 5
```

Note Three handlers — httpGet, tcpSocket, exec. readinessProbe failure removes the Pod from Service endpoints without restarting it; livenessProbe failure restarts the container; startupProbe blocks the other two until the app is initialised.

Liveness, Readiness & Startup Probes

✓ Test it

Deleting the index page makes the liveness probe fail three times and the container restarts (RESTARTS increments), while the TCP and startup Pods reach READY 1/1.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Health Verification Methods

Any verification method can be used by any of the probe types

Method	Option	Description
Custom Command	<code>exec.command</code>	Executes a command inside of the container e.g. a cat command and checks its exit code. Kubernetes considers an zero exit code to be successful. A non-zero exit code indicates an error.
HTTP GET Request	<code>httpGet</code>	Sends an HTTP GET request to an endpoint exposed by the application. A HTTP response code in the range of 200 and 399 indicates success. Any other response code is regarded as an error.
TCP socket connection	<code>tcpSocket</code>	Tries to open a TCP socket connection to a port. If the connection could be established, the probing attempt was successful. The inability to connect is accounted for as an error.

Health Check Attributes

Any verification method can be used by any of the probe types

Attribute	Default Value	Description
<code>initialDelaySeconds</code>	0	Delay in seconds until first check is executed.
<code>periodSeconds</code>	10	Interval for executing a check (e.g., every 20 seconds).
<code>timeoutSeconds</code>	1	Maximum number of seconds until the check operation times out.
<code>successThreshold</code>	1	Number of successful check attempts until probe is considered successful after a failure.
<code>failureThreshold</code>	3	Number of failures for check attempts before probe is marked failed and takes action.
<code>terminationGracePeriodSeconds</code>	30	Grace period before forcing a container to stop upon failure.

Using a Named Port

Referencing a port by assigned name in multiple probes

```
ports:
- name: http-port
  containerPort: 8080

readinessProbe:
  httpGet:
    path: /healthz
    port: http-port

livenessProbe:
  httpGet:
    path: /healthz
    port: http-port
```

A diagram consisting of a vertical line on the right side of the code block. Three horizontal arrows point from this line to the 'http-port' values in the 'ports', 'readinessProbe', and 'livenessProbe' sections, illustrating that the same named port is referenced by multiple different Kubernetes probes.



TOPIC 2

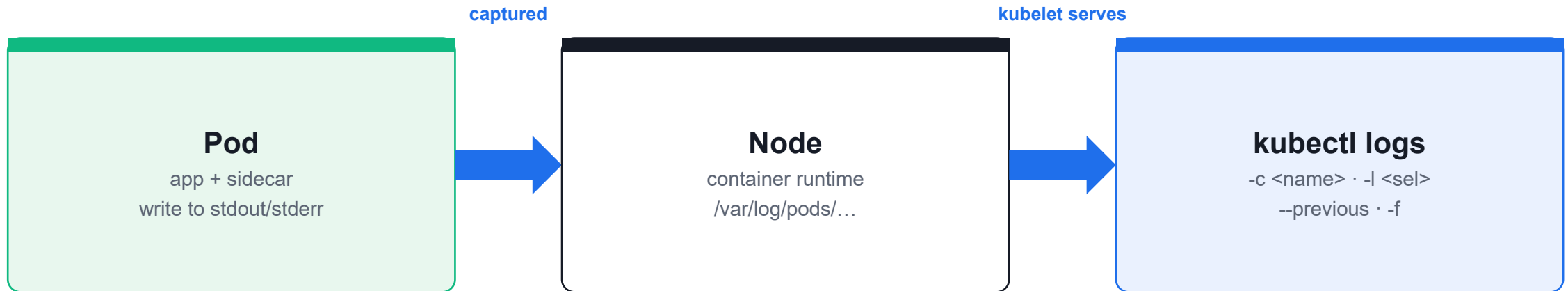
Logging

kubectl logs — your first debugging tool

Reading Container Logs

- Logs are whatever the container writes to stdout/stderr; the kubelet captures and serves them.
- `--tail=N` for recent lines; `-f` to stream live (like `tail -f`).
- `--previous` reads the prior terminated container — the key to diagnosing `CrashLoopBackOff`.
- `-c <container>` selects one container; `-l <selector>` aggregates logs across matching Pods.

Where Logs Come From



Logs are whatever the container prints to **stdout/stderr**. Use **--previous** for a crashed container.

Container Logging

LAB 17

Logging

kubectl logs is the first debugging tool on the exam. Read logs from single and multi-container Pods, follow a live stream, retrieve logs from a crashed container, and aggregate logs across a Deployment.

You'll build

Read and tail logs, stream with -f, recover crash logs with --previous, then aggregate by container and label.

Key commands: `k logs · » `--tail=N``

Container Logging

Step 1 — Read recent lines and follow a live stream

```
k logs noisy --tail=10
k logs -f noisy &          # -f streams like tail -f
kill %1
```

Step 2 — Retrieve logs from a crashed container

```
k logs crasher --previous | head -5
```

Step 3 — Multi-container Pod logs

```
k logs multi -c app | head -3
k logs multi -c sidecar | head -3
k logs multi --all-containers=true --prefix=true | head -8
```

Container Logging (cont.)

Step 4 — Aggregate logs across a Deployment

```
k logs deployment/fleet --tail=3  
k logs -l app=fleet --prefix=true --tail=2
```

Note --tail=N for recent lines, -f to stream, --previous for crash loops, -c <name> for a specific container, -l <selector> to aggregate matching Pods. --prefix=true labels each line with [pod/container].

Container Logging

✓ Test it

`k logs crasher --previous` shows the message printed before the crash, and `k logs -l app=fleet --prefix=true` interleaves lines from all three Pods.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Tea Break

15 minutes



TOPIC 3

Metrics

kubectl top & the metrics-server add-on

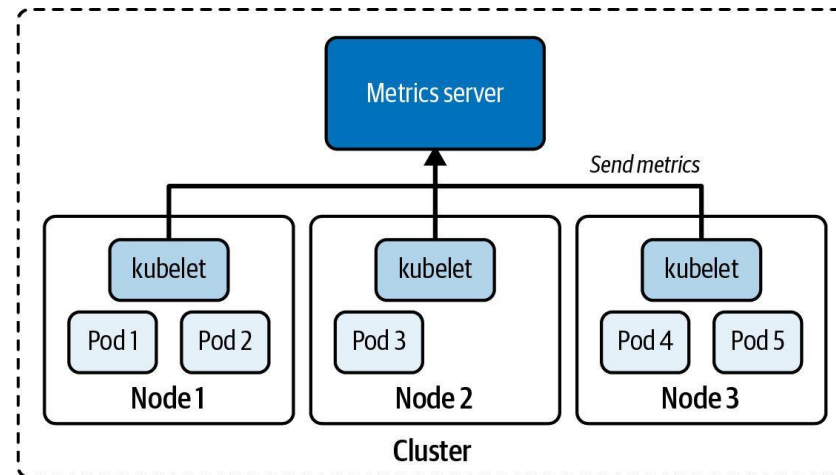
Resource Usage at a Glance

- `kubectl top` shows live CPU and memory for nodes and Pods — it requires metrics-server.
- metrics-server scrapes each kubelet's cAdvisor and serves the aggregated Metrics API.
- Sort with `--sort-by=cpu` / `--sort-by=memory`; span all namespaces with `-A`.
- On Killercode's self-signed certs, patch the Deployment with `--kubelet-insecure-tls`.

What is the Metrics Server?

Cluster-wide resource metrics aggregator

- Gathers resource consumption metrics on nodes, e.g. CPU and memory. The kubelet sends those metrics to the Kubernetes API server and exposes them through the [Metrics API](#).
- The metrics server stores data in memory and does not persist data over time.



kubectl top and Metrics Server

LAB 18

Metrics

kubectl top shows real-time CPU and memory for nodes and Pods. It requires the metrics-server add-on. CKAD tests installing metrics-server, reading node/Pod metrics, and sorting by resource usage.

You'll build

Install metrics-server, read node metrics, generate CPU load, then sort Pod metrics by CPU and memory.

Key commands: `kubectl apply · k top · k run · » `kubectl`

kubectl top and Metrics Server

Step 1 — Install metrics-server (and trust kubelet certs on Killercode)

```
kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/releases/latest/download/components.yaml
kubectl patch -n kube-system deployment metrics-server --type=json -p='
[{"op": "add", "path": "/spec/template/spec/containers/0/args/-", "value": "--kubelet-insecure-tls"}]'
until kubectl top node 2>/dev/null; do sleep 5; done
```

Step 2 — Top nodes

```
k top node          # CPU cores/% and memory bytes/% per node
```

Step 3 — Generate load, then top Pods sorted

```
k run cpu-burner --image=busybox -- sh -c 'while true; do :; done'
sleep 30
k top pod --sort-by=cpu
k top pod --sort-by=memory
k top pod -A --sort-by=cpu | head -10
```

kubectl top and Metrics Server (cont.)

Note kubectl top needs metrics-server running. --kubelet-insecure-tls is required on Killercode's self-signed certs. cpu-burner should top the CPU sort; -A spans all namespaces.

kubectl top and Metrics Server

✓ Test it

After ~30s, k top node reports CPU/memory and k top pod --sort-by=cpu lists cpu-burner at the top of the usage ranking.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Querying for Metrics

Metrics for nodes and Pods can be rendered with the `top` command

```
$ kubectl top nodes
```

NAME	CPU (cores)	CPU%	MEMORY (bytes)	MEMORY%
controlplane	193m	9%	1239Mi	32%
node01	119m	5%	889Mi	23%

```
$ kubectl top pods
```

NAME	CPU (cores)	MEMORY (bytes)
frontend	38m	85Mi
backend	29m	59Mi

Kubernetes Features that Tap into Metrics

Where else does the metrics server come into play? ([docs](#))

- A Horizontal Pod Autoscaler (HPA) can only determine if Deployment replicas need to be scaled up or down based on available resource consumption determined by metrics server. If the current resource consumption cannot be determined, it is indicated by `<unknown>` in the `TARGETS` column of an HPA.
- Pod scheduling on cluster nodes is based on the definition of resource requests. The scheduler looks at the resources capacity available on a cluster node and decides if the workload can be scheduled for the Pod's resource request.



TOPIC 4

Debugging

Diagnose and fix broken workloads under time pressure

The Three Failures You Will See

- ImagePullBackOff — wrong image name/tag; confirm in describe → Events, then fix and recreate.
- CrashLoopBackOff — container exits non-zero; read `kubectl logs --previous`.
- OOMKilled — exceeded the memory limit; raise `resources.limits.memory`.
- `kubectl get events --sort-by=.lastTimestamp` and `kubectl debug` (ephemeral container) are your fast tools.

Common Pod Error Statuses

Poorly-documented in Kubernetes docs, encountered often in practice

Status	Root Cause	Potential Fix
ImagePullBackOff or ErrImagePull	Image could not be pulled from registry.	Check correct image name, check that image name exists in registry, verify network access from node to registry, ensure proper authentication.
CrashLoopBackOff	Application or command run in container crashes.	Check command executed in container, ensure that image can properly execute (e.g., by creating a container with Docker Engine).
CreateContainerConfigError	ConfigMap or Secret referenced by container cannot be found.	Check correct name of the configuration object, verify the existence of the configuration object in the namespace.

Debugging Pods and Events

LAB 19

Debugging

CKAD regularly includes broken workloads you must diagnose and fix under time pressure. Identify and resolve ImagePullBackOff, CrashLoopBackOff and OOMKilled, and use `kubectl debug` for live triage.

You'll build

Diagnose three classic failures with `describe/logs/events`, then inject an ephemeral debug container.

Key commands: `k describe · k get · » ImagePullBackOff`

Debugging Pods and Events

Step 1 — ImagePullBackOff (wrong image name)

```
k describe pod typo | tail -15      # 'Failed to pull image'  
k delete pod typo --force --grace-period=0  
k run typo --image=nginx:1.25      # corrected
```

Step 2 — CrashLoopBackOff (container exits non-zero)

```
k describe pod crash | grep -A3 "Last State"  
k logs crash --previous
```

Step 3 — OOMKilled (exceeds memory limit)

```
k describe pod hungry | grep -A4 "Last State"  # Reason: OOMKilled  
# fix: raise resources.limits.memory
```

Debugging Pods and Events (cont.)

Step 4 — Cluster-wide events + ephemeral debug container

```
k get events -A --sort-by=.lastTimestamp | tail -20  
k get events --field-selector type=Warning  
k debug crash -it --image=busybox --target=crash -- sh
```

Note ImagePullBackOff → wrong image/tag. CrashLoopBackOff → use logs --previous. OOMKilled → raise the memory limit. kubectl debug injects an ephemeral container sharing the target's namespaces — vital for distroless images with no shell.

Debugging Pods and Events

✓ Test it

k describe reveals the correct failure reason for each Pod (ImagePullBackOff, CrashLoopBackOff, OOMKilled), and k debug opens a shell inside the running Pod.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>



TOPIC 5

API Deprecations

Find the right apiVersion with kubectl

APIs Change — Know How to Look Them Up

- Old API versions are deprecated and eventually removed (e.g. extensions/v1beta1 Ingress, gone in 1.22).
- `kubectl api-resources` shows the correct `apiVersion` for every resource in its `APIVERSION` column.
- `kubectl api-versions` lists every version the cluster serves.
- `kubectl explain <resource>.<field> --recursive` documents fields without leaving the terminal.

Listing Available API version

kubectl can discover API version useable in manifests

```
$ kubectl api-versions
```

```
admissionregistration.k8s.io/v1
```

```
apiextensions.k8s.io/v1
```

```
apiregistration.k8s.io/v1 apps/v1
```

```
authentication.k8s.io/v1
```

```
authorization.k8s.io/v1 autoscaling/v1
```

```
autoscaling/v2
```

```
...
```

API Deprecations & kubectl explain

LAB 20

API Deprecations

The Kubernetes API evolves — old versions are deprecated and removed. CKAD tests kubectl explain, api-resources and api-versions. You must find the correct apiVersion for any resource on exam day using only kubectl.

You'll build

List resources and versions, explore schemas with explain, detect a removed API, and look up the current version.

Key commands: `k api-resources · k explain · Pod/Service/ConfigMap/Secret/SA/Quota/LimitRange -> · » `kubectl`

API Deprecations & kubectl explain

Step 1 — List resources, groups and served versions

```
k api-resources --api-group=apps
k api-resources --api-group=batch
k api-versions | sort
```

Step 2 — Explore a schema with explain

```
k explain deployment.spec.strategy.rollingUpdate
k explain pod.spec.containers.resources --recursive | head -30
```

Step 3 — Detect a removed API and find the replacement

```
# extensions/v1beta1 Ingress was removed in 1.22
k explain ingress --api-version=networking.k8s.io/v1 | head -10
k api-resources | grep -i ingress
```

API Deprecations & kubectl explain (cont.)

Step 4 — Stable apiVersions to memorise (v1.35)

```
Pod/Service/ConfigMap/Secret/SA/Quota/LimitRange -> v1
Deployment/StatefulSet/DaemonSet/ReplicaSet      -> apps/v1
Job/CronJob                                       -> batch/v1
Ingress/NetworkPolicy                           -> networking.k8s.io/v1
Role/RoleBinding/ClusterRole(+Binding)          -> rbac.authorization.k8s.io/v1
HorizontalPodAutoscaler                         -> autoscaling/v2
```

Note kubectl api-resources shows the correct apiVersion in its APIVERSION column — the exam-legal way to look it up. kubectl explain <res>.<field> documents fields without leaving the terminal.

API Deprecations & kubectl explain

✓ Test it

`k api-resources | grep -i ingress` shows `networking.k8s.io/v1`, and `k explain deployment.spec.strategy.rollingUpdate` prints the `maxSurge/maxUnavailable` fields.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>



Custom Resource Definitions (CRDs)

Extending the Kubernetes API by custom primitives

Deprecation Policy

Some APIs will be deprecated and eventually removed

- The Kubernetes project releases 3 versions per year (see [release cadence](#)).
- With each release, an API can become deprecated which means that is scheduled for removal or replacement. You can find more information about the details in the [Kubernetes Deprecation Policy](#).
- The use of a deprecated API renders a warning message when creating the object that uses it.
- The [Deprecated API Migration Guide](#) shows a list of deprecated APIs and the scheduled releases that will remove support for the API.

Deprecated HPA API Warning Message

kubectl renders a warning message, but creates object

```
$ kubectl create -f hpa.yaml
```

```
Warning: autoscaling/v2beta2 HorizontalPodAutoscaler is deprecated
```

```
↵in v1.23+, unavailable in v1.26+; use autoscaling/v2 ↵
```

```
HorizontalPodAutoscaler
```


Removed ClusterRole API

Uses [removed API](#) with Kubernetes version 1.22

```
apiVersion:  
kind: ClusterRole  
metadata:  
  name: pod-reader  
rules:  
- apiGroups: [""]  
  resources: ["pods"]  
  verbs: ["get", "watch", "list"]
```

The **rbac.authorization.k8s.io/v1beta1** API version of ClusterRole, ClusterRoleBinding, Role, and RoleBinding is no longer served as of v1.22.

Removed ClusterRole API Error Message

kubectl renders an error message, as API does not exist anymore

```
$ kubectl create -f clusterrole.yaml
error: resource mapping not found for name: "pod-reader" namespace:
← "" from "clusterrole.yaml": no matches for kind "ClusterRole" in
  ← version "rbac.authorization.k8s.io/v1beta1"
```

Using the ClusterRole API Replacement

Usage of replacement API

```
apiVersion:  
kind: ClusterRole  
metadata:  
  name: pod-reader  
rules:  
- apiGroups: [""]  
  resources: ["pods"]  
  verbs: ["get", "watch", "list"]
```

- Migrate manifests and API clients to use the **rbac.authorization.k8s.io/v1** API version, available since v1.8.
- All existing persisted objects are accessible via the new APIs
- No notable changes

Lunch Break

1 hour



DOMAIN 4

Configuration & Security

ConfigMaps · Secrets · securityContext · ServiceAccounts · RBAC · quotas (25%)



TOPIC 6

ConfigMaps

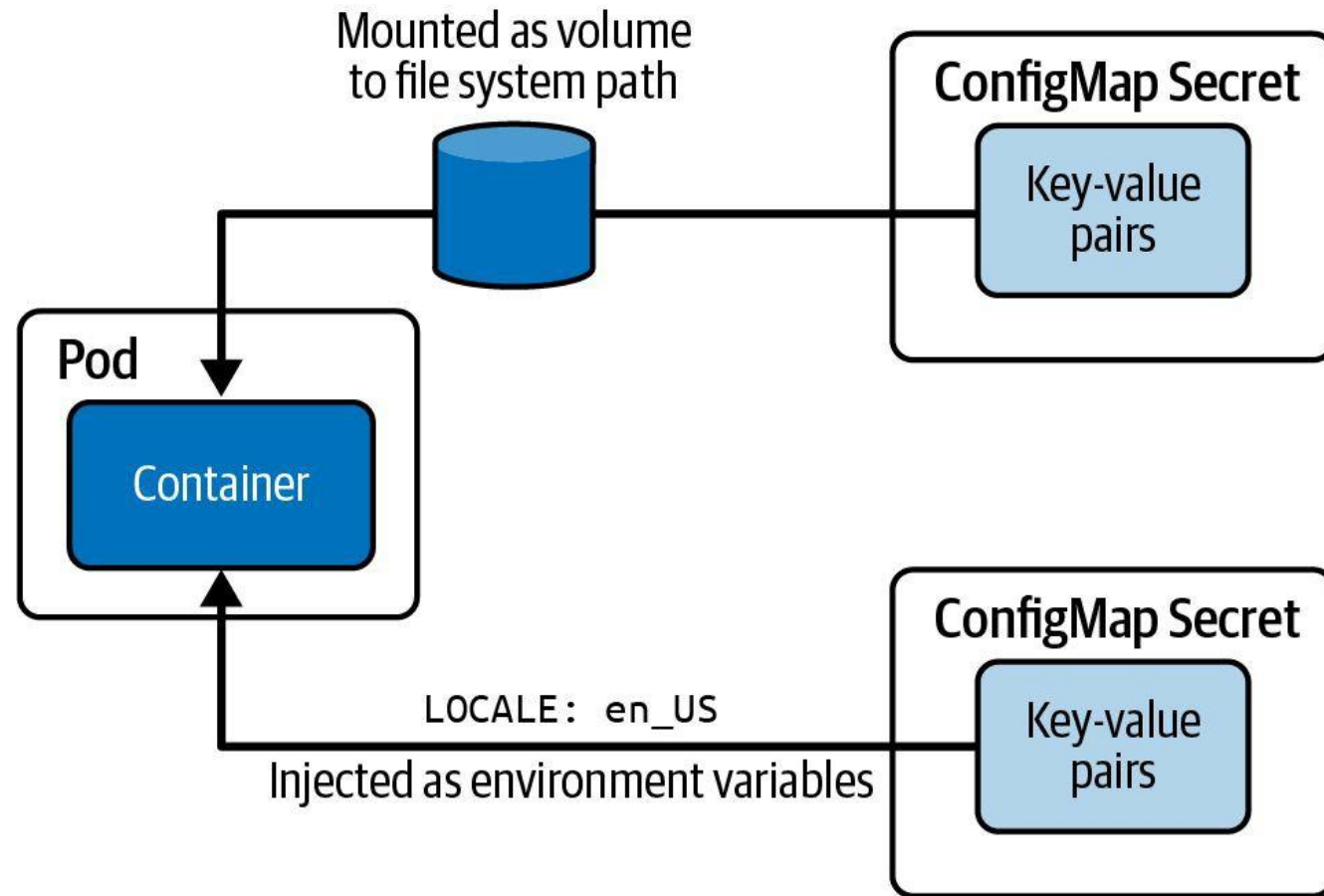
Inject non-secret configuration into Pods

Decouple Config from Image

- A ConfigMap holds non-secret key/value data; create it with `--from-literal`, `--from-file` or `--from-env-file`.
- Inject one key with `configMapKeyRef`, all keys with `envFrom`, or mount it as a file volume.
- Volume-mounted ConfigMaps update live (~60s); env-var injections are fixed at Pod startup.
- Keep config out of the image so the same image runs in every environment.

Consuming Configuration Data

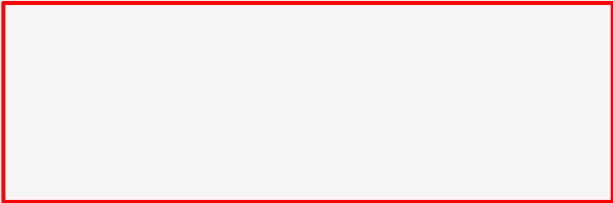
Applicable to ConfigMap and Secret



Consuming a ConfigMap as Environment Variables

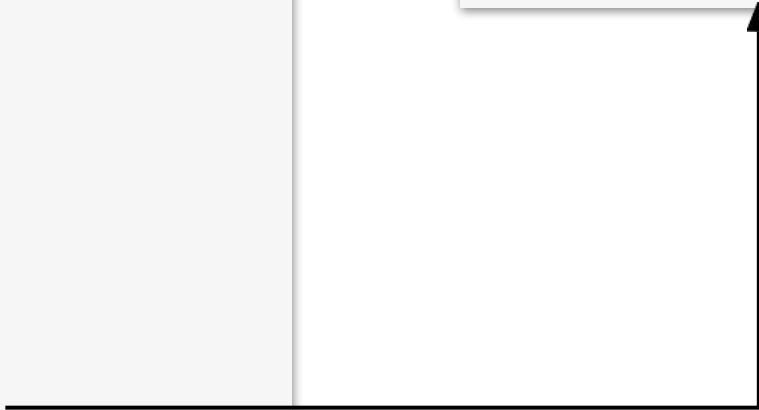
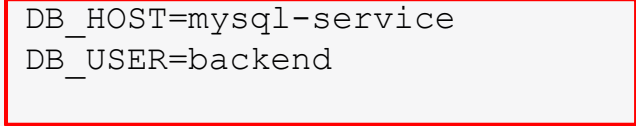
Controlling runtime behavior with the help of simple values

```
apiVersion: v1 kind: Pod
metadata:
  name: backend
spec:
  containers:
  - image: bmuschko/web-app:1.0.1
    name: backend
    envFrom:
    - configMapRef: name: db-
      config
```



```
$ kubectl exec -it backend -- env
```

```
DB_HOST=mysql-service
DB_USER=backend
```



ConfigMaps — Env & Volume Injection

LAB 21

ConfigMaps

ConfigMaps inject non-secret configuration into Pods. CKAD tests all three injection styles: a single env var (configMapKeyRef), bulk env vars (envFrom), and file-based volume mounts — and that only file mounts update live.

You'll build

Create ConfigMaps three ways, inject one key, inject all keys with envFrom, mount as files, then update live.

Key commands: `k create · env: · envFrom: · volumeMounts: · » Creation:`

ConfigMaps — Env & Volume Injection

Step 1 — Create ConfigMaps three ways

```
k create configmap app-cfg --from-literal=COLOR=blue --from-literal=GREETING=hello
k create configmap app-conf --from-file=app.conf
k create configmap app-env --from-env-file=env.list
```

Step 2 — Inject a single key as an env var

```
env:
- name: COLOR
  valueFrom:
    configMapKeyRef: {name: app-cfg, key: COLOR}
```

Step 3 — Inject all keys with envFrom

```
envFrom:
- configMapRef: {name: app-cfg}
- configMapRef: {name: app-env}
```

ConfigMaps — Env & Volume Injection (cont.)

Step 4 — Mount a ConfigMap as a file volume

```
volumeMounts:  
- {name: cfg, mountPath: /etc/app}  
volumes:  
- name: cfg  
  configMap: {name: app-conf}
```

Note Creation: --from-literal, --from-file, --from-env-file. Consumption: configMapKeyRef (one key), envFrom (all keys), volume mount (file per key). Volume-mounted ConfigMaps update live (~60s); env-var injections are fixed at Pod startup and need a restart.

ConfigMaps — Env & Volume Injection

✓ Test it

The single-key Pod logs `COLOR=blue`, the envFrom Pod shows all four variables, and the mounted ConfigMap appears as files under `/etc/app`.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Creating a ConfigMap with Imperative Approach

Key-value pairs can be parsed from different sources

```
$ kubectl create configmap db-config --from-literal=db=staging  
configmap/db-config created
```

```
$ kubectl create configmap db-config --from-env-file=config.env  
configmap/db-config created
```

```
$ kubectl create configmap db-config --from-file=config.txt  
configmap/db-config created
```

```
$ kubectl create configmap db-config --from-file=app-config  
configmap/db-config created
```

Source Options for Data Parsed by a ConfigMap

Configuration options and example values

Option	Description	Example Value
<code>--from-literal</code>	Literal values, which are key-value pairs as plain text.	<code>--from-literal=locale=en_US</code>
<code>--from-env-file</code>	A file that contains key-value pairs and expects them to be environment variables.	<code>--from-env-file=config.env</code>
<code>--from-file</code>	Huge page resource type.	<code>--from-file=app-config.json</code>
<code>--from-file</code>	A directory with one or many files.	<code>--from-file=config-dir</code>

ConfigMap - Imperative Approach

Configuration options and example values

```
kubectl create configmap color-config-map --from-literal=APP_COLOR="black"
```

Create configmap command

Name of configmap

Key-value pair of configmap

ConfigMap YAML Manifest

Key-value pair definition under `data` attribute

```
1  apiVersion: v1
2  kind: ConfigMap
3  metadata:
4    name: hello-configmap
5  data:
6    APP_COLOR: black
7    demo_greeting: Welcome to Kubernetes 101
```

Data of configmap

ConfigMap YAML Manifest

Key-value pair definition under `data` attribute

```
1  apiVersion: v1
2  kind: ConfigMap
3  metadata:
4    name: hello-configmap
5  data:
6    APP_COLOR: black
7    demo_greeting: Welcome to Kubernetes 101
```

```
kubectl create -f configmap.yaml
```

ENV Variables - Define all key-value pairs in configmap

Key-value pair definition under `data` attribute

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: sample-pod-1
5  spec:
6    containers:
7      - name: sample-pod-1
8        image: python:3.6-alpine
9        envFrom:
10          - configMapRef:
11              name: color-configmap
```

```
envFrom:
- configMapRef:
    name: color-configmap
```

Environment Variables from:
color-configmap ConfigMap

ENV Variables - Define via Multiple ConfigMaps

Key-value pair definition under `data` attribute

```
env:  
  - name: color  
    valueFrom:  
      configMapKeyRef:  
        name: color-configmap  
        key: APP_COLOR  
  - name: demo_greeting
```

```
    valueFrom:  
      configMapKeyRef:  
        name: hello-configmap-file  
        key: demo_greeting
```

From color-configmap

From hello-configmap-file

Environment:

```
color:      <set to the key 'APP_COLOR' of config map 'color-configmap'>  
demo_greeting: <set to the key 'demo_greeting' of config map 'hello-configmap-file'>
```



Consuming Changed ConfigMap Data

Containers will not refresh consumed data upon a change

- The key-value pairs of a ConfigMap can be changed for a live object.
- Changed ConfigMap key-value pairs consumed by containers will not be reloaded automatically for an already-running Pod.
- The application code needs to implement logic to either reload the configuration data periodically or on-demand.



TOPIC 7

Secrets

Sensitive configuration, guarded by RBAC

Secrets vs ConfigMaps

- Secrets work like ConfigMaps but values are base64-encoded and access is more tightly controlled.
- Types: generic, tls (kubernetes.io/tls) and docker-registry (for imagePullSecrets).
- Inject with envFrom: secretRef (all keys) or secretKeyRef (one key); mount with defaultMode: 0400.
- Base64 is not encryption — restrict access with RBAC and enable encryption at rest.

Consuming a Secret as Environment Variables

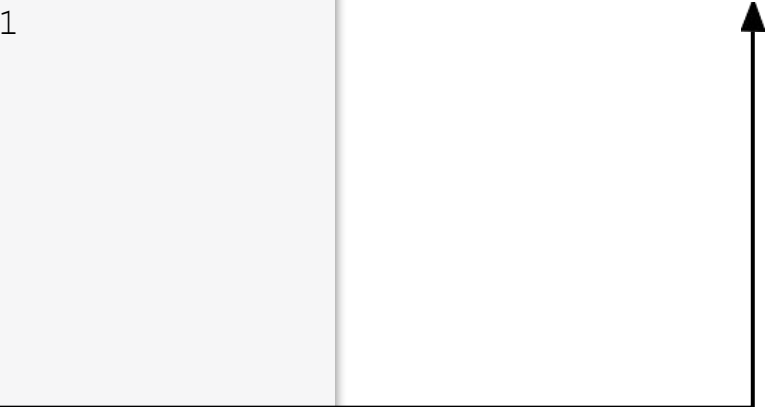
Accessed values are base64-decoded

```
apiVersion: v1 kind: Pod
metadata:
  name: backend
spec:
  containers:
  - image: bmuschko/web-app:1.0.1
    name: backend
```

```
envFrom:
- secretRef:
  name: mysecret
```

```
$ kubectl exec -it backend -- env
```

```
pwd=s3cre!
```



Consuming a Secret as a Volume

A good choice for processing a machine-readable configuration file

- `apiVersion: v1 kind: Pod metadata:`
- `name: backend`
- `spec:`
- `containers:`
 - `- name: backend`
- `image:`
- `bmuschko/web-app:1.0.1`
- `volumeMounts:`
 - `- name: ssh-volume mountPath:`
`/var/app readOnly: true`

```
$ kubectl exec -it backend -- /bin/sh
# ls -l /var/app ssh-privatekey
```

```
# cat /var/app/ssh-privatekey
-----BEGIN RSA PRIVATE KEY----- Proc-Type:
4, ENCRYPTED
DEK-Info:
AES-128-CBC,8734C9153079F2E8497C8075289E BBF1
...
-----END RSA PRIVATE KEY-----
```

Secrets

LAB 22

Secrets

Secrets are like ConfigMaps but base64-encoded with stricter access. CKAD tests generic, tls and docker-registry types, env-var injection, file mounts with defaultMode, and decoding values. Secrets are not encrypted — protect them with RBAC.

You'll build

Create a generic Secret, decode it, inject as env vars, mount with 0400 mode, then create tls and docker-registry Secrets.

Key commands: `k create · envFrom: · volumes: · »` Types:

Secrets

Step 1 — Create a generic Secret and decode a value

```
k create secret generic db-cred \
  --from-literal=DB_USER=admin --from-literal=DB_PASS=s3cr3t
k get secret db-cred -o jsonpath='{.data.DB_PASS}' | base64 -d; echo
```

Step 2 — Inject Secret keys as env vars

```
envFrom:
- secretRef: {name: db-cred}
```

Step 3 — Mount a Secret as files with restrictive mode

```
volumes:
- name: s
  secret:
    secretName: db-cred
    defaultMode: 0400
```

Secrets (cont.)

Step 4 — TLS and docker-registry Secrets

```
k create secret tls demo-tls --cert=tls.crt --key=tls.key
k create secret docker-registry myreg \
  --docker-server=registry.example.com --docker-username=u \
  --docker-password=p --docker-email=u@example.com
```

Note Types: generic, tls (kubernetes.io/tls), docker-registry. envFrom: secretRef injects all keys; secretKeyRef injects one. Reference a registry Secret via spec.imagePullSecrets. Base64 ≠ encryption — restrict with RBAC.

Secrets

✓ Test it

base64 -d on .data.DB_PASS returns s3cr3t, the mounted Secret files are mode 0400, and the tls Secret reports type kubernetes.io/tls.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Creating a Secret with Imperative Approach

Parsed values are base64-encoded automatically

```
kubectl create secret generic system-secret --from-literal=system_admin=p@ssword999
```

Create secret command

Name of secret

Value of secret

```
apiVersion: v1
data:
  system_admin: cEBzc3dvcmQ5OTk=
kind: Secret
metadata:
  creationTimestamp: "2020-08-21T05:50:27Z"
  name: system-secret
  namespace: default
  resourceVersion: "1042130"
  selfLink: /api/v1/namespaces/default/secrets/system-secret
  uid: 50f8462b-1125-4f5b-ba82-4eeabf11c274
```

Source Options for Data Parsed by a Generic Secret

Configuration options and example values

Option	Description	Example Value
<code>--from-literal</code>	Literal values, which are key-value pairs as plain text.	<code>--from-literal=password=secret</code>
<code>--from-env-file</code>	A file that contains key-value pairs and expects them to be environment variables.	<code>--from-env-file=config.env</code>
<code>--from-file</code>	Huge page resource type.	<code>--from-file=id_rsa=~/.ssh/id_rsa</code>
<code>--from-file</code>	A directory with one or many files.	<code>--from-file=config-dir</code>

Create Secret from File

Configuration options and example values

secrets.properties

```
1 DB_Admin: admin
2 DB_Password: p@ssword
```

```
kubectl create secret generic credentials-secret-file --from-file=credentials_secret.properties
```

Create secret command

Name of secret

Filepath of the secret

Declarative Approach

Configuration options and example values

- Declarative way: create yaml file

```
1  apiVersion: v1
2  kind: Secret
3  metadata:
4    name: credentials-secret
5  data:
6    DB_Admin: admin
7    DB_User: root
8    DB_Password: p@ssword
```

```
Error from server (BadRequest): error when creating "credentials-secret.yaml": Secret in version "v1" cannot be handled
as a Secret: v1.Secret.Data: decode base64: illegal base64 data at input byte 4, error found in #10 byte of ...|n":"ad
min","DB_Passw|..., bigger context ...|{"apiVersion":"v1","data":{"DB_Admin":"admin","DB_Password":"p@ssword","DB_User"
:"root"},"kind"|...
```

Declarative Approach

Encoding Secrets

```
DB_Admin: admin
DB_User: root
DB_Password: p@ssword
```



```
DB_Admin: YWRtaW4=
DB_User: cm9vdA==
DB_Password: cEBzc3dvcmQ=
```

```
echo -n 'admin' | base64
```



```
YWRtaW4=
```

```
echo -n 'root' | base64
```



```
cm9vdA==
```

```
echo -n 'p@ssword' | base64
```



```
cEBzc3dvcmQ=
```

Declarative Approach

Decoding Secrets

```
DB_Admin: YWRtaW4=  
DB_User: cm9vdA==  
DB_Password: cEBzc3dvcmQ=
```



```
DB_Admin: admin  
DB_User: root  
DB_Password: p@ssword
```

```
echo 'YMRtaW4==' | base64 --decode
```



admin

```
echo 'cm9vdA==' | base64 --decode
```



root

```
echo 'cEBzc3dvcmQ=' | base64 --decode
```



p@ssword

Describing Secrets

```
kubectl describe secret credentials-secret
```

```
Name:          credentials-secret
Namespace:     default
Labels:        <none>
Annotations:   <none>

Type:  Opaque

Data
====
DB_Admin:      5 bytes
DB_Password:   8 bytes
DB_User:       4 bytes
```

Viewing Secrets

```
kubectl get secret credentials-secret -o yaml
```

```
apiVersion: v1
data:
  DB_Admin: YWRtaW4=
  DB_Password: cEBzc3dvcmQ=
  DB_User: cm9vdA==
kind: Secret
metadata:
  creationTimestamp: "2020-06-19T06:31:51Z"
  name: credentials-secret
  namespace: default
  resourceVersion: "29616"
  selfLink: /api/v1/namespaces/default/secrets/credentials-secret
  uid: 276b2049-8588-418f-87a4-91a1851a9c88
type: Opaque
```


Secrets Are Not Really Secret

Techniques that help with making Secrets more secure

- Secret objects are stored in etcd in unencrypted form by default. Encryption of data in etcd is [configurable](#).
- Define RBAC permissions to only allow developers to create, view, and modify Secret objects in dedicated namespaces. An even stricter policy could only allow administrators to manage Secrets.
- Use Kubernetes-external, third-party Secrets services for managing sensitive data, e.g. AWS Secrets Manager or HashiCorp Vault. You can consume the data with the help of the [External Secrets Operator](#).

Source Options for Data Parsed by a Generic Secret

Configuration options and example values

Option	Description	Example Value
<code>--from-literal</code>	Literal values, which are key-value pairs as plain text.	<code>--from-literal=password=secret</code>
<code>--from-env-file</code>	A file that contains key-value pairs and expects them to be environment variables.	<code>--from-env-file=config.env</code>
<code>--from-file</code>	Huge page resource type.	<code>--from-file=id_rsa=~/.ssh/id_rsa</code>
<code>--from-file</code>	A directory with one or many files.	<code>--from-file=config-dir</code>

Tea Break

15 minutes



TOPIC 8

SecurityContext

Control container identity & privileges

Run Containers Safely

- securityContext sets the user, group and privileges a container runs with.
- Pod-level applies to all containers; container-level overrides one container.
- runAsNonRoot: true blocks any image whose default user is root.
- Hardened pattern: readOnlyRootFilesystem + capabilities.drop:[ALL] + allowPrivilegeEscalation:false.

Defining a Security Context

Pod- and/or container-level definition

```
apiVersion: v1 kind: Pod
metadata:
  name: secured-pod
spec:
  securityContext:
    fsGroup: 3000
  volumes:
    - name: data-vol
      emptyDir: {}
  containers:
    - image: nginx:1.18.0 name: secured-container
      volumeMounts:
        - name: data-vol
          mountPath: /data/app
```

Defined on the Pod-level (applies to all containers)

SecurityContext

LAB 23

SecurityContext

securityContext controls the identity and privileges of containers. CKAD asks you to enforce non-root execution, a read-only root filesystem and dropped Linux capabilities — at Pod level (all containers) or container level (one override).

You'll build

Run as a set user/group, enforce runAsNonRoot, lock the root filesystem, then drop all capabilities.

Key commands: securityContext: · » Pod-level

SecurityContext

Step 1 — Run as a specific user and group (Pod level)

```
securityContext:
  runAsUser: 1000
  runAsGroup: 3000
  fsGroup: 2000    # applies to volume mounts
```

Step 2 — Enforce runAsNonRoot (blocks root images)

```
securityContext:
  runAsNonRoot: true
# nginx runs as root -> Pod refuses to start
```

Step 3 — Read-only root filesystem (container level)

```
securityContext:
  readOnlyRootFilesystem: true
# mount an emptyDir for any writable path (e.g. /tmp)
```

SecurityContext (cont.)

Step 4 — Drop Linux capabilities (hardened container)

```
securityContext:
  capabilities:
    drop: ["ALL"]
    add: ["NET_BIND_SERVICE"]
  allowPrivilegeEscalation: false
```

Note Pod-level securityContext applies to all containers; container-level overrides one. The CKAD hardened pattern = runAsNonRoot: true + readOnlyRootFilesystem: true + capabilities.drop: [ALL] + allowPrivilegeEscalation: false.

SecurityContext

✓ Test it

The non-root Pod logs uid=1000 gid=3000, the runAsNonRoot nginx Pod refuses to start, and writes to the read-only root filesystem are rejected.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Overriding a Value on the Container-Level

Container-level definition wins

```
apiVersion: v1
kind: Pod
metadata:
  name: non-root-success
spec:
  securityContext:
    runAsNonRoot: false
  containers:
    - image: bitnami/nginx:1.23.1
      name: nginx
      securityContext:
        runAsNonRoot: true
```

← Takes precedence



TOPIC 9

ServiceAccounts

The identity a Pod presents to the API

Pod Identity

- Every Pod runs as a ServiceAccount — the default one has minimal permissions.
- `spec.serviceAccountName` attaches a dedicated ServiceAccount to a Pod.
- A token, CA cert and namespace are projected into `/var/run/secrets/kubernetes.io/serviceaccount/`.
- `automountServiceAccountToken: false` enforces least privilege; `kubectl create token` mints short-lived tokens.

Service Account as Subject

Pod assign the Service Account by name



ServiceAccounts

LAB 24

ServiceAccounts

Every Pod runs as a ServiceAccount; the default one has minimal RBAC. CKAD tests creating dedicated ServiceAccounts, attaching them to Pods, disabling the auto-mounted token, and requesting short-lived tokens with `kubectl create token`.

You'll build

Create a ServiceAccount, attach it to a Pod, inspect the projected token, disable auto-mount, then mint a token.

Key commands: `k create · k exec · k patch · TOKEN=$(k create · » `spec.serviceAccountName``

ServiceAccounts

Step 1 — Create a ServiceAccount and attach it to a Pod

```
k create serviceaccount app-sa
# Pod spec:
spec:
  serviceAccountName: app-sa
k get pod app -o jsonpath='{.spec.serviceAccountName}'; echo
```

Step 2 — Inspect the auto-mounted token inside the Pod

```
k exec app -- ls /var/run/secrets/kubernetes.io/serviceaccount/
```

Step 3 — Disable token auto-mount (least privilege)

```
k patch sa app-sa -p '{"automountServiceAccountToken": false}'
```

ServiceAccounts (cont.)

Step 4 — Request a short-lived ad-hoc token

```
TOKEN=$(k create token app-sa --duration=1h)
echo $TOKEN | cut -c1-60
```

Note spec.serviceAccountName attaches a custom SA. Token, CA and namespace are projected under /var/run/secrets/kubernetes.io/serviceaccount/. automountServiceAccountToken: false enforces least privilege; kubectl create token replaces the old Secret-backed tokens (removed in 1.24+).

ServiceAccounts

✓ Test it

`k get pod app -o jsonpath='{.spec.serviceAccountName}'` returns `app-sa`, and `kubectl create token app-sa` prints a short-lived JWT.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>



TOPIC 10

RBAC

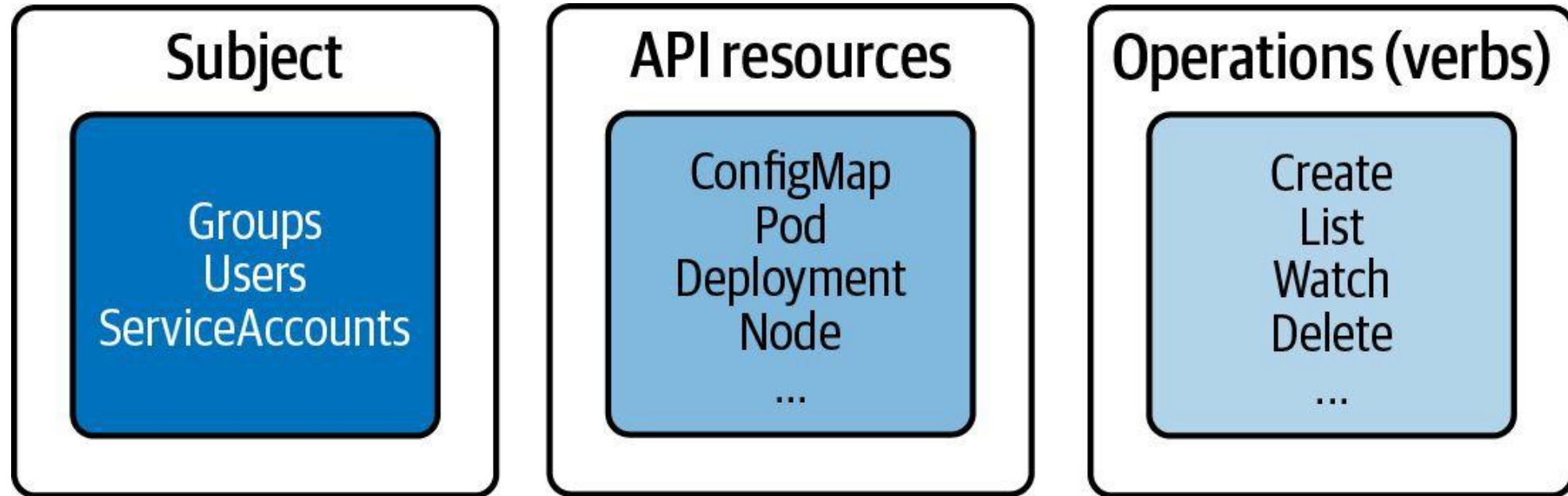
Who can do what, on which resources, where

Role-Based Access Control

- A Role/ClusterRole lists allowed verbs (get, list, create, delete...) on resources.
- A RoleBinding/ClusterRoleBinding ties that Role to a subject (ServiceAccount, user or group).
- Role + RoleBinding = one namespace; ClusterRole + ClusterRoleBinding = cluster-wide.
- ClusterRole + RoleBinding limits a cluster role's verbs to a single namespace.
- Validate without logging in: `kubectl auth can-i <verb> <resource> --as=<identity>`.

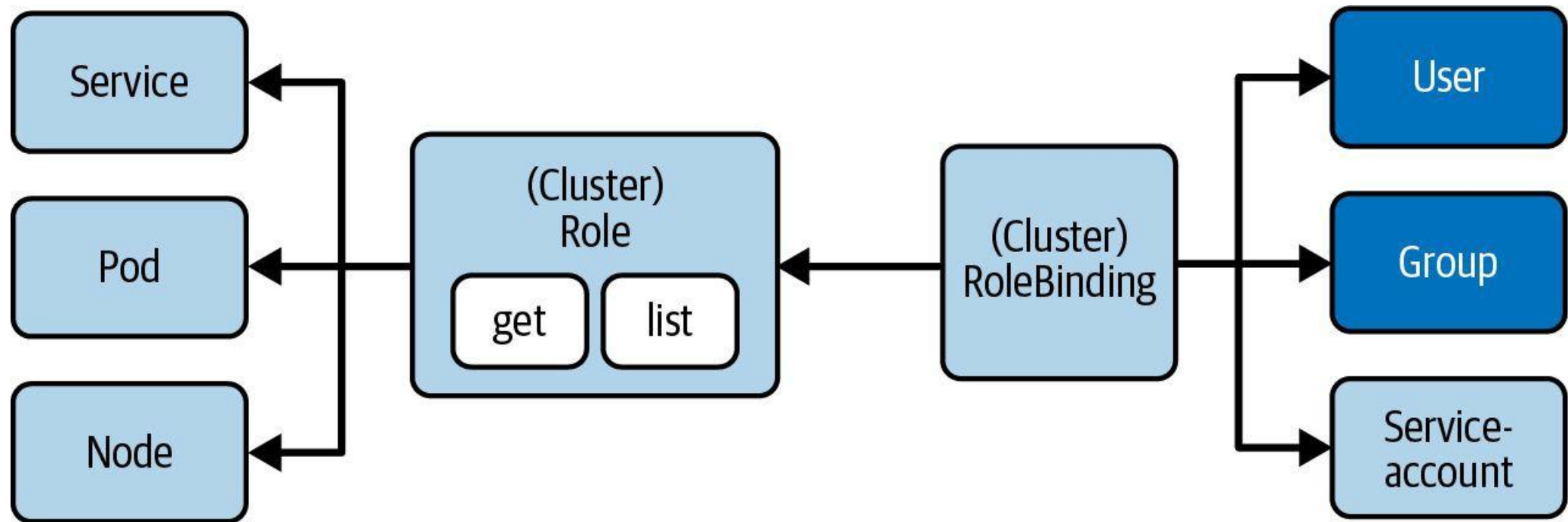
RBAC High-Level Overview

Three key elements for understanding concept



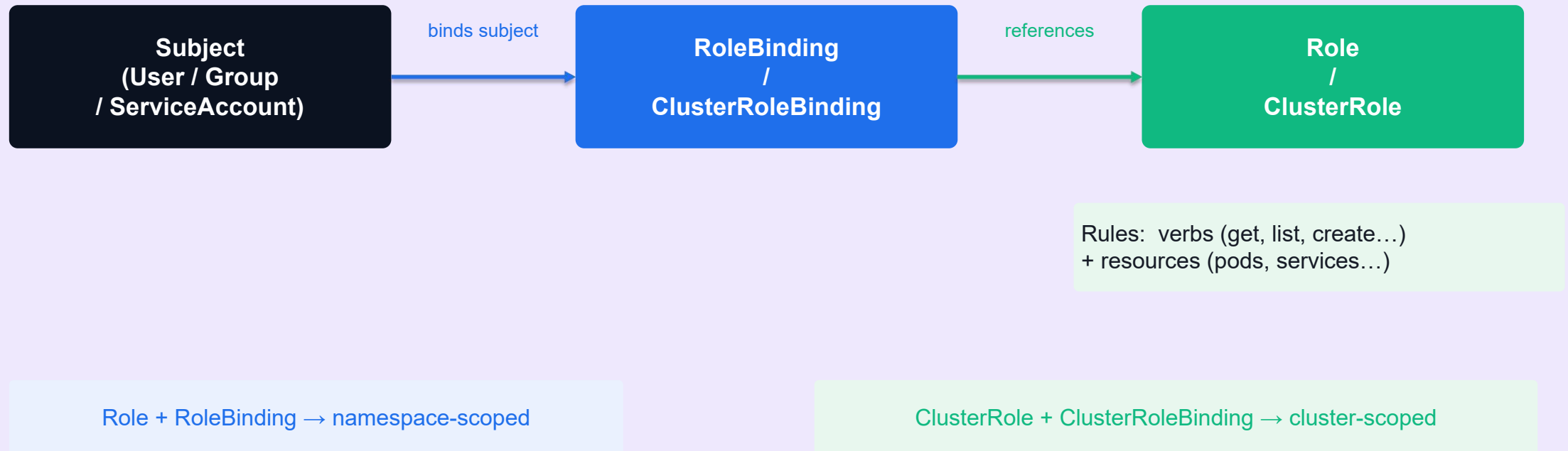
Involved RBAC Primitives

Restrict access to certain operations of API resources for subjects



RBAC — Role-Based Access Control Model

Kubernetes Cluster



Role vs ClusterRole

Role (Namespace-scoped)

- Scoped to a single namespace
- `kubectl get roles -n <ns>`
- Common uses: app-specific read/write access
 - Example: allow listing pods in 'default' namespace only
- Paired with RoleBinding in the same namespace

ClusterRole (Cluster-scoped)

- Applies across all namespaces (or non-namespaced resources)
- `kubectl get clusterroles`
- Common uses: admin, monitoring, ingress controller
 - Example: allow listing nodes, PVs, namespaces cluster-wide
- Paired with ClusterRoleBinding or namespaced RoleBinding

Defining a Role and RoleBinding

Role: grant read access to pods in the 'dev' namespace

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  namespace: dev
  name: pod-reader
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "list", "watch"]
```

RoleBinding: bind the Role to a ServiceAccount

```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: read-pods
  namespace: dev
subjects:
- kind: ServiceAccount
  name: my-app-sa
  namespace: dev
roleRef:
  kind: Role
  name: pod-reader
  apiGroup: rbac.authorization.k8s.io
```

ClusterRole and ClusterRoleBinding

ClusterRole: read access to nodes (non-namespaced resource)

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: node-reader
rules:
- apiGroups: [""]
  resources: ["nodes"]
  verbs: ["get", "list"]
```

ClusterRoleBinding: bind to a user

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: read-nodes-global
subjects:
- kind: User
  name: jane
  apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: ClusterRole
  name: node-reader
  apiGroup: rbac.authorization.k8s.io
```

RBAC — Essential kubectl Commands

Create a Role imperatively (quick in exam)

```
kubectl create role pod-reader \  
  --verb=get,list,watch \  
  --resource=pods \  
  --namespace=dev
```

Bind the Role to a ServiceAccount

```
kubectl create rolebinding read-pods \  
  --role=pod-reader \  
  --serviceaccount=dev:my-app-sa \  
  --namespace=dev
```

Check permissions for a ServiceAccount (auth can-i)

```
kubectl auth can-i list pods \  
  --as=system:serviceaccount:dev:my-app-sa \  
  --namespace=dev  
# yes
```


RBAC — Roles & RoleBindings

LAB 25

RBAC

RBAC controls who can do what on which resources. CKAD tests creating Roles, ClusterRoles, RoleBindings and ClusterRoleBindings imperatively, and validating with `kubectl auth can-i --as`. Know namespace- vs cluster-scope.

You'll build

Create a Role + RoleBinding for a ServiceAccount, validate with can-i, then a ClusterRole bound two ways.

Key commands: `k create · k auth · » Role`

RBAC — Roles & RoleBindings

Step 1 — Role: namespace-scoped permissions

```
k create role pod-reader \  
  --verb=get,list,watch --resource=pods -n dev
```

Step 2 — RoleBinding: link Role to a ServiceAccount

```
k create rolebinding viewer-binding \  
  --role=pod-reader --serviceaccount=dev:viewer -n dev
```

Step 3 — Validate with kubectl auth can-i

```
k auth can-i list pods    -n dev      --as=system:serviceaccount:dev:viewer # yes  
k auth can-i delete pods -n dev      --as=system:serviceaccount:dev:viewer # no  
k auth can-i list pods    -n default --as=system:serviceaccount:dev:viewer # no
```

RBAC — Roles & RoleBindings (cont.)

Step 4 — ClusterRole bound cluster-wide vs to one namespace

```
k create clusterrole node-reader --verb=get,list --resource=nodes
k create clusterrolebinding nodes-binding \
  --clusterrole=node-reader --serviceaccount=dev:nodes-sa # all namespaces
k create rolebinding dev-node-reader \
  --clusterrole=node-reader --serviceaccount=dev:viewer -n dev # only dev
```

Note Role + RoleBinding = namespace-scoped. ClusterRole + ClusterRoleBinding = cluster-wide. ClusterRole + RoleBinding = the cluster role's verbs limited to one namespace. `kubectl auth can-i <verb> <res> --as=<identity>` validates without logging in.

RBAC — Roles & RoleBindings

✓ Test it

k auth can-i list pods -n dev --as=system:serviceaccount:dev:viewer returns yes while delete pods and cross-namespace list return no.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>



SECURITY

Authentication and Authorization

Access control for the Kubernetes API

Processing a Request to the API Server

- Every client request to the API server passes four sequential phases:
 - Authentication — who is the caller?
 - Authorization — is the caller allowed to perform this action?
 - Admission Control — is the request well-formed and valid?
 - Validation — schema validation of the requested resource
- All four phases must pass for the request to be processed.

API Request Processing Phases

- **Authentication:** Validates the identity of the caller using a supported strategy, e.g. client certificates or bearer tokens.
- **Authorization:** Determines if the authenticated identity may perform the requested verb + HTTP path. Typically implemented with RBAC.
- **Admission Controller:** Verifies the request is well-formed and may mutate or reject it before processing. Ensures the resource definition is valid.

Authentication via Credentials in Kubeconfig

The kubeconfig file at `$HOME/.kube/config` is used by `kubectl`

```
apiVersion: v1
kind: Config
clusters:
- cluster:
    certificate-authority: /home/user/.minikube/ca.crt
    server: https://127.0.0.1:58936
  name: minikube
contexts:
- context:
    cluster: minikube
    user: bmuschko
  name: bmuschko
current-context: bmuschko
users:
- name: bmuschko
  user:
    client-certificate-data: REDACTED
    client-key-data: REDACTED
```


Managing Kubeconfig and Current Context

View the full kubeconfig

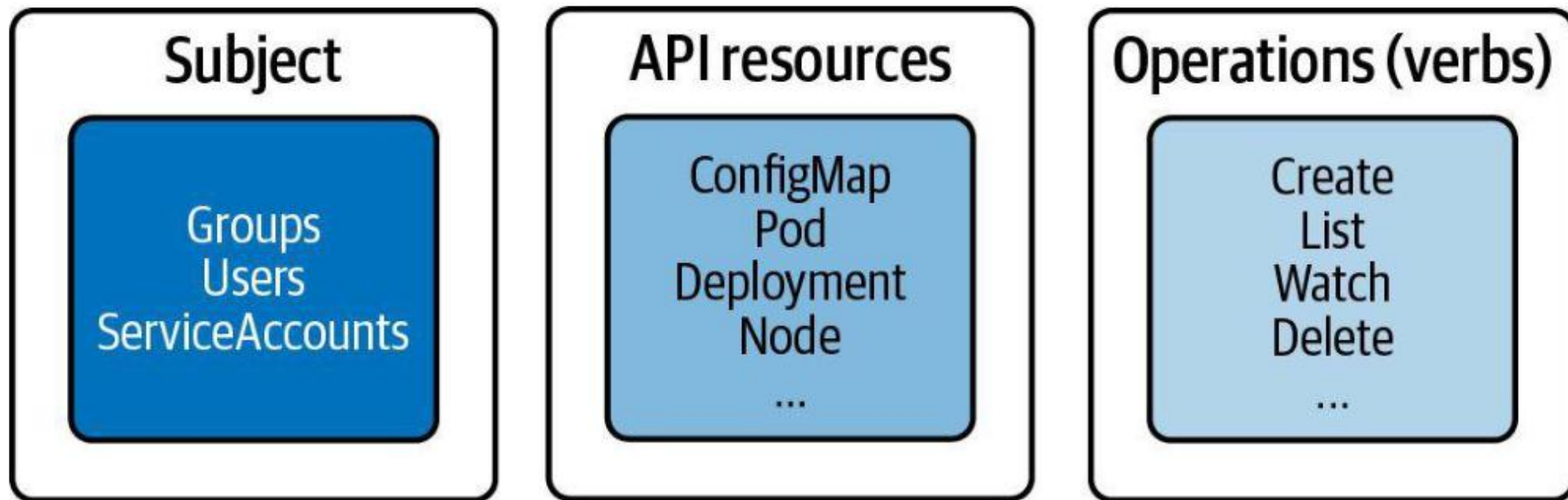
```
kubectl config view  
apiVersion: v1  
kind: Config  
current-context: bmuschko  
...
```

Show and switch the active context

```
kubectl config current-context  
bmuschko  
  
kubectl config use-context johndoe  
Switched to context "johndoe".  
  
kubectl config current-context  
johndoe
```

RBAC High-Level Overview

Three key elements for understanding concept



Namespace and Cluster-Wide Permissions

Defining permission scopes

- The **Role** maps API resources to verbs for a **single namespace**.
- The **RoleBinding** maps a reference of a Role to one or many subjects for a **single namespace**.
- To apply permissions across all namespaces of a cluster or cluster-wide resources, use the corresponding API primitives **ClusterRole** and **ClusterRoleBinding**. The configuration options of the manifests are the same.

Namespace and Cluster-Wide Permissions

Defining permission scopes

Term	Scope
Role	Namespace specific
Rolebinding	Namespace specific
ClusterRole	Cluster-wide
ClusterRoleBinding	Cluster-wide

Namespace and Cluster-Wide Permissions

Term	Scope	Use Case
Role	Namespace specific	Allow user to list pods in dev namespace Role + Rolebinding
Rolebinding	Namespace specific	Allow user to list pods in dev namespace Role + Rolebinding
ClusterRole	Cluster-wide	Allow a monitoring agent to list pods in any namespace, ClusterRole + RoleBinding
ClusterRoleBinding	Cluster-wide	Allow a user to manage nodes cluster-wide ClusterRole +

Creating a Role with Imperative Approach

Defines verbs and resources in comma-separated list

```
$ kubectl create role read-only --verb=list,get,watch  
↵  
  --resource=pods,deployments,services  
role.rbac.authorization.k8s.io/read-only created
```

Resources: Primitives the operations should apply to

Operations: Only allow listing, getting, watching

Role YAML Manifest

Can define a list of rules in an array

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: read-only
rules:
- apiGroups: [""]
  resources: ["pods", "services"]
  verbs: ["list", "get", "watch"]
- apiGroups: ["apps"]
  resources: ["deployments"]
  verbs: ["list", "get", "watch"]
```

Resources with groups need to be spelled out explicitly (in this case `apps/Deployment`)

Getting Role Details

Maps objects of a Kubernetes primitive to verbs

```
$ kubectl describe role read-only
```

```
Name:          read-only
```

```
Labels:        <none>
```

```
Annotations:   <none>
```

```
PolicyRule:
```

Resources	Non-Resource URLs	Resource Names	Verbs
-----	-----	-----	-----
pods	[]	[]	[list get watch]
services	[]	[]	[list get watch]
deployments.apps	[]	[]	[list get watch]

Creating a RoleBinding with Imperative Approach

Maps subject to Role

```
$ kubectl create rolebinding read-only-binding --role=read-only  
←   
--user=johndoe  
rolebinding.rbac.authorization.k8s.io/read-only-binding created
```

Subject: Binds a user to
the Role

Role: Reference the name of
the object

RoleBinding YAML Manifest

Roles can be mapped to multiple subjects if needed

```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: read-only-binding
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: Role
  name: read-only
subjects:
- apiGroup: rbac.authorization.k8s.io
  kind: User
  name: johndoe
```

← Reference to Role

← Reference to User

Getting RoleBinding Details

Only shows mapping between Role and subjects

```
$ kubectl describe rolebinding read-only-binding
```

```
Name:          read-only-binding
```

```
Labels:         <none>
```

```
Annotations:    <none>
```

```
Role:
```

```
  Kind: Role
```

```
  Name: read-only
```

```
Subjects:
```

```
  Kind  Name  Namespace
```

```
-----
```

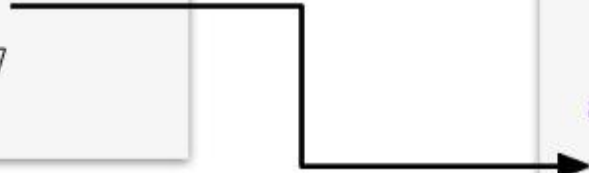
```
User   johndoe
```

Service Account YAML Manifest

In a Pod manifest, refer to the Service Account by name

```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: sa-api
  namespace: k97
```

```
apiVersion: v1
kind: Pod
metadata:
  name: list-pods
  namespace: k97
spec:
  serviceAccountName: sa-api
  ...
```



Accessing Service Account Authentication Token

Automounted at specific Volume mount path

```
$ kubectl describe pod list-pods -n k97
...
Containers
  : pods:
    ...
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from
      kube-api-access-xnkwd (ro)
...

$ kubectl exec -it list-pods -n k97 -- /bin/sh
# cat
```

RoleBinding YAML Manifest

Default RBAC policies don't span beyond kube-system

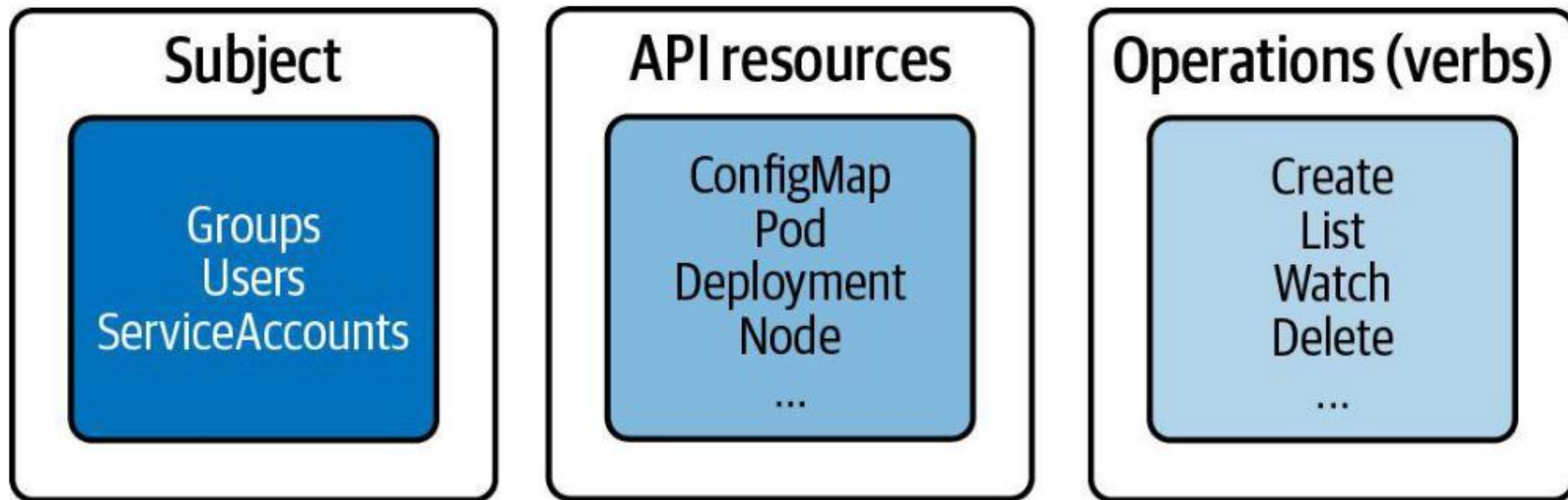
```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: serviceaccount-pod-rolebinding
  namespace: k97
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: list-pods-clusterrole
subjects:
- kind: ServiceAccount
  apiGroup: rbac.authorization.k8s.io
  name: sa-api
```

Maps the Service
Account as a
subject



RBAC High-Level Overview

Three key elements for understanding concept



EXTENDING KUBERNETES

Custom Resource Definitions (CRDs)

Extending the Kubernetes API with custom primitives

What is a Custom Resource Definition (CRD)?

- Some use cases with custom requirements cannot be fulfilled with built-in Kubernetes primitives.
- A CRD allows you to define, create, and persist objects for custom primitives and expose them via the Kubernetes API.
- CRDs are combined with controllers to implement custom logic. The combination is called the Operator pattern.

The Operator Pattern

- An Operator = CRD + Controller
- The CRD extends the Kubernetes API with a new object type (schema, validation, versioning).
- The Controller is a reconciliation loop that watches for CRD instances via the API server and implements the custom business logic.
- Operators are deployed as Pods in the cluster and interact with the API server continuously.
- Example operators: Prometheus Operator, cert-manager, Kafka Operator.

Example CRD — Smoke Testing

- Requirement: A web-based application stack deployed via a Deployment. A Service routes traffic to the Pods.
- Desired functionality: After deployment, run a quick smoke test against the Service's DNS name. Send the result to an external dashboard.
- Goal: Implement the Operator pattern (CRD + controller) for this use case.
- Custom Kind: SmokeTest — attributes: service, path, timeout, retries.

CRD YAML Manifest

Defines the schema for the custom SmokeTest object

```
apiVersion: apiextensions.k8s.io/v1
kind: CustomResourceDefinition
metadata:
  name: smoketests.stable.bmuschko.com
spec:
  group: stable.bmuschko.com
  versions:
    - name: v1
      served: true
      storage: true
      schema:
        openAPIV3Schema:
          type: object
          properties:
            spec:
              type: object
              properties:
                service: {type: string}
                path: {type: string}
                timeout: {type: integer}
                retries: {type: integer}
  scope: Namespaced
  names:
    plural: smoketests
    singular: smoketest
    kind: SmokeTest
    shortNames: [st]
```

Custom Resource Object YAML Manifest

An instance of the SmokeTest custom kind

```
apiVersion: stable.bmuschko.com/v1
kind: SmokeTest
metadata:
  name: backend-smoke-test
spec:
  service: backend
  path: /health
  timeout: 600
  retries: 3
```

Creating the CRD and Custom Object

Create the CRD first, then create instances

```
kubectl apply -f smoketest-crd.yaml  
customresourcedefinition.apiextensions.k8s.io/smoketests.stable.bmuschko.com created
```

```
kubectl apply -f backend-smoke-test.yaml  
smoketest.stable.bmuschko.com/backend-smoke-test created
```

Discovering CRDs

List all installed CRDs

```
kubectl get crds
NAME                                CREATED AT
smoketests.stable.bmuschko.com    2023-05-04T14:49:40Z
```

Discover custom resource types via api-resources

```
kubectl api-resources --api-group=stable.bmuschko.com
NAME          SHORTNAMES  APIVERSION          NAMESPACE  KIND
smoketests    st          stable.bmuschko.com/v1  true       SmokeTest
```

Controller Implementation

- You can CRUD custom resource objects, but no smoke-test logic is initiated without a controller.
- A controller acts as a reconciliation process: it polls for new SmokeTest objects via the API server and executes the smoke test, sending results to the external service.
- Controllers are typically written in Go or Python using a Kubernetes client library (e.g. client-go, kubernetes-client).



EXERCISE

Defining and Interacting with a CRD

Exam Essentials

- You are not expected to implement a CRD schema yourself. Know how to discover and use them.
- Use `kubectl get crds` to list installed CRDs, and `kubectl api-resources --api-group=<group>` to see their types.
- Create objects from a CRD schema exactly as you would for built-in resources: `kubectl apply -f <manifest>.yaml`.
- Controller implementations are out-of-scope for the CKAD exam.

Tea Break

15 minutes



TOPIC 11

Quotas & Limits

Cap namespace usage & per-container resources

Govern Resource Consumption

- ResourceQuota caps the aggregate resources a namespace can consume (pods, requests, limits).
- LimitRange enforces per-container defaults, defaultRequests and maximums.
- Without a LimitRange, Pods with no resource block are rejected in a quota-enforced namespace.
- `kubectl describe quota` shows Used vs Hard — the primary troubleshooting view.

ResourceQuota YAML Manifest

Limits # of objects and aggregate container resource consumption

```
apiVersion: v1 kind:
ResourceQuota metadata:
  name: awesome-quota
  namespace: team-awesome
spec:

  hard:
    pods: 2
    requests.cpu: "1"
    requests.memory: 1024Mi
    limits.cpu: "4" limits.memory:
4096Mi
```

What is a LimitRange?

Constraints or defaults the resource allocations for an object type

- Enforces minimum and maximum compute resources usage per Pod or container in a namespace.
- Enforces minimum and maximum storage request per PersistentVolumeClaim in a namespace.
- Enforces a ratio between request and limit for a resource in a namespace.
- Set default requests/limits for compute resources in a namespace and automatically injects them into containers at runtime.

ResourceQuota & LimitRange

LAB 26

Quotas & Limits

ResourceQuota caps the total resources used across a namespace; LimitRange enforces per-container defaults and maximums. CKAD tests both — write the YAML, apply them, and understand the rejection error messages.

You'll build

Apply a ResourceQuota and a LimitRange, watch defaults get injected, then violate the max and the quota.

Key commands: `spec: · k run · for i · » ResourceQuota`

ResourceQuota & LimitRange

Step 1 — ResourceQuota: namespace-wide ceilings

```
spec:
  hard:
    pods: "5"
    requests.cpu: "1"
    requests.memory: 1Gi
    limits.cpu: "2"
    limits.memory: 2Gi
```

Step 2 — LimitRange: per-container defaults and maximums

```
spec:
  limits:
    - type: Container
      default: {cpu: 200m, memory: 256Mi}
      defaultRequest: {cpu: 100m, memory: 128Mi}
      max: {cpu: 500m, memory: 512Mi}
```

ResourceQuota & LimitRange (cont.)

Step 3 — Defaults injected; max enforced

```
k run a --image=nginx:1.25 -n team-a
k get pod a -n team-a -o jsonpath='{.spec.containers[0].resources}'; echo
# a container requesting cpu:1 is rejected: 'maximum cpu per Container is 500m'
```

Step 4 — Exceed the quota

```
for i in 1 2 3 4; do k run quota-$i --image=nginx:1.25 -n team-a; done
k run quota-5 --image=nginx:1.25 -n team-a    # 'exceeded quota: team-a-quota'
k describe quota team-a-quota -n team-a      # Used vs Hard
```

Note ResourceQuota limits aggregate usage; exceeding it rejects new Pods. LimitRange enforces per-container min/max/defaults. Without a LimitRange, Pods with no resource block cannot be created in a quota-enforced namespace.

ResourceQuota & LimitRange

✓ Test it

A Pod with no resource block gets requests/limits injected by the LimitRange, the 6th Pod is rejected with exceeded quota, and describe quota shows Used vs Hard.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Managing Resources for Objects and Namespaces

- Containers should declare a minimum amount of resources needed (requests) and a maximum allowed (limits).
- Application developers should right-size resources with load tests or by monitoring production consumption.
- ResourceQuota enforces resource consumption at the namespace level.
- LimitRange constrains or defaults resource allocations for individual objects (Pod, PVC).

Resource Units in Kubernetes

- CPU is measured in millicores (m). Example: 600m = 0.6 CPU cores.
- Memory is measured in bytes using fixed-point numbers or power-of-two suffixes:
 - Ki = kibibytes • Mi = mebibytes • Gi = gibibytes
 - K = kilobytes • M = megabytes • G = gigabytes
- Example: 256Mi = 268,435,456 bytes. See the official docs for full details.

Container Resource Requests

- Containers declare a minimum amount of resources via `spec.containers[].resources.requests`.
- Resource types: CPU, memory, huge pages, ephemeral storage.
- The scheduler uses requests to decide which node can run the Pod.
- If no node has enough capacity → the Pod stays Pending with status `PodExceedsFreeCPU` or `PodExceedsFreeMemory`.

Example Scheduling Scenario

- The scheduler finds a node whose available CPU and memory $\geq$ the Pod's requests.
- Example: Pod requests 1 CPU + 256 Mi memory. Node must have at least that available.
- Requests do not cap actual usage — they are scheduling guarantees only.
- Once scheduled, actual usage can exceed requests (up to the limit).

Container Resource Requests — Options

- `resources.requests.cpu` — minimum CPU in cores or millicores (e.g. "1" or "500m")
- `resources.requests.memory` — minimum memory in bytes with suffix (e.g. "256Mi")
- `resources.requests.ephemeral-storage` — minimum local ephemeral storage
- `resources.requests.hugepages-<size>` — huge page allocation
- Values are per container, not per Pod.

Container Resource Requests YAML Manifest

Minimum resource amounts declared for a container

```
apiVersion: v1
kind: Pod
metadata:
  name: rate-limiter
spec:
  containers:
    - name: business-app
      image: bmuschko/nodejs-business-app:1.0.0
      ports:
        - containerPort: 8080
      resources:
        requests:
          memory: "256Mi"
          cpu: "1"
```

Container Resource Limits

- Containers declare a maximum amount of resources via `spec.containers[].resources.limits`.
- Resource types: CPU, memory, huge pages, ephemeral storage.
- The container runtime enforces limits at runtime. Behaviour when exceeded:
 - CPU limit exceeded → process is throttled (not killed)
 - Memory limit exceeded → container may be OOMKilled (Out Of Memory)

Container Resource Limits — Options

- `resources.limits.cpu` — maximum CPU in cores or millicores (e.g. "2" or "1000m")
- `resources.limits.memory` — maximum memory in bytes with suffix (e.g. "512Mi")
- `resources.limits.ephemeral-storage` — maximum local ephemeral storage
- `resources.limits.hugepages-<size>` — maximum huge pages
- Best practice: always set both requests and limits for every container.

Container Resource Limits YAML Manifest

Maximum resource amounts declared for a container

```
apiVersion: v1
kind: Pod
metadata:
  name: rate-limiter
spec:
  containers:
    - name: business-app
      image: bmuschko/nodejs-business-app:1.0.0
      ports:
        - containerPort: 8080
      resources:
        limits:
          memory: "512Mi"
          cpu: "2"
```

Pod Scheduling Details

Find the node a Pod was scheduled on, then inspect its resource usage

```
kubectl get pod rate-limiter -o yaml | grep nodeName:
  nodeName: worker-3

kubectl describe node worker-3
...
Non-terminated Pods:
  Namespace   Name           CPU Requests  CPU Limits   Memory Requests  Memory Limits
  default     rate-limiter   1250m (62%)   0 (0%)       256Mi (6%)       512Mi (13%)
```

Container Resource Requests & Limits — Combined

It is best practice to define both requests and limits for every container

```
spec:
  containers:
  - name: business-app
    image: bmuschko/nodejs-business-app:1.0.0
    resources:
      requests:
        memory: "256Mi"
        cpu: "1"
      limits:
        memory: "512Mi"
        cpu: "2"
```



EXERCISE

Defining Container Resource Requests and Limits

No Requests/Limits Definition

Scheduler rejects the creation of the object

```
$ kubectl create -f nginx-pod.yaml -n team-awesome
```

```
Error from server (Forbidden): error when creating "nginx-pod.yaml":  
␣ pods "nginx" is forbidden: failed quota: awesome-quota: must  
specify ␣limits.cpu,limits.memory,requests.cpu,requests.memory
```


Exceeds Requests/Limits Definition

Scheduler rejects the creation of the object

```
$ kubectl create -f nginx-pod.yaml -n team-awesome
```

```
Error from server (Forbidden): error when creating "nginx-pod.yaml":  
␣pods "nginx" is forbidden: exceeded quota: awesome-quota,  
requested: ␣pods=1,requests.cpu=500m,requests.memory=512Mi, used:  
pods=2, ␣requests.cpu=1,requests.memory=1024Mi, limited: pods=2, ␣  
requests.cpu=1,requests.memory=1024Mi
```

LimitRange YAML Manifest

Sets defaults, minimum and maximum CPU allocation for containers

```
apiVersion: v1
kind: LimitRange
metadata:
  name:
cpu-resource-constraint
spec:
  limits:
  - default:
      cpu: 500m
    defaultRequest:
      cpu: 500m
    max:
      cpu: "1"
    min:
      cpu: 100m
    type: Container
```

Defines default limits

Defines default requests

Minimum and maximum
resource range

Automatically Uses Container Defaults for Pod

Container doesn't provide any resource requests/limits

```
$ kubectl create -f pod.yaml
pod/rate-limiter created

$ kubectl describe pod rate-limiter
...
Containers:
  business-app
  :
  ...
```

Rejected Pod Creation for Exceeded Limits

Requests a higher maximum for CPU resources than defined by LimitRange

```
$ kubectl create -f pod.yaml
```

```
Error from server (Forbidden): error when creating "pod.yaml": pods
↳ "rate-limiter" is forbidden: maximum cpu usage per Container is 1,
↳ but limit is 2
```

WRAP-UP

Day 3 done — you can observe, configure and secure applications.

Tomorrow (half day): Services & Networking, then the final assessment.

COURSE SLIDES

Certified Kubernetes Application Developer (CKAD)

WSQ Course Code: TGS-2025053212

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

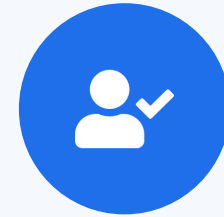
DAY 4 (½ day)

Services & Networking

Services · Service DNS · Ingress with TLS · NetworkPolicy (CKAD Domain 5 — 20%) + Final Assessment

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your phone camera and submit your attendance.
- A minimum of 75% attendance is required for assessment and funding.



**Minimum 75%
attendance required**

Recap & Today

- So far: design & build (D1), deployment (D2), observability and config & security (D3).
- This morning — Domain 5 Services & Networking (20%): how Pods are reached and isolated.
- Services and DNS for stable in-cluster addressing; Ingress for HTTP/HTTPS; NetworkPolicy for isolation.
- From 1:00 PM — mock-exam practice; the final written assessment starts at 2:00 PM.



TOPIC 1

Services

Stable virtual IPs & DNS for a set of Pods

Why Services?

- Pod IPs are ephemeral — a Service gives a stable virtual IP and DNS name in front of matching Pods.
- ClusterIP (default) is in-cluster only; NodePort opens a port on every node; LoadBalancer asks the cloud for an external IP.
- A Service finds its Pods by label selector; the matching set is tracked as EndpointSlices.
- Empty Endpoints almost always means a selector/label mismatch — the most common Service bug.

Choosing a Service Type

ClusterIP

- In-cluster only
- Stable virtual IP + DNS
- Default type

NodePort

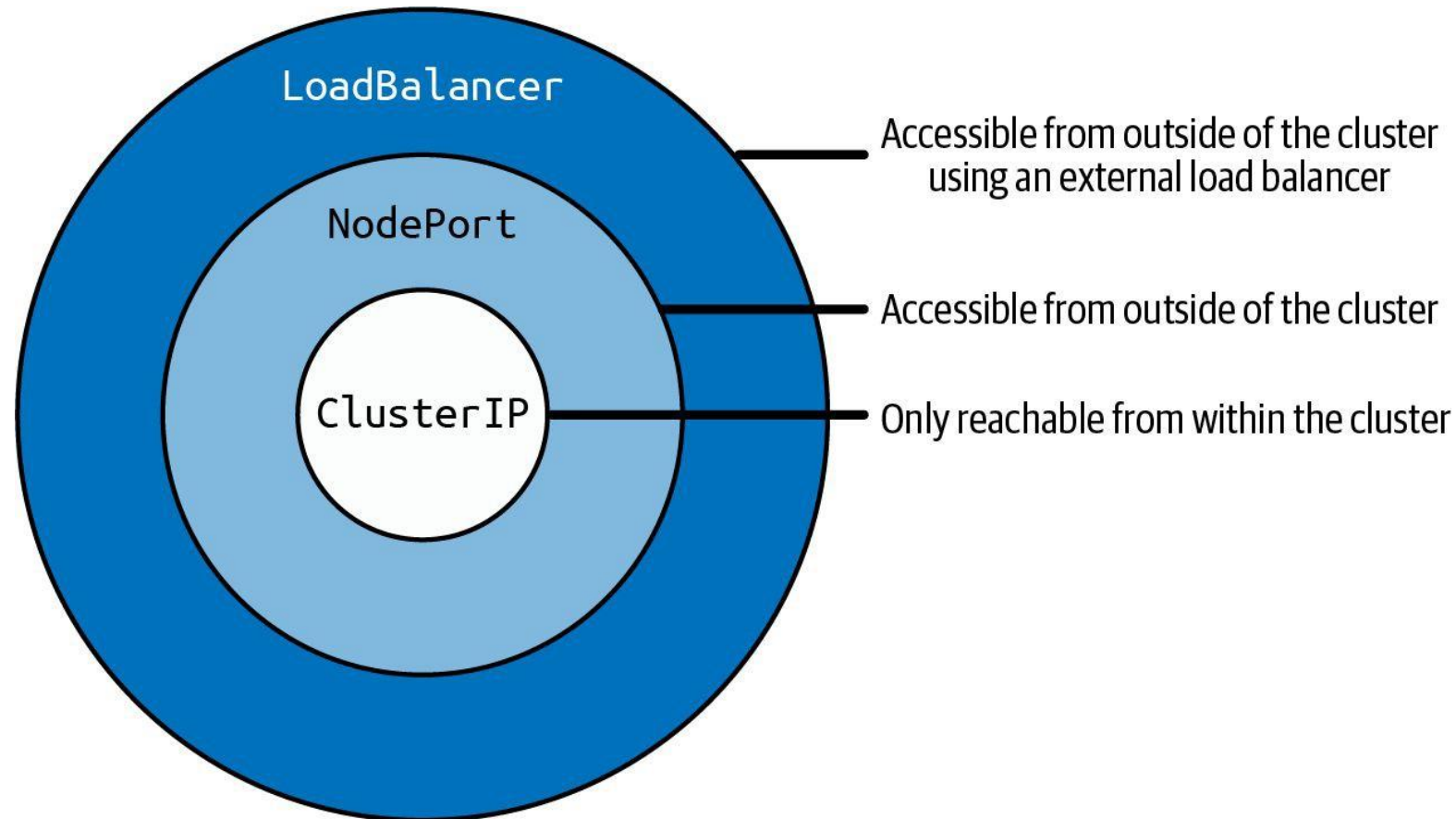
- Port on every node
- 30000–32767
- Reachable from outside

LoadBalancer

- Cloud external IP
- Builds on NodePort
- One Service, public entry

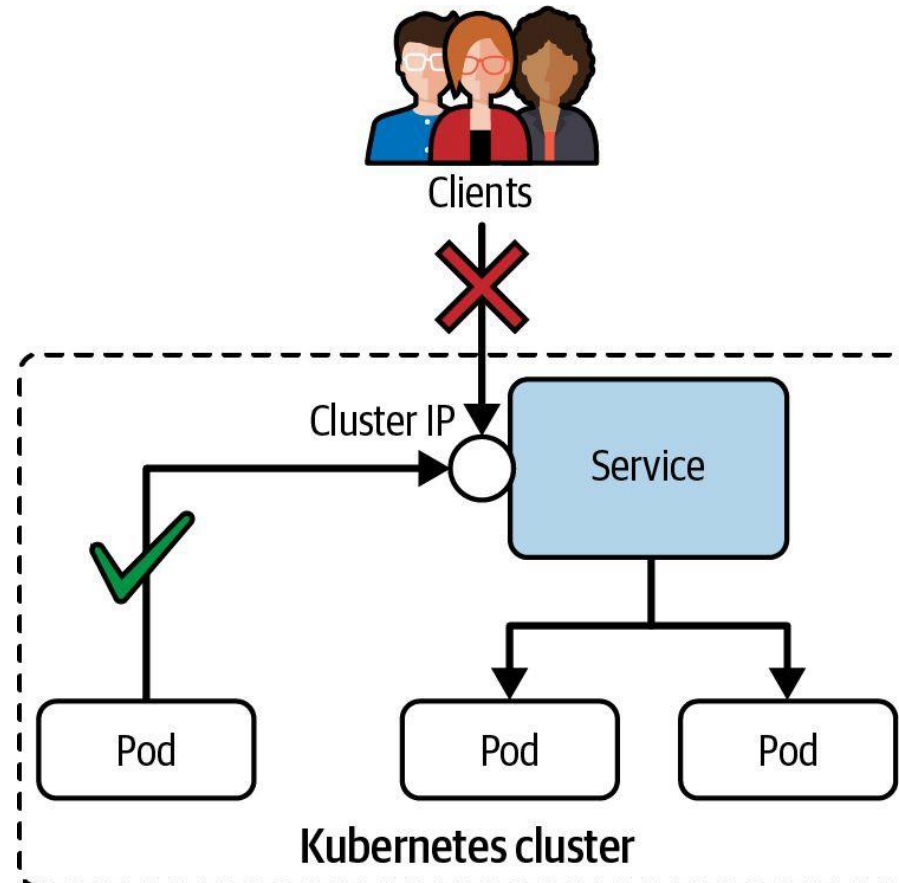
Service Type Inheritance

Services build on top of each other



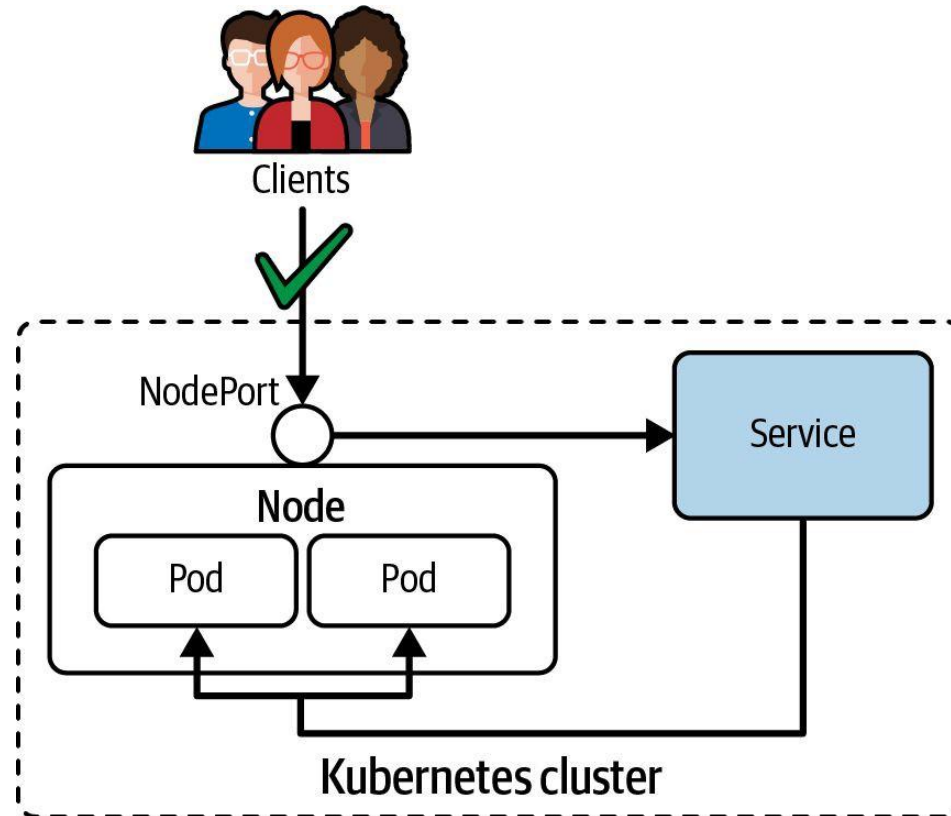
ClusterIP Service Type

Only reachable from within the cluster, e.g. another Pod



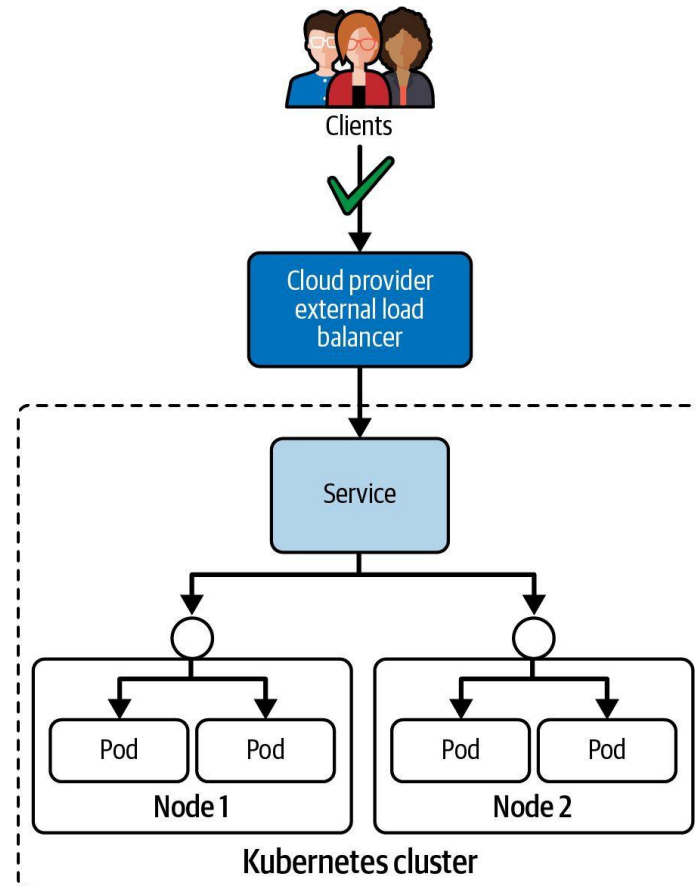
NodePort Service Type

Can be resolved from outside of the Kubernetes cluster



LoadBalancer Service Type

Routes traffic from existing, external load balancer to Service



Service Port Mapping — port, targetPort, nodePort

Three port fields: port (Service), targetPort (Pod container), nodePort (Node)

```
apiVersion: v1
kind: Service
metadata:
  name: web-svc
spec:
  type: NodePort
  selector:
    app: web
  ports:
    - port: 80          # port clients use to reach the Service (ClusterIP)
      targetPort: 8080  # port the container listens on
      nodePort: 30080   # port opened on every Node (30000-32767)
                        # omit nodePort for ClusterIP; not needed for LoadBalancer
```

Expose a deployment as a Service in one command

```
kubectl expose deployment web --port=80 --target-port=8080 --type=NodePort
kubectl get svc web
```

NAME	TYPE	CLUSTER-IP	PORT(S)	AGE
web	NodePort	10.96.1.50	80:30080/TCP	5s

Service DNS — How Pods Resolve Services

<service-name>.<namespace>.svc.cluster.local

web-svc

service
name

production

namespace

svc

resource
type

cluster.local

cluster
domain

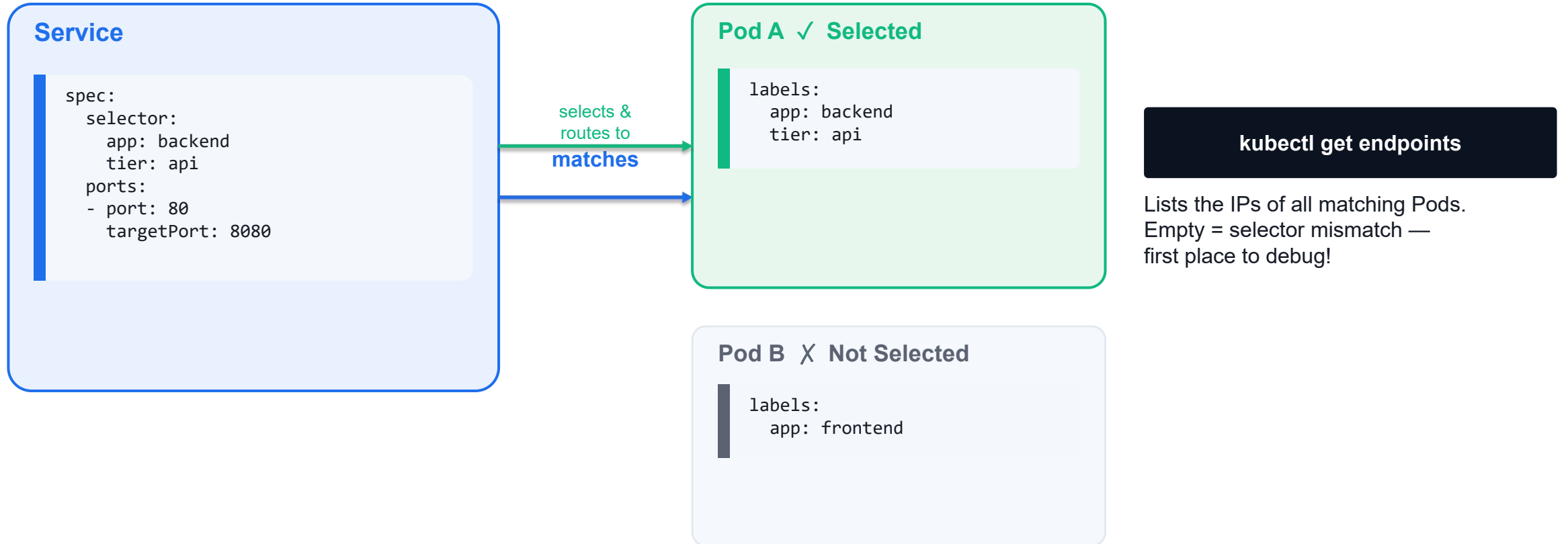
Short forms (within same namespace)

```
# From a pod in the SAME namespace – just use the service name:  
curl http://web-svc/api
```

```
# From a pod in a DIFFERENT namespace – include the namespace:  
curl http://web-svc.production/api
```

```
# Full FQDN always works:  
curl http://web-svc.production.svc.cluster.local/api
```

Service Selector Matches Pod Labels



Services — Quick Reference

Service Types

- ClusterIP (default): internal VIP, cluster-only
 - `kubectl create service clusterip my-svc --tcp=80:8080`
- NodePort: extends ClusterIP + opens port 30000-32767 on every node
 - `kubectl create service nodeport my-svc --tcp=80:8080 --node-port=30080`
- LoadBalancer: extends NodePort + provisions cloud LB
- ExternalName: DNS CNAME alias to external hostname

When to Use

- Use ClusterIP when only pods inside the cluster need access
- Use NodePort for simple external access without a cloud provider
- Use LoadBalancer in managed cloud environments (GKE, EKS, AKS)
- Headless service (clusterIP: None): no VIP — direct pod DNS records
 - `kubectl create service clusterip my-svc --clusterip=None`
- Check: `kubectl get svc -A` | `kubectl describe svc <name>`

Services — ClusterIP, NodePort, LoadBalancer

LAB 27

Services

A Service gives a stable virtual IP and DNS name to a set of Pods. CKAD requires fluency with ClusterIP, NodePort and LoadBalancer types, kubectl expose, endpoint debugging and the selector-mismatch pattern.

You'll build

Expose a Deployment as ClusterIP, NodePort and LoadBalancer, inspect EndpointSlices, then debug a selector mismatch.

Key commands: `k create · k expose · k patch · » ClusterIP`

Services — ClusterIP, NodePort, LoadBalancer

Step 1 — Backend Deployment + ClusterIP Service

```
k create deployment web --image=nginx:1.25 --replicas=3
k expose deployment web --port=80 --target-port=80 --name=web-cip
k describe svc web-cip | grep -E "IP:|Endpoints:"
```

Step 2 — NodePort (a port on every node)

```
k expose deployment web --port=80 --target-port=80 --type=NodePort --name=web-np
PORT=$(k get svc web-np -o jsonpath='{.spec.ports[0].nodePort}')
curl -s http://localhost:$PORT | head -3 # 30000-32767
```

Step 3 — LoadBalancer + EndpointSlices

```
k expose deployment web --port=80 --type=LoadBalancer --name=web-lb
k get endpointslices -l kubernetes.io/service-name=web-cip
```

Services — ClusterIP, NodePort, LoadBalancer (cont.)

Step 4 — Debug a selector mismatch (the #1 Service bug)

```
k patch svc web-cip -p '{"spec":{"selector":{"app":"does-not-exist"}}}'  
k get endpoints web-cip          # empty = no Pod matches  
k patch svc web-cip -p '{"spec":{"selector":{"app":"web"}}}'
```

Note ClusterIP = in-cluster only; NodePort = a port on every node; LoadBalancer = a cloud LB. Empty Endpoints almost always means the selector does not match Pod labels. EndpointSlices (v1.21+) succeed Endpoints.

Services — ClusterIP, NodePort, LoadBalancer

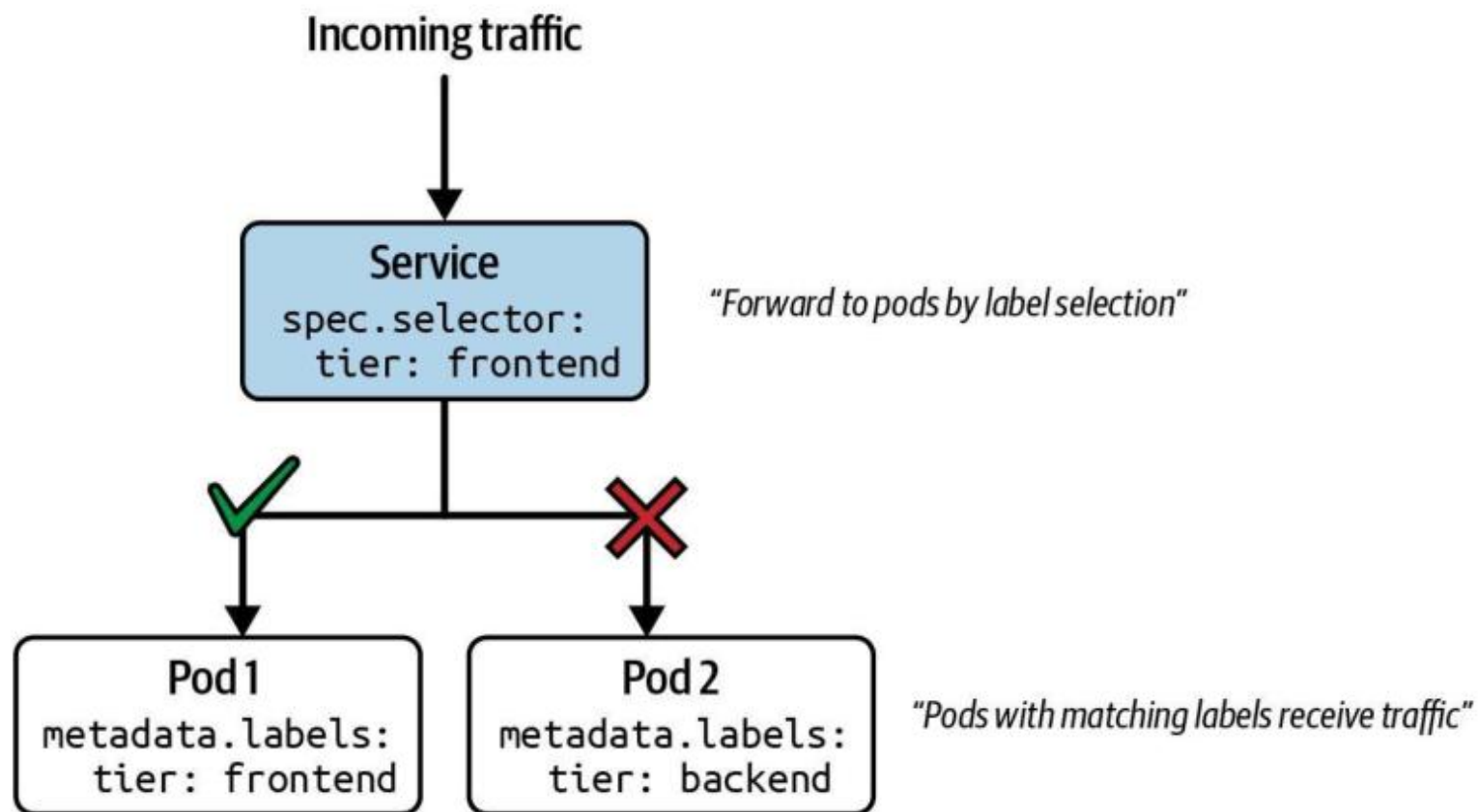
✓ Test it

k describe svc web-cip lists three endpoint IPs, the NodePort serves the page on localhost:\$PORT, and a bad selector empties the Endpoints until it is fixed.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Request Routing

"How does a Service decide which Pod to forward the request to?"



Creating a Service with Imperative Approach

Needs to provide the service type and ports

```
$ kubectl run echoserver↵  
  --image=k8s.gcr.io/echoserver:1.10  
  --port=8080  
pod/echoserver  
created
```

Exposed container port

```
$ kubectl create↵  
  service  
  --tcp=80:8080  
service/echoserver created
```

Incoming and outgoing port

```
$ kubectl run echoserver↵  
  --image=k8s.gcr.io/echoserver:1.10  
  --expose  
  --port=8080↵  
service/echoserver
```

Shortcut for creating a Pod + Service

Service YAML Manifest

Needs to provide the service type and ports

```
apiVersion: v1
kind: Service
metadata:
  name: echoserver
spec:
  selector:
    app: echoserver
  ports:
  - port: 80
    targetPort: 8080
    protocol: TCP
```

← Label selection for Pods

← Mapping of incoming to outgoing port

ClusterIP Service YAML Manifest at Runtime

ClusterIP service type has been assigned if not already provided

```
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: my-service
5  spec:
```

```
6    type: ClusterIP
```

```
7    ports:
```

```
8      - port: 80
```

```
9        targetPort: 80
```



Type: ClusterIP (default)



Port of the service

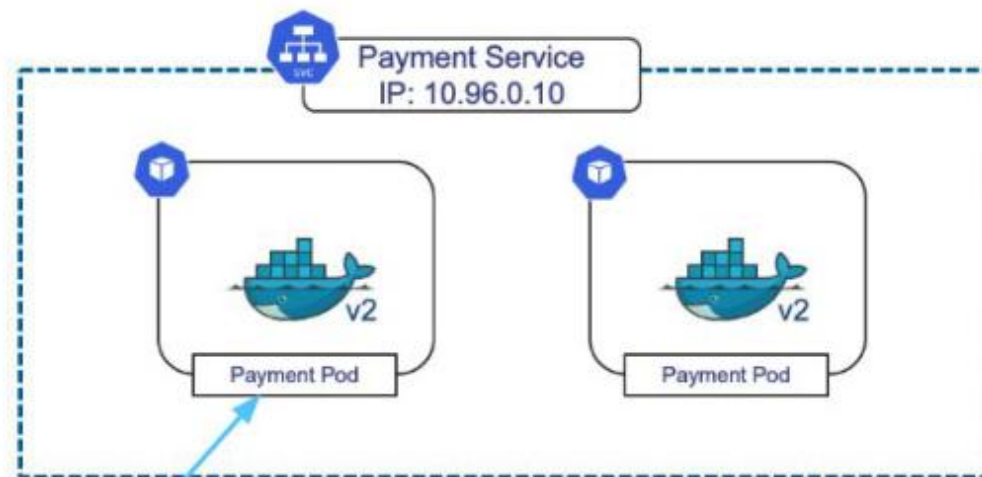


Port that the pod is listening on

ClusterIP Service YAML Manifest at Runtime

ClusterIP service type has been assigned if not already provided

```
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: my-service
5  spec:
6    type: ClusterIP
7    ports:
8      - port: 80
9      targetPort: 80
```



`kubectl create -f my-service.yaml`

ClusterIP - Binding Pods to Services

```
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: my-service
5  spec:
6    type: ClusterIP
7    ports:
8      - port: 80
9        targetPort: 80
10   selector:
11     env: dev
```

```
1  apiVersion: v1
2  kind: Pod
3  metadata:
4    name: service-pod-1
5    labels:
6      env: dev
7  spec:
8    containers:
9      - name: nginx
10        image: nginx
11        ports:
12          - containerPort: 80
```

NodePort Service YAML Manifest at Runtime

Service type is NodePort, node port has been assigned

```
apiVersion: v1
kind: Service
metadata:
  name: my-service
spec:
```

```
  type: NodePort
```

← Type: *NodePort

```
  ports:
```

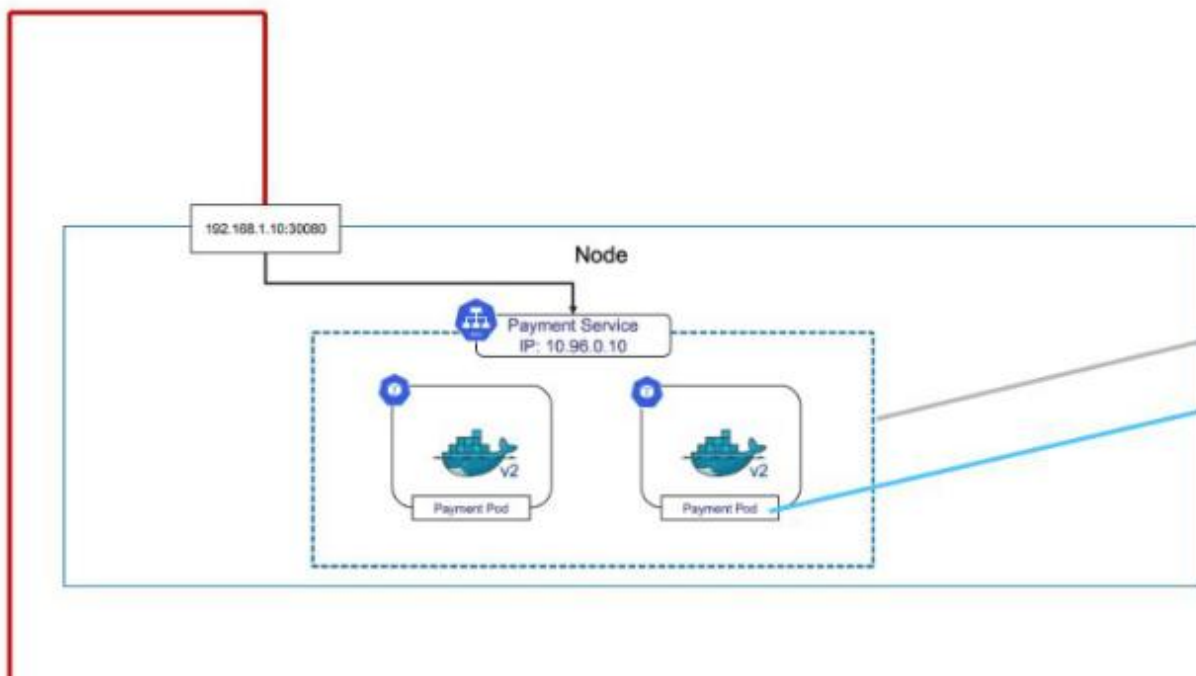
```
    - targetPort: 80
      port: 80
```

```
    NodePort: 30080
```

← NodePort Number

Accessing a NodePort Service

Use the combination of node IP and statically-assigned port



```
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: nodeport-service
5  spec:
6    type: NodePort
7    ports:
8      - port: 80
9        targetPort: 80
10       nodePort: 30080
```




TROUBLESHOOTING

Troubleshooting Services

Built-in mechanisms for identifying the root cause of a failure

Checking Service Connectivity

Identify the Service type and attempt to connect to it

```
kubectl get service echoserver
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
echoserver	ClusterIP	10.96.254.0	<none>	8080/TCP	8s

Run a temporary Pod and attempt to reach the Service

```
kubectl run tmp --image=busybox --restart=Never -it --rm \
  -- wget echoserver:8080
Connecting to echoserver (10.96.254.0:8080)
wget: can't connect to remote host (10.96.254.0:8080): Connection refused
pod "tmp" deleted
```

Verifying Endpoints

No endpoints = label selector does not match any Pod

```
kubectl get endpoints echoserver
NAME           ENDPOINTS   AGE
echoserver     <none>      63s
```

Correct configuration exposes Pod IP addresses

```
kubectl get endpoints echoserver
NAME           ENDPOINTS                                     AGE
echoserver     10.244.0.3:8080,10.244.0.4:8080             63s
```

Checking Label Selection and Port Assignment

Inspect the Service — verify Selector and TargetPort

```
kubectl describe service echoserver
Name:      echoserver
Selector:  app=echoserver
Type:      ClusterIP
IP:        10.111.148.223
Port:      8080/TCP
TargetPort: 8080/TCP
Endpoints: 10.244.0.3:8080,10.244.0.4:8080
```

Compare with Pod labels — they must match the Selector

```
kubectl describe pod echoserver
Labels:  app=echoserver
Containers:
  echoserver:
    Port: 8080/TCP
```

Checking Network Policies

Check for NetworkPolicies that may block ingress traffic

```
kubectl get networkpolicies
No resources found in default namespace.

# If policies exist, inspect them:
kubectl get networkpolicies
NAME                POD-SELECTOR  AGE
default-deny-ingress <none>        9s
```

A default-deny-ingress policy blocks ALL ingress — may need an allow rule

Ensuring Pod Functionality

Check Pod status — even 'Running' may hide container problems

```
kubectl get pods
NAME          READY  STATUS      RESTARTS  AGE
echoserver    0/1    ErrImagePull 0          2s
```

After fix:

```
kubectl get pods
NAME          READY  STATUS   RESTARTS  AGE
echoserver    1/1    Running  0          5s
```

Exec into the Pod or use logs to verify the application is actually serving

```
kubectl logs echoserver
kubectl exec -it echoserver -- wget -qO- localhost:8080
```



EXERCISE

Troubleshooting a Service

Exam Essentials

- Service misconfiguration is a common failure. Any misconfiguration prevents traffic from reaching the target Pods.
- Common causes: incorrect label selector (Service selector must match Pod labels), wrong port or targetPort assignment.
- Use `kubectl get endpoints <service>` to verify the Service has resolved Pod IP:port pairs.
- Check for NetworkPolicies that may block ingress traffic to the Pod.
- Verify the Pod itself is healthy: check status, logs, and exec a test request from inside the cluster.



TOPIC 2

Service DNS

How CoreDNS resolves Services and Pods

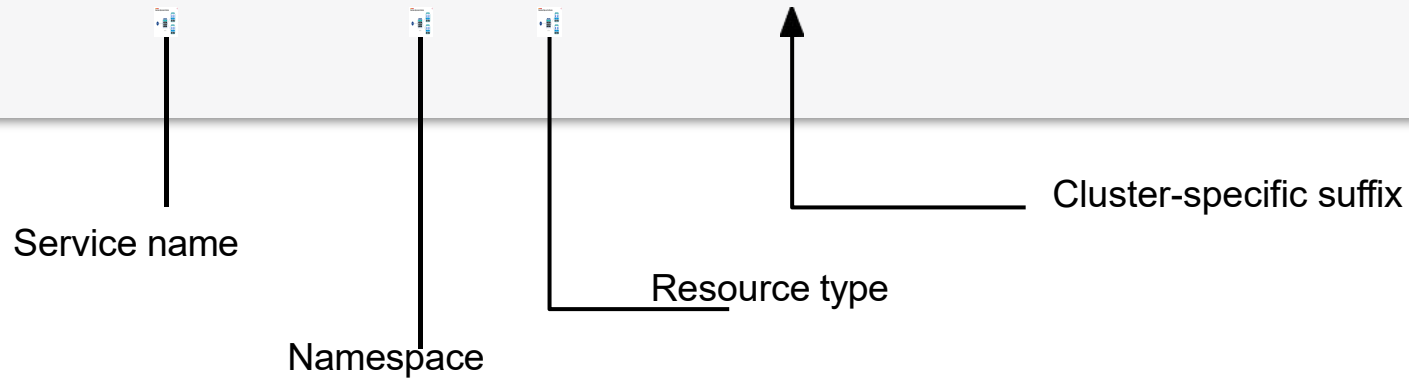
Names, Not IPs

- CoreDNS gives every Service a name: `<service>.<namespace>.svc.cluster.local`.
- Within the same namespace the short name resolves via the Pod's search list.
- Cross-namespace you need at least `<service>.<namespace>`; the full FQDN always works.
- A headless Service (clusterIP: None) returns every Pod IP instead of a single virtual IP.

Discovering the Service by DNS Lookup

Kubernetes' DNS service maps cluster IP address to Service name

```
$ kubectl run tmp --image=busybox --restart=Never -it --rm <␣  
-- wget echoserver.default.svc.cluster.local:8080  
Connecting to echoserver.default.svc.cluster.local:8080 (10.96.254.0:8080)  
...
```



Service DNS

LAB 28

Service DNS

CoreDNS gives every Service a stable DNS name: <service>.<namespace>.svc.cluster.local. CKAD tests cross-namespace resolution, Pod DNS records, headless Services, and reading /etc/resolv.conf to understand ndots:5.

You'll build

Resolve a Service by short name and FQDN, cross namespaces, read a Pod DNS record and /etc/resolv.conf, then a headless Service.

Key commands: `k -n nslookup web · cat /etc/resolv.conf · » FQDN`

Service DNS

Step 1 — Resolve from the same namespace (short name)

```
k -n app run client --image=busybox --restart=Never -it --rm -- \
  sh -c 'nslookup web; wget -qO- web | head -3'
```

Step 2 — Resolve from a different namespace

```
nslookup web           # fails cross-namespace
nslookup web.app       # works
nslookup web.app.svc.cluster.local
```

Step 3 — Pod DNS record and resolv.conf

```
# Pod A-record: <dashed-ip>.<namespace>.pod.cluster.local
cat /etc/resolv.conf
# search app.svc.cluster.local svc.cluster.local cluster.local
# ndots:5
```

Service DNS (cont.)

Step 4 — Headless Service returns all Pod IPs

```
k -n app create service clusterip headless --clusterip="None" --tcp=80:80  
nslookup headless.app    # returns every Pod IP, not one VIP
```

Note FQDN = <service>.<namespace>.svc.cluster.local. Short names resolve via the search list within the namespace; cross-namespace needs <service>.<namespace>. ndots:5 is why short names check the search list before an absolute lookup. Headless (clusterIP: None) returns per-Pod IPs.

Service DNS

✓ Test it

nslookup web works in-namespace but needs web.app cross-namespace, and the headless Service resolves to all backing Pod IPs.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Headless Services

Headless Service — a Service with **clusterIP: None**. No virtual IP; DNS resolves directly to Pod IPs.

Regular ClusterIP Service

DNS → single virtual IP (kube-proxy load-balances)

Client never sees individual Pod IPs

Best for: stateless services

Headless Service (clusterIP: None)

DNS → returns A record per Pod IP

Client discovers and connects to individual Pods

Best for: StatefulSets, databases, peer discovery

Headless Service — YAML and DNS

Set clusterIP: None to create a headless service

```
apiVersion: v1
kind: Service
metadata:
  name: mysql-headless
spec:
  clusterIP: None          # <-- this makes it headless
  selector:
    app: mysql
  ports:
    - port: 3306
```

DNS — each Pod gets its own A record (used with StatefulSets)

```
# StatefulSet pod DNS pattern:
# <pod-name>.<service-name>.<namespace>.svc.cluster.local
mysql-0.mysql-headless.default.svc.cluster.local
mysql-1.mysql-headless.default.svc.cluster.local
```

Tea Break

15 minutes



TOPIC 3

Ingress & TLS

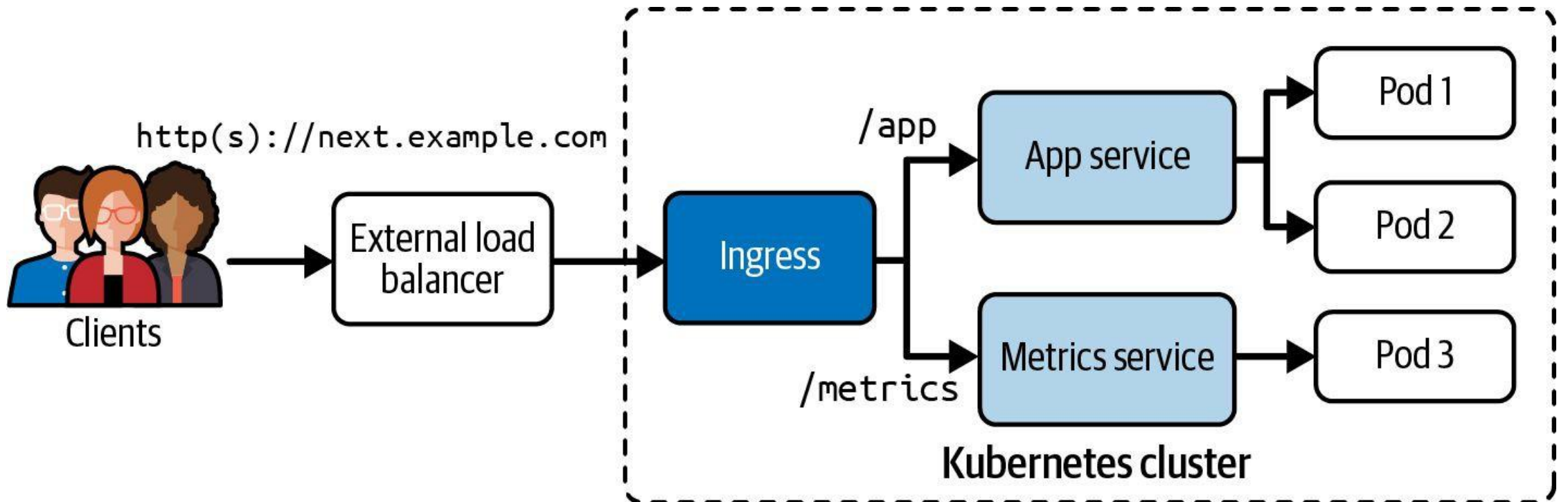
Layer-7 HTTP/HTTPS routing into the cluster

One Entry Point, Many Rules

- An Ingress defines host- and path-based HTTP/HTTPS rules that forward to backend Services.
- An Ingress controller (e.g. ingress-nginx) must be running; ingressClassName selects it.
- TLS terminates at the controller — tls.secretName points at a kubernetes.io/tls Secret.
- pathType: Prefix matches a path and everything below it; Exact requires an exact match.
- apiVersion is networking.k8s.io/v1 (extensions/v1beta1 was removed in 1.22).

Ingress Traffic Routing

The context path is mapped to a backend (Service name and port)



Why Ingress? Services vs Ingress

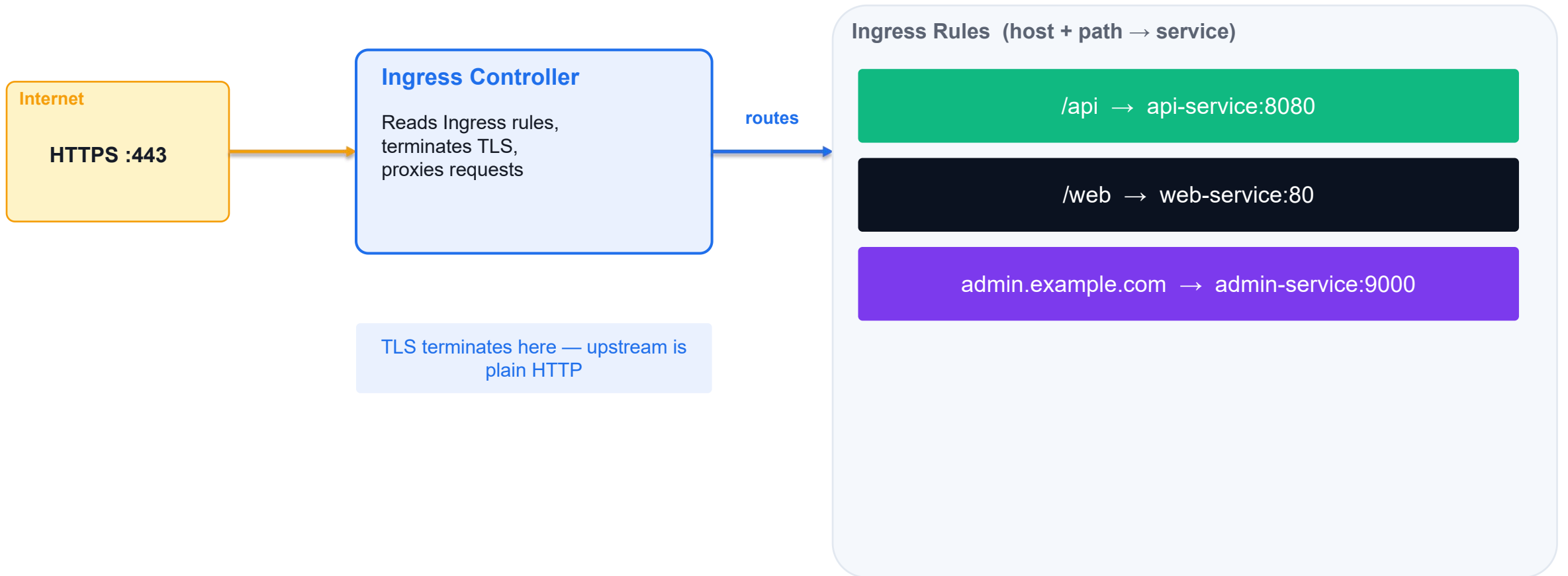
Without Ingress (Services only)

- One Service = one IP/port
- LoadBalancer Service = one cloud LB per service (expensive)
- No SSL/TLS termination at Service level
- No host/path-based routing with Services alone
- Many services = many public IPs to manage

With Ingress

- Single entry point for all HTTP(S) traffic
- L7 routing: route by hostname or URL path
 - example.com/api → api-service
 - example.com/web → web-service
- TLS termination: one cert handles all services
- Requires an IngressController (nginx, traefik, AWS ALB...)

Ingress — L7 Routing Architecture



Ingress YAML — Anatomy

Path-based routing for two services on the same host

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: app-ingress
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /
spec:
  ingressClassName: nginx          # which IngressController to use
  tls:
    - hosts: [example.com]
      secretName: tls-secret        # kubectl create secret tls
  rules:
    - host: example.com
      http:
        paths:
          - path: /api
            pathType: Prefix
            backend:
              service: {name: api-svc, port: {number: 8080}}
          - path: /web
            pathType: Prefix
            backend:
              service: {name: web-svc, port: {number: 80}}
```


IngressClass and IngressController

- An IngressController is a running Pod (e.g. nginx, traefik) that reads Ingress rules and configures its proxy.
- An IngressClass links an Ingress resource to a specific IngressController.
 - `spec.ingressClassName: nginx` → handled by the nginx IngressController
- If only one IngressClass exists it can be marked as default (`isDefaultClass: true`).
- Check available classes: `kubectl get ingressclass`
- CKAD exam: IngressClass is usually pre-configured — just reference it in your Ingress spec.
- Troubleshoot: `kubectl describe ingress <name>` — check Events for routing errors.

Ingress with TLS

LAB 29

Ingress & TLS

An Ingress provides Layer-7 HTTP/HTTPS routing: hostname and path rules forwarding to backend Services. CKAD tests installing a controller, TLS Secrets, host routing and path-based routing with pathType.

You'll build

Install ingress-nginx, create a TLS Secret, write a host+TLS Ingress, test HTTPS, then add a path-based rule.

Key commands: `k create · apiVersion: networking.k8s.io/v1 · curl -k · » Ingress`

Ingress with TLS

Step 1 — Backend Service + TLS Secret

```
k create deployment web --image=hashicorp/http-echo --replicas=2 -- -text=hello-ingress
k expose deployment web --port=5678 --target-port=5678
k create secret tls demo-tls --cert=tls.crt --key=tls.key
```

Ingress with TLS (cont.)

Step 2 — Ingress with TLS and host routing

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata: {name: demo}
spec:
  ingressClassName: nginx
  tls:
    - hosts: [demo.local]
      secretName: demo-tls
  rules:
    - host: demo.local
      http:
        paths:
          - path: /
            pathType: Prefix
            backend: {service: {name: web, port: {number: 5678}}}}
```

Ingress with TLS (cont.)

Step 3 — Test HTTPS routing

```
curl -k --resolve demo.local:$HTTPS_PORT:127.0.0.1 \  
https://demo.local:$HTTPS_PORT/      # hello-ingress
```

Step 4 — Add path-based routing

```
# add a second rule: path /v2 -> service v2:5678 (pathType: Prefix)  
curl -k --resolve demo.local:$HTTPS_PORT:127.0.0.1 \  
https://demo.local:$HTTPS_PORT/v2      # hello-v2
```

Note Ingress apiVersion is networking.k8s.io/v1. ingressClassName selects the controller; tls.secretName must point at a kubernetes.io/tls Secret; pathType: Prefix matches a path and everything below it. --resolve + -k let curl test without real DNS or a valid cert.

Ingress with TLS

✓ Test it

`curl https://demo.local:$HTTPS_PORT/` returns `hello-ingress` over TLS, and the `/v2` path returns `hello-v2` after the second rule is added.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>

Creating an Ingress with Imperative Approach

The rule definition requires intricate knowledge of syntax

```
$ kubectl create  
  ingress corellian  
  --rule="star-alliance.com/corellian/api=corellian:8080"  
ingress.networking.k8s.io/corellian created
```

Ingress YAML Manifest

Allows for defining multiple rules that map to backend

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: corellian
spec:
  r host: star-alliance.com
  http:
    paths:
      - backend:
          service:
            name: corellian
            port:
              number: 8080
          path: /corellian/api
          pathType: Exact
```

The host definition is optional

Path Types

Incoming requests match based on assigned type in rule

Path Type	Rule	Incoming Request
Exact	<code>/corellian/api</code>	Matches <code>/corellian/api</code> but does not match <code>/corellian/test</code> or <code>/corellian/api/</code> .
Prefix	<code>/corellian/api</code>	Matches <code>/corellian/api</code> and <code>/corellian/api/</code> but does not match <code>/corellian/test</code> .

Listing and Describing Ingresses

Renders hosts, IP addresses, and ports

```
$ kubectl get
    ingress
NAME          CLASS    HOSTS          ADDRESS          PORTS    AGE
corellian     <none>   star-alliance.com 192.168.64.15    80       10m
$ kubectl describe ingress corellian
Name:                corellian
Namespace:           default
Address:             192.168.64.15
Default backend:     default-http-backend:80 (<error: \
Rules:               endpoints "default-http-backend" not found>)
  Host              Path  Backends
  ----             -
  star-alliance.com /corellian/api  corellian:8080 (172.17.0.5:8080)
Annotations:        <none>
Events:             <none>
```

Configuring DNS for an Ingress

Making an Ingress convenient to use by end users

- To resolve the Ingress, you'll need to configure DNS entries to the external address.
- You'll need to either configure an A record, or a CNAME record. The [ExternalDNS](#) cluster add-on can help with managing those DNS records for you.
- If you have no domain or are using a local Kubernetes solution, you can add the mapping between IP address and domain name to `/etc/hosts`.

Accessing an Ingress

Use host name, port, and context path

```
$ wget star-alliance.com/coreellian/api --timeout=5 --tries=1
--2021-11-30 19:34:57-- http://star-alliance.com/coreellian/api
Resolving star-alliance.com (star-alliance.com)... 192.168.64.15
Connecting to star-alliance.com (star-alliance.com)|192.168.64.15|:80...
↪connected.
HTTP request sent, awaiting response... 200 OK
...
$ wget star-alliance.com/coreellian/api/ --timeout=5 --tries=1
--2021-11-30 15:36:26-- http://star-alliance.com/coreellian/api/
Resolving star-alliance.com (star-alliance.com)... 192.168.64.15
Connecting to star-alliance.com (star-alliance.com)|192.168.64.15|:80...
↪connected.
HTTP request sent, awaiting response... 404 Not Found
2021-11-30 15:36:26 ERROR 404: Not Found.
```



TOPIC 4

NetworkPolicy

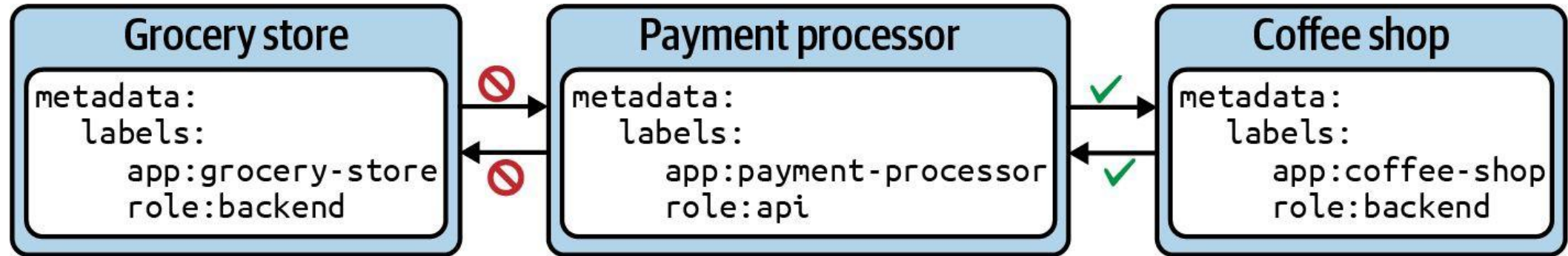
Allow-list Pod-to-Pod traffic

Restrict the Default Open Network

- By default every Pod can talk to every other Pod — NetworkPolicies allow-list specific traffic.
- Default-deny: podSelector: {} with policyTypes:[Ingress] and no ingress rules blocks all inbound.
- Allow by podSelector (Pods by label) or namespaceSelector (whole namespaces) in from/to.
- Egress works the same way — the classic exam pattern is 'deny all egress except DNS (port 53)'.
- Policies are additive: multiple policies combine with logical OR on their allow rules.

Example Network Policy

Restricting access to the payment processor Pod from other Pods



Default Deny Network Policies

Network policies are additive

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny-all
spec:
```

```
  podSelector: {}
```

```
  policyTypes:
```

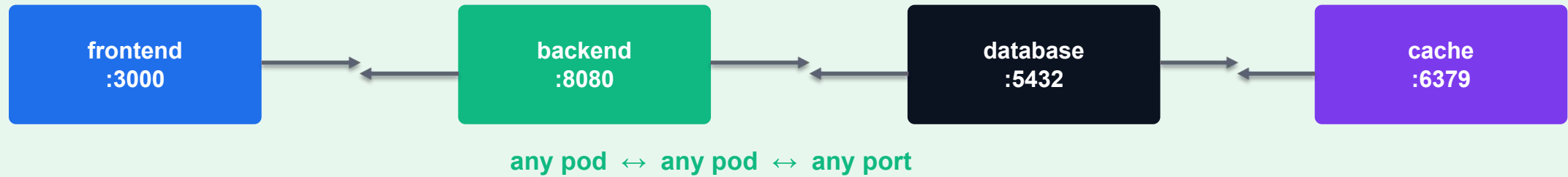
- Ingress
- Egress

Selects all Pods in the namespace

Applies in incoming and outgoing traffic

Default: All Pods Can Communicate

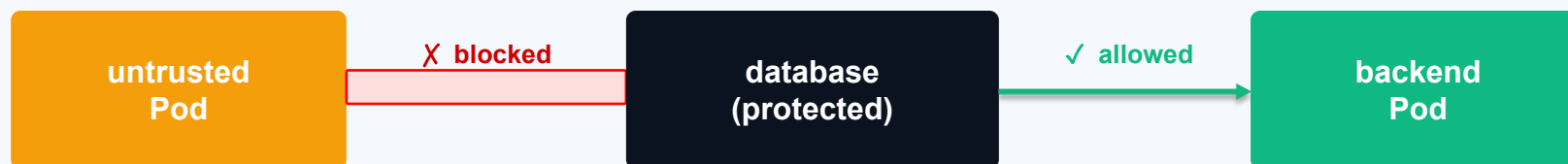
Namespace (no NetworkPolicy)



Without NetworkPolicy: every Pod can reach every other Pod on any port — no isolation

NetworkPolicy — Restrict Ingress Traffic

Namespace: production



```
kind: NetworkPolicy
spec:
  podSelector: {matchLabels: {app: database}}
  policyTypes: [Ingress]
  ingress:
    - from:
      - podSelector: {matchLabels: {role: backend}}
```

```
    ports:
      - port: 5432
```

NetworkPolicy — Restrict Egress Traffic

Allow backend to reach only the database (port 5432) — block everything else

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: backend-egress
  namespace: production
spec:
  podSelector:
    matchLabels:
      role: backend
  policyTypes:
    - Egress
  egress:
    - to:
        - podSelector:
            matchLabels:
              app: database
      ports:
        - protocol: TCP
          port: 5432
    - ports: # allow DNS
        - protocol: UDP
          port: 53
```

NetworkPolicy Patterns

- Default-deny ingress: select all pods, no ingress rules → blocks all incoming traffic.
 - `kind: NetworkPolicy spec: podSelector: {} policyTypes: [Ingress] ingress: []`
- Default-deny egress: select all pods, no egress rules → blocks all outgoing traffic.
 - `kind: NetworkPolicy spec: podSelector: {} policyTypes: [Egress] egress: []`
- Always allow DNS (port 53 UDP) in egress rules, or pods cannot resolve service names.
- `namespaceSelector`: target all pods in a specific namespace (e.g. monitoring).
- `ipBlock`: allow/deny traffic to/from external CIDRs (e.g. 192.168.0.0/16).

NetworkPolicy

LAB 30

NetworkPolicy

By default every Pod can talk to every other Pod. NetworkPolicies allow-list specific sources and destinations. CKAD regularly includes a NetworkPolicy question — you must write a default-deny and a selective-allow policy from scratch.

You'll build

Apply default-deny ingress, a selective podSelector allow, an egress DNS-only lockdown, then a cross-namespace allow.

Key commands: `spec: · ingress: · » Default-deny`

NetworkPolicy

Step 1 — Default deny: block all ingress

```
spec:
  podSelector: {}
  policyTypes:
  - Ingress
```

Step 2 — Selective allow: only role=allowed reaches the backend

```
spec:
  podSelector:
    matchLabels: {app: backend}
  policyTypes: [Ingress]
  ingress:
  - from:
    - podSelector:
        matchLabels: {role: allowed}
    ports:
    - {protocol: TCP, port: 5678}
```

NetworkPolicy (cont.)

Step 3 — Egress lockdown: allow DNS only

```
spec:
  podSelector: {matchLabels: {role: blocked}}
  policyTypes: [Egress]
  egress:
  - to:
    - namespaceSelector: {}
      podSelector: {matchLabels: {k8s-app: kube-dns}}
    ports: [{protocol: UDP, port: 53}, {protocol: TCP, port: 53}]
```

Step 4 — Cross-namespace allow (namespaceSelector)

```
ingress:
- from:
  - namespaceSelector:
      matchLabels: {purpose: trusted}
```

NetworkPolicy (cont.)

Note Default-deny = podSelector: {} + policyTypes: [Ingress] with no ingress list. podSelector in from matches Pods by label in the same namespace; namespaceSelector matches whole namespaces. Policies are additive — multiple policies combine with logical OR.

NetworkPolicy

✓ Test it

After the allow policy, client-ok (role=allowed) reaches the backend while client-bad stays blocked, proving the NetworkPolicy is enforced.

Killercoda: <https://killercoda.com/playgrounds/scenario/kubernetes>



What is an Network Policy Controller?

Evaluating Network Policy rules

- A Network Policy cannot work without a Network Policy controller. The Network Policy controller evaluates the collection of rules defined by a Network Policy.
- One option for such a Network Policy controller is [Cilium](#).

Network Policy YAML Manifest

Key-value pairs can be parsed from different sources

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: api-allow
spec:
  podSelector:
    matchLabels:
      app: payment-processor
      role: api
  ingress:
    - from:
        - podSelector:
            matchLabels:
              app: coffeeshop
```

Selects the Pod the policy should apply to by label selection

Allows incoming traffic from the Pod with matching labels within the same namespace

Network Policy Rules

Attributes that form the rules

Attribute	Description
<code>podSelector</code>	Selects the Pods in the namespace to apply the network policy to.
<code>policyTypes</code>	Defines the type of traffic (i.e., ingress and/or egress) the network policy applies to.
<code>ingress</code>	Lists the rules for incoming traffic. Each rule can define <code>from</code> and <code>ports</code> sections.
<code>egress</code>	Lists the rules for outgoing traffic. Each rule can define <code>to</code> and <code>ports</code> sections.

Behavior of `to` and `from` Selectors

Ingress and egress sections define the same attributes

Attribute	Description
<code>podSelector</code>	Selects Pods by label(s) in the same namespace as the Network Policy which should be allowed as ingress sources or egress destinations.
<code>namespaceSelector</code>	Selects namespaces by label(s) for which all Pods should be allowed as ingress sources or egress destinations.
<code>namespaceSelector</code> and <code>podSelector</code>	Selects Pods by label(s) within namespaces by label(s).

Rendering Network Policy Details

Ingress and egress rules can be inspected in details

```
$ kubectl describe networkpolicy api-allow
```

```
Name:          api-allow
```

```
Namespace:     default
```

```
Created on:    2020-09-26 18:02:57 -0600 MDT
```

```
Labels:        <none>
```

```
Annotations:   <none>
```

```
Spec:
```

```
  PodSelector:    app=payment-processor,role=api
```

```
  Allowing ingress traffic:
```

```
    To Port: <any> (traffic allowed to all ports)
```

```
    From:
```

```
      PodSelector: app=coffeeshop
```

```
  Not affecting egress traffic
```

```
  Policy Types:  Ingress
```

Verifying the Runtime Behavior

Communication from Pods not matching with labels will be blocked

```
$ kubectl run grocery-store --rm -it --image=busybox
```

```
↵
```

```
  -l app=grocery-store,role=backend -- /bin/sh
```

```
/ # wget --spider --timeout=1 10.0.0.51
```

```
Connecting to 10.0.0.51 (10.0.0.51:80)
```

```
wget: download timed out
```

```
/ # exit
```

```
pod "grocery-store" deleted
```

```
$ kubectl run coffeeshop --rm -it --image=busybox ↵
```

```
  -l app=coffeeshop,role=backend -- /bin/sh
```

```
/ # wget --spider --timeout=1 10.0.0.51
```

```
Connecting to 10.0.0.51 (10.0.0.51:80)
```

```
remote file exists
```

```
/ # exit
```

```
pod "coffeeshop" deleted
```


Restricting Access to Ports

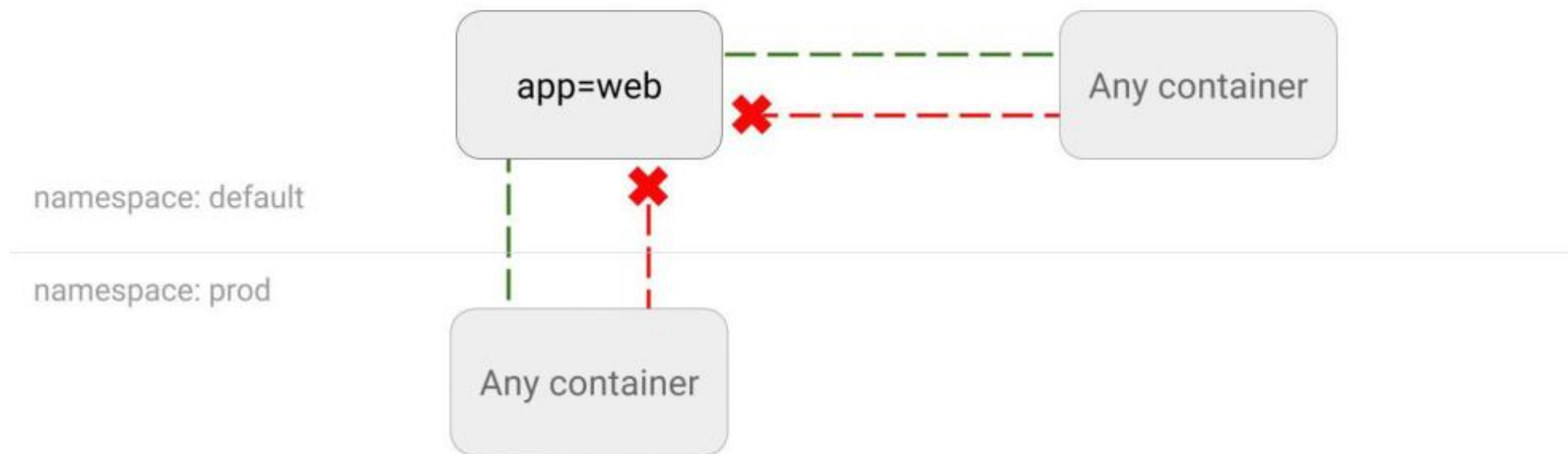
By default, all ports are accessible

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: port-allow
spec:
  podSelector:
    matchLabels:
      app: backend
  ingress:
  - from:
    - podSelector:
      ports:
      - protocol: TCP
        port: 8080
```

Only allow incoming
traffic to port 8080

Scenario-Based Learning Resource

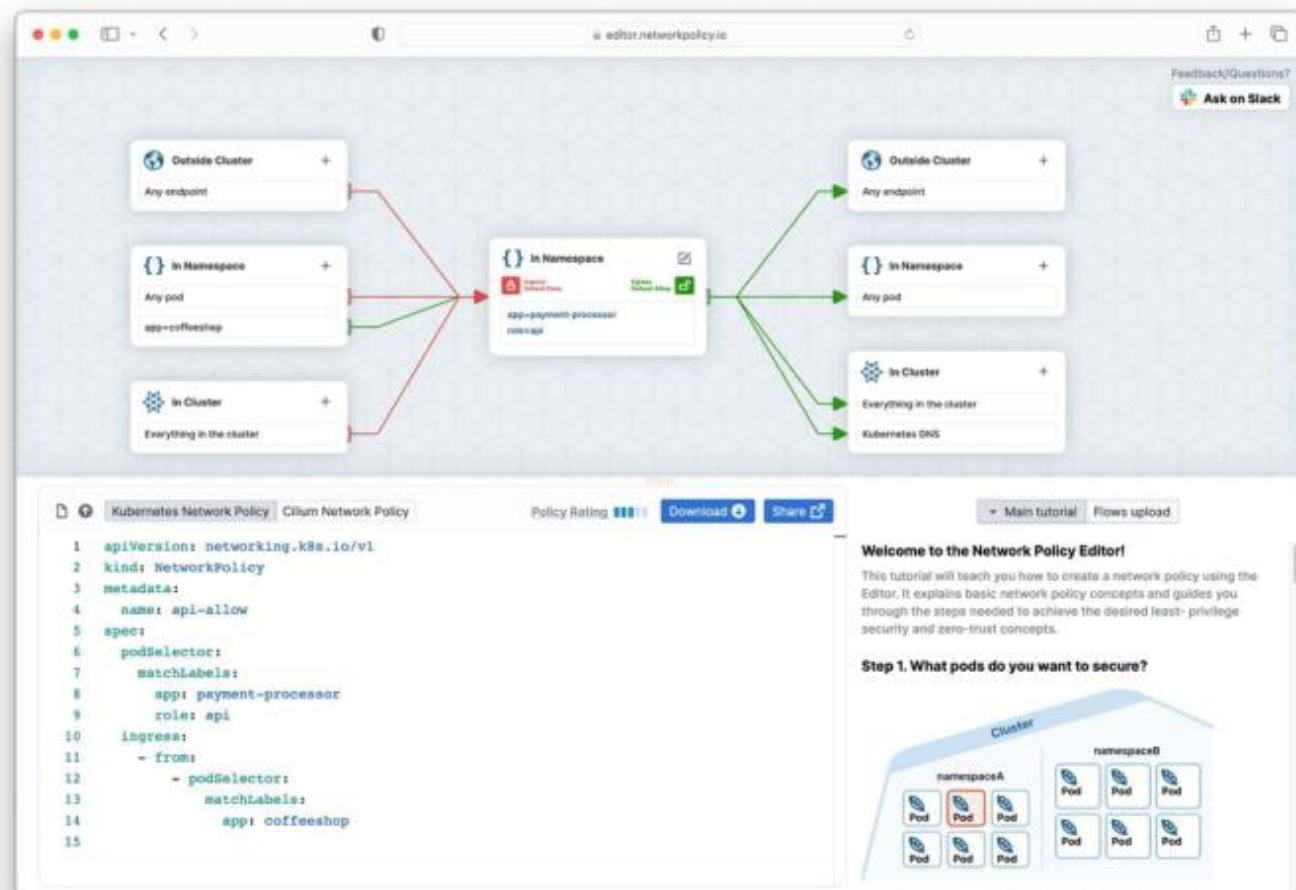
Exercising use cases for Network Policies is helpful with understanding concept



<https://github.com/ahmetb/kubernetes-network-policy-recipes>

Visualizing and Analysing a Network Policy

The page networkpolicy.io provides policy editor



COURSE COMPLETE

All five CKAD domains complete.

Design & build · Deployment · Observability · Config & Security · Services & Networking.

CKAD Exam Environment

PSI Secure Browser — Remote Desktop | 2 hours | ~15–20 questions | 66% pass mark

- Single monitor; 1 additional tab allowed (kubernetes.io docs only)
- No notes, no bookmarks — docs must be navigated fresh
- Allowed: kubernetes.io/docs, kubernetes.io/blog, helm.sh/docs
- Remote desktop — use built-in terminal and browser only
- Candidate ID verification required before exam starts
- All questions are practical — live cluster
- Multiple contexts — always switch context per question
- Partial credit possible — attempt every question
- Paste YAML from docs or use `--dry-run=client -o yaml`
- Flag difficult questions and return — time management critical

kubectl Shortcuts for the Exam

Set alias and enable auto-completion (pre-configured in exam)

```
alias k=kubectl
source <(kubectl completion bash)
complete -o default -F __start_kubectl k
```

```
k get po      # pods
k get svc     # services
k get deploy  # deployments
k get ns      # namespaces
```

Generate YAML with --dry-run (fastest way to create manifests)

```
k run nginx --image=nginx --dry-run=client -o yaml > pod.yaml
k create deploy myapp --image=myapp:1.0 --dry-run=client -o yaml
k create svc clusterip my-svc --tcp=80:80 --dry-run=client -o yaml
k apply -f pod.yaml
k delete -f pod.yaml --force --grace-period=0
```

Working in the Right Context & Namespace

Each exam question specifies a cluster context — switch before solving

```
kubectl config get-contexts  
kubectl config use-context <context-name>  
kubectl config current-context
```

Set the default namespace to avoid typing -n every time

```
kubectl config set-context --current --namespace=<namespace>  
kubectl config view --minify | grep namespace  
kubectl get pods -n kube-system # one-shot without changing default
```

Exam Strategy & Time Management

- Read the question fully — note context, namespace, and exact requirements
- Switch context FIRST before any kubectl command
- Use `--dry-run=client -o yaml` to generate YAML skeletons, then edit
 - Faster than writing YAML from scratch every time
- Use 'kubectl explain' for YAML field names (no internet needed)
 - `kubectl explain pod.spec.containers.livenessProbe`
- Flag and skip — if a question takes >5 min, flag and return
- Verify after each task: `kubectl get / describe` to confirm result
- Use `kubectl api-resources` to find short names and API groups
- Practice: KillerCoda.com and killer.sh mock exams are the best prep



Kubernetes Version Used in Exam

Suffix in curriculum file name indicates version, changes ~ every 3 months

zuzannapn Archive CKAD v.1.26		2216e88 2 weeks ago	🕒 152 commits
📁 kcna	add freeCodeCamp KCNA resource (#57)		last year
📁 old-versions	Archive CKAD v.1.26		2 weeks ago
📄 .DS_Store	Archive CKAD v.1.26		2 weeks ago
📄 CKAD_Curriculum_v1.27.pdf	Updating exam curriculum to V1.27		last month
📄 CKA_Curriculum_v1.27.pdf	Updating exam curriculum to V1.27		last month
📄 CKS_Curriculum_v1.27.pdf	Updating exam curriculum to V1.27		2 weeks ago
📄 KCNA_Curriculum.pdf	KCNA Exam Curriculum		2 years ago
📄 PCA_Curriculum.pdf	PCA Exam Curriculum		10 months ago
📄 README.md	Update README.md		8 months ago

Using Auto-Completion for kubectl

Preconfigured in exam environment

```
$ kubectl cre<tab>
```

\$ kubectl create



<https://kubernetes.io/docs/reference/kubectl/cheatsheet/#kubectl-autocomplete>

Lunch Break

1 hour



MOCK EXAM

Mock-Exam Practice — from 1:00 PM

Practise on [killer.sh](#) / [Killercoda](#) before the graded assessment · ask questions freely

Mock-Exam Practice (1:00 PM)

- Warm-up before the graded assessment — work hands-on questions across all five domains.
- Use `killer.sh` and `Killercoda`; rehearse the alias `k`, `$do` dry-run and `kubectl` explain workflow.
- Time yourself and flag weak areas — the real CKAD is 2 hours, 100% hands-on.
- Trainer is available for questions; the final assessment begins at 2:00 PM.

Take the Online CKAD Practice Exams

- Reinforce all five CKAD domains with realistic, exam-style questions before exam day.
- 3 full practice exams · 65 questions · 120-minute timer · 66% pass mark — aligned to CNCF CKAD v1.35.
- Practice mode reveals the answer and a full explanation after each question; Exam mode mirrors the real timed test.
- Start free with the practice teaser — no credit card required; sign in to unlock the full bundle.

Sign up and practise at:

exams.tertiaryinfotech.com/practice-exams/linuxfoundation/linuxfoundation-ckad

The screenshot shows the Tertiary Exams website interface for the Certified Kubernetes Application Developer (CKAD) practice exam. The header includes the Tertiary Exams logo, navigation links (Practice Exams, Vendors, About Us, FAQ), and a Sign in button with a Get started button. The main content area is titled 'All exams / Linux Foundation' and 'Linux Foundation CKAD Associate'. The exam title is 'Certified Kubernetes Application Developer (CKAD)'. Below the title, it states: 'All 3 CKAD practice exams in one bundle — covering application design & build, deployment, observability & maintenance, environment/configuration/security, and services & networking, aligned to CNCF CKAD v1.35.' The exam details are listed in a table: Questions (65), Duration (120 min), and Pass score (66%). The exam mode options are 'Practice mode' (See the correct answer and full explanation immediately after each question. Best for studying.) and 'Exam mode' (120-minute timer, no answer reveal until you submit. Auto-saves and auto-submits — like the real exam.). The domains covered are listed in a table: Application Design and Build (20%), Application Deployment (20%), Application Observability and Maintenance (15%), Application Environment, Configuration and Security (25%), and Services and Networking (20%). The pricing section shows 'Practice Exam' for \$20.00 and 'Exam Voucher (practice exams included)' for \$395.00. A note states: 'Heads up: exam vouchers take 3-5 business days to deliver — you'll get an email with your code when it's ready. Practice access is unlocked immediately on purchase.' The bottom right corner has a Sign in to buy button and a WhatsApp icon.

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Certified Kubernetes Application Developer (CKAD)

All 3 CKAD practice exams in one bundle — covering application design & build, deployment, observability & maintenance, environment/configuration/security, and services & networking, aligned to CNCF CKAD v1.35.

Questions	Duration	Pass score
65	120 min	66%

Practice mode

See the correct answer and full explanation immediately after each question. Best for studying.

Exam mode

120-minute timer, no answer reveal until you submit. Auto-saves and auto-submits — like the real exam.

Domains covered	
Application Design and Build	20%
Application Deployment	20%
Application Observability and Maintenance	15%
Application Environment, Configuration and Security	25%
Services and Networking	20%

2026 PRACTICE EXAM DETAILS

Exam code	CKAD
Level	Associate
Duration	120 min
Pass score	66%
Question count	65

FREE

Try our free practice teaser

No credit card required.

[Start free practice exam](#)

PICK A PLAN

☒ Practice Exam	\$20.00
☐ Exam Voucher (practice exams included)	\$395.00

Heads up: exam vouchers take 3-5 business days to deliver — you'll get an email with your code when it's ready. Practice access is unlocked immediately on purchase.

[Sign in to buy](#)



ASSESSMENT

Final Assessment — from 2:00 PM

Written / MCQ on the LMS · open book · covers all five domains and the hands-on labs

Assessment



Final Assessment

- Written / MCQ assessment on the LMS
- Mock-exam practice from 1:00 PM; final assessment from 2:00 PM
- Covers all five CKAD domains and the hands-on labs
- Open book — slides, Learner Guide & kubernetes.io
- Must be attempted for SSG funding



Funding & Competency

Minimum 75% attendance

Based on SSG digital attendance records.

Assessed as Competent

Pass the written / MCQ assessment.

Briefing for Assessment



Clear your space

Keep only approved materials on the table.



No recording

No photos of the assessment.



Work individually

No discussion during the assessment.



Use the LMS

The assessment is delivered online.



Open book

Slides, Learner Guide & kubernetes.io are allowed.



Time's up

Submit when time is up.

THANK YOU

Thank you — and good luck with the CKAD exam!

Keep practising on [killer.sh](#) and [Killercoda](#); the exam is 2 hours, 100% hands-on.